

0001. 1. In case of conflicting information you can always trust your Seat-of-the-Pants-Sense.
 2. In case of conflicting information between the sensory organs and the instruments you must believe the instruments.
- a) 1 is false , 2 is correct.
b) 1 and 2 are false.
c) 1 and 2 are correct.
d) 1 is correct , 2 is false.

Answer ;

- a) 1 is false , 2 is correct.
-

0002. A barotrauma of the middle ear ;
- a) is to be expected during rapid decompressions , but an emergency descent immediately following the decompression will eliminate the problem.
b) is only caused by large pressure changes during climb.
c) is more likely , when the pilot is flying with a respiratory infection and during descent.
d) causes severe pain in the sinuses.

Answer ;

- c) is more likely , when the pilot is flying with a respiratory infection and during descent.
-

0003. A barotrauma of the middle ear is ;
- a) an acute or chronic trauma to the middle ear caused by a difference of pressure on either side of the eardrum.
b) a dilatation of the Eustachian tube.
c) an infection of the middle ear caused by rapid decompression.
d) a bacterial infection of the middle ear.

Answer ;

- a) an acute or chronic trauma to the middle ear caused by a difference of pressure on either side of the eardrum.
-

0004. A Captain running a “Laisser-faire” cockpit will ;
- a) be an efficient crew member.
b) make few suggestions or decisions.
c) will encourage a “sterile” cockpit.
d) fill a vacuum.

Answer ;

- b) make few suggestions or decisions.
-

0005. A Captain running an Autocratic cockpit is normally ;
- a) under-worked.
b) a good leader.
c) efficient.
d) over-loaded.

Answer ;

- d) over-loaded.
-

0006. A copilot has passed an upgrading course to become a captain. Which psychological consequence is most likely ?
- a) His/her self-concept is going to change because of new roles and tasks which have to be incorporated.
b) His/her self -concept is going to be stabilized because of the higher status as a captain.
c) An upgrading never has psychological consequences.
d) The increased command authority leads to a higher professionalism.

Answer ;

- a) His/her self-concept is going to change because of new roles and tasks which have to be incorporated.
-

0007. A “Graveyard-Spin” is ;
- a) a fatal accident caused by a state of spinning the fluid in the brain.
b) a spin maneuver that should only be performed by well trained pilots.
c) an illusion caused by misperception of being higher than you really are when performing a spin. Thus the recovery tends to be dangerously low.
d) a spin in which the pilot , on recovery , tends to re-enter the spin due to the somatogyral illusion where the vestibular system no longer senses radial acceleration.

Answer ;

- d) a spin in which the pilot , on recovery , tends to re-enter the spin due to the somatogyral illusion where the vestibular system no longer senses radial acceleration.
-

0008. A high degree of cockpit automation may alter the traditional tasks of the pilots in a way , that ;
- a) the attention of the cockpit crew will become reduced as a consequence of “being out of the loop”.
 - b) it is guaranteed that the crew always maintains situational awareness.
 - c) Crew Coordination can be neglected on long haul flights without compromising safety.
 - d) the crew can pay more attention to problem-solving in an abnormal situation without monitoring the automatic systems.

Answer ;

- a) the attention of the cockpit crew will become reduced as a consequence of “being out of the loop”.
-

0009. A human breathing 100% oxygen at 33.700 ft is the equivalent of breathing air at ;

- a) 8.000 ft.
- b) 10.000 ft.
- c) sea level.
- d) 21.300 ft.

Answer ;

- c) sea level.
-

0010. A large number of medical preparations can be bought without a doctor’s prescription. In relation to using these preparations , which of the following is correct ?

- a) They will cause a condition of over-arousal.
- b) The side effects of these types of preparations are sufficiently negligible as to be ignored by pilots.
- c) They have no side effects which would give problems to a pilot during flight.
- d) A pilot using any of these preparations should get professional advice from an aviation medical specialist if he intends to fly and self-medicate at the same time.

Answer ;

- d) A pilot using any of these preparations should get professional advice from an aviation medical specialist if he intends to fly and self-medicate at the same time.
-

0011. A leader who demonstrates a democratic and co-operative style will deal with conflict by ;

- a) trying to clarify the reasons and causes of the conflict with all persons involved.
- b) deciding what to do and pushes his own opinion through.
- c) trying to reconcile all persons involved in the conflict and tries to re-establish a pleasant and friendly atmosphere within the team.
- d) keeping a neutral position and does not participate in arguing.

Answer ;

- a) trying to clarify the reasons and causes of the conflict with all persons involved.
-

0012. A non synergistic cockpit ;

- a) is not very dangerous as each person checks everything personally.
- b) is characterised by withdrawn crewmembers and unclear communication.
- c) is characterised by a highly efficient crew , communicating appropriately with the outside.
- d) always results from an over-relaxed atmosphere.

Answer ;

- b) is characterised by withdrawn crewmembers and unclear communication.
-

0013. A passenger complains about a painful inflated belly at 8.000 feet. You advise him to ;

- 1. unbuckle and massage the belly.
 - 2. stand up and let go the gases out of the intestines.
 - 3. eat less gas forming food and avoid carbonhydrated beverages before flight in the future.
 - 4. drink a lot of water throughout the flight.
- a) 2 , 3 and 4 are correct.
 - b) 1 , 2 and 3 are correct.
 - c) 1 and 3 not advisable.
 - d) only 4 is correct.

Answer ;

- b) 1 , 2 and 3 are correct.
-

0014. A person being exposed to extreme or prolonged stress factors can perceive ;

- a) coping stress.
- b) distress.
- c) eustress.
- d) stressors.

Answer ;

- b) distress.
-

0015. A person's attitudes define a set of learned dispositions (likes and dislikes) which ;
- a) is exactly the same as a person's personality.
 - b) do not depend on a person's past experiences.
 - c) cannot be modified through training.
 - d) influence a person's reaction and response to people , situations , etc.

Answer ;

- d) influence a person's reaction and response to people , situations , etc.
-

0016. A pilot , accelerating or decelerating in level flight may get ;
- a) the illusion to turning.
 - b) the feeling of rotation.
 - c) the illusion of climbing or descending.
 - d) the impression of stationary objects moving to the right or left.

Answer ;

- c) the illusion of climbing or descending.
-

0017. A pilot accustomed to landing on a wide runway may find , when approaching to a narrow runway , that he/she is at a ;
- a) greater height than he actually is with the tendency to land short.
 - b) lower height and the impression of landing slow.
 - c) greater height and the impression of landing short.
 - d) lower than actual height with the tendency to overshoot.

Answer ;

- a) greater height than he actually is with the tendency to land short.
-

0018. A pilot approaching a runway which is narrower than normal may feel he is at a greater height than he actually is. To compensate he may fly a ;

- a) compensatory glide path and land long.
- b) flatter than normal approach with the tendency to undershoot.
- c) compensatory glide path and stall out.
- d) higher than normal approach with the tendency to overshoot.

Answer ;

- b) flatter than normal approach with the tendency to undershoot.
-

0019. A pilot approaching an upsloping runway ;
- a) may feel that he is higher than actual. This illusion may cause him to land short.
 - b) is performing a steeper than normal approach , landing long.
 - c) establishes a higher than normal approach speed.
 - d) establishes a slower than normal approach speed with the risk of stalling out.

Answer ;

- a) may feel that he is higher than actual. This illusion may cause him to land short.
-

0020. A pilot becomes skilled when he/she ;
- 1. trains or practises regularly.
 - 2. knows how to manage himself/herself.
 - 3. possesses all the knowledge associated with his aircraft.
 - 4. knows how to keep resources in reserve for coping with the unexpected.

- a) 1 , 2 , 4.
- b) 1 , 2 , 3 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2.

Answer ;

- a) 1 , 2 , 4.
-

0021. A pilot can be described as being proficient , when he/she ;
- a) has automated a large part of the necessary flight deck routine operations in order to free his/her cognitive resources.
 - b) knows how to off-load all his/her resources to the automation of tasks.
 - c) is capable of maintaining a high level of arousal during a great bulk of the flight.
 - d) is able to reduce his/her arousal to a low level during the entire flight.

Answer ;

- a) has automated a large part of the necessary flight deck routine operations in order to free his/her cognitive resources.
-

0022. A pilot can overcome hyperventilation by ;
- a) controlling the rate and depth of breathing and/or breathing into a bag.
 - b) depending on instruments.
 - c) the use of drugs stabilizing blood pressure.
 - d) increasing the rate and depth of breathing to eliminate harmful carbon dioxide.

Answer ;

- a) controlling the rate and depth of breathing and/or breathing into a bag.

0023. A pilot can prevent hypoxia by ;
- a) relying on the body's built in warning system recognizing any stage of hypoxia.
 - b) not exceeding a cabin pressure altitude of 20000 ft.
 - c) swallowing , yawing and applying the Valsalva method.
 - d) using additional oxygen when flying above 10000 ft.

Answer ;

- d) using additional oxygen when flying above 10000 ft.

0024. A pilot , climbing in a non-pressurised aircraft and without using supplemental oxygen will pass the “critical threshold” at approximately ;

- a) 16 000 ft.
- b) 38 000 ft.
- c) 18 000 ft.
- d) 20 000 ft.

Answer ;

- d) 20 000 ft.

0025. A pilot experiences difficulties in the control of his aircraft and consequently omits to monitor the fuel level. This is caused by ;

- a) a lack of attention due to distraction.
- b) both a lack of attention and a lack of alertness.
- c) this pilot manages his attention well.
- d) a lack of alertness.

Answer ;

- a) a lack of attention due to distraction.

0026. A pilot , height 1.7 m and weighing 57 kg. She is ;

- a) overweight.
- b) normal.
- c) underweight.
- d) obese.

Answer ;

- b) normal.

$$\text{BMI} = \text{Body mass (Kilo)} : \text{Height}^2 \text{ (Boy}^2 \text{)}$$
$$\text{BMI} = 57 : 1.7^2 = 57 : 2.9584 = 19.26$$

0027. A pilot is 1.7 metres tall and weighs 57 kgs. She is ;

- a) normal.
- b) obese.
- c) underweight.
- d) overweight.

Answer ;

- a) normal.

BMI		Category
Male	Female	
< 20	< 19	Underweight
20 – 25	19 – 24	Acceptable
> 25 – 30	>24 - 29	Overweight
> 30	> 29	Obese

0028. A pilot is used to land on small and narrow runways only. Approaching a larger and wider runway can lead to ;

- a) a steeper than normal approach dropping low.
- b) a flatter than normal approach with the risk of “ducking under”.
- c) the risk to land short of the overrun.
- d) an early or high “round out”.

Answer ;

- d) an early or high “round out”.

0029. A pilot may get the illusion of low altitude on approach although the aircraft is on the correct glidepath ;

- a) depending on the approach speed of the aircraft.
- b) when the runway is unusually narrow.
- c) on night approaches only.
- d) when the runway is wider than he is used to.

Answer ;

- d) when the runway is wider than he is used to.

0030. A pilot may suffer from hypoxia ;

- a) after decompression at high altitude and not using additional oxygen in time.
- b) if he/she is flying an unpressurised airplane at an altitude of 15 000 feet and breathing 100 % oxygen.
- c) after decompression to 30 000 feet and using 100 % oxygen via an oxygen mask.
- d) if his rate of climb exceeds 5 000 ft/min.

Answer ;

- a) after decompression at high altitude and not using additional oxygen in time.
-

0031. A pilot meets the ground engineer in charge on the apron. Although the pilot has not been through flight preparation yet , he nevertheless instructs the engineer as to the amount of fuel to be loaded , since he is used to flying this route. This behaviour results from ;

- a) a bias in selecting data.
- b) a decision bias due to group (social) pressure.
- c) a confirmation bias.
- d) a decision bias due to habits (frequency bias).

Answer ;

- d) a decision bias due to habits (frequency bias).
-

0032. A pilot should not fly immediately after donating blood because ;

- a) your blood-pressure is too low after blood-donation.
- b) your heart rate is too low after blood-donation.
- c) you have an increased susceptibility to fainting.
- d) the chance you get the bends is higher after blood-donation.

Answer ;

- c) you have an increased susceptibility to fainting.
-

0033. A pilot successfully completes a difficult and stressful landing at an aerodrome. The next time a landing is attempted under the same conditions and at the same aerodrome , is the pilot likely to experience ;

- a) the pilot should only attempt a landing at the same aerodrome if the conditions are improved.
- b) the same stress level as the first landing.
- c) a higher level of stress.
- d) a lower level of stress.

Answer ;

- d) a lower level of stress.
-

0034. A pilot , trying to pick up a fallen object from the cockpit floor during a tight turn , experiences ;

- a) autokinetic illusion.
- b) pressure vertigo.
- c) coriolis illusion.
- d) barotrauma.

Answer ;

- c) coriolis illusion
-

0035. A pilot using a checklist is an example of the interaction within the SHELL Concept of ;

- a) L and L.
- b) L and S.
- c) L and H.
- d) L and E.

Answer ;

- b) L and S.
-

0036. A pilot who is hyperventilating for a prolonged period of time may even become unconscious. Hyperventilation is likely to occur , when ;

- a) he/she is flying a tight turn.
- b) the pilot is emotionally aroused.
- c) there is an increased blood flow to the brain.
- d) there is a low CO₂-pressure in the blood.

Answer ;

- b) the pilot is emotionally aroused.
-

0037. A pilot who is trying to pick up a fallen object from the cockpit floor during a tight turn , may experience ;

- a) pressure vertigo.
- b) autokinetic illusion.
- c) coriolis illusion.
- d) barotrauma.

Answer ;

c) coriolis illusion.

0038. A pilot who smokes will lose some of his capacity to transport oxygen combined with haemoglobin. Which percentage of his total oxygen transportation capacity would he give away when he smokes one pack of cigarettes a day ?

- a) 12 - 18%.
- b) 5 - 8%.
- c) 20 - 25%.
- d) 0.5 - 2%.

Answer ;

b) 5 - 8%.

0039. A pressurized cabin helps to prevent ;

- 1. decompression sickness.
 - 2. the problem of expansion of gases in the intestines.
 - 3. hypoxia.
 - 4. coronary disease.
- a) 1 , 2 and 3 are correct.
 - b) 2 , 3 and 4 are correct.
 - c) 1 , 3 and 4 are correct.
 - d) 1 , 2 and 4 are correct.

Answer ;

a) 1 , 2 and 3 are correct.

0040. A selective attentional mechanism is required ;

- a) because of the limitations of the sense organs.
- b) because of the limited capacity of the central decision maker and working memory.
- c) because of limitations in our store of motor programmes.
- d) because the capacity of the long term memory is limited.

Answer ;

b) because of the limited capacity of the central decision maker and working memory.

0041. A shining light is fading out (i.e. when flying into fog , dust or haze). What kind of illusion could the pilot get ?

- a) The source of light moves away from him.
- b) The source of light is approaching him with increasing speed.
- c) The light source will make the pilot believe , that he is climbing.
- d) The source of light stands still.

Answer ;

a) The source of light moves away from him.

0042. A stereotype and involuntary reaction of the organism on stimulation of receptors is called ;

- a) reflex.
- b) change of stimulation level.
- c) control system.
- d) data processing.

Answer ;

a) reflex.

0043. A stress reaction is ;

- a) the specific stimuli causing a human body to respond.
- b) the specific response of the body to demands placed on a person.
- c) the non-specific stimuli causing a human body to respond.
- d) the non-specific response of the body to demands placed on a person.

Answer ;

d) the non-specific response of the body to demands placed on a person.

0044. Accommodation , which enables a clear image to be obtained , is accomplished by which of the following ?

- a) The retina.
- b) The cones.
- c) The rods.
- d) The crystalline lens.

Answer ;

d) The crystalline lens.

0045. According to the different phases of the “General Adaptation Syndrome” which of the following statement/s is/are correct ?

- 1. During the alarm phase stress hormones (i.e. adrenalin) will cause a massive release of glucose into the blood , an acceleration of pulse and blood pressure as well as an increase in the rate and depth of breathing.
 - 2. During the resistance phase the parasympathetic system uses a different type of hormone (cortisol) assisting for the conversion of fat into sugar.
 - 3. During the exhaustion phase the body has to be given time to eliminate the waste products which have been generated excessively during the two preceding phases.
- a) 2 and 3 are correct , 1 is false.
 - b) 1 and 2 are correct , 3 is false.
 - c) 1 , 2 and 3 are correct.
 - d) only 1 is correct.

Answer ;

c) 1 , 2 and 3 are correct.

0046. According to Rasmussen’s model , errors are of the following type(s) in skill-based behaviour ;

- a) routine errors.
- b) creative errors.
- c) handling errors.
- d) knowledge errors.

Answer ;

a) routine errors.

0047. According to Rasmussen’s model , errors in rule-based control mode are of the following type(s) ;

- a) creative errors.
- b) handling errors.
- c) errors of technical knowledge.
- d) routine errors.

Answer ;

c) errors of technical knowledge.

0048. According to the different phases of the “General Adaptation Syndrome” which of the following statement/s is/are correct ?

- 1. During the alarm phase stress hormones (i.e. adrenalin) will cause a massive release of glucose into the blood , an acceleration of pulse and blood pressure as well as an increase in the rate and depth of breathing.
 - 2. During the resistance phase the parasympathetic system uses a different type of hormone (cortisol) assisting for the conversion of fat into sugar.
 - 3. During the exhaustion phase the body has to be given time to eliminate the waste products which have been generated excessively during the two preceding phases.
- a) 2 and 3 are correct , 1 is false.
 - b) only 1 is correct.
 - c) 1 and 2 are correct , 3 is false.
 - d) 1 , 2 and 3 are correct.

Answer ;

d) 1 , 2 and 3 are correct.

0049. Action plans (SOPs) in a cockpit must ;

- a) only follow the manufacturers proposals and not reflect individual operators cockpit philosophies.
- b) be shared by the members of the crew and updated at each modification in order to maintain maximum synergy.
- c) be tailored to the individual pilot’s needs in order to facilitate the normal operation of the aircraft.
- d) only be tailored to the type of aircraft , regardless of current MCC procedures.

Answer ;

b) be shared by the members of the crew and updated at each modification in order to maintain maximum synergy.

0050. Active errors/failures are committed at ;

- a) the system/system interface and have a delayed effect.
- b) the human/system interface and have a delayed effect.
- c) the system/system interface and have an immediate effect.
- d) the human/system interface and have an immediate effect.

Answer ;

- d) the human/system interface and have an immediate effect.**
-

0051. Acute stress quickly leads to ;

- a) a decrease in the amount of resources mobilized to face the situation.
- b) the mobilization of resources required to cope with the stressor.
- c) a permanent state of incapacitation.
- d) a state of overactivation beyond the control of willpower.

Answer ;

- b) the mobilization of resources required to cope with the stressor.**
-

0052. Adaptation is ;

- a) the reflection of the light at the cornea.
- b) the change of the diameter of the pupil.
- c) the adjustment of the eyes to high or low levels of illumination.
- d) the adjustment of the crystalline lens to focus light on the retina.

Answer ;

- c) the adjustment of the eyes to high or low levels of illumination.**
-

0053. Adverse effects of carbon monoxide increase as ;

- a) relative humidity decreases.
- b) altitude decreases.
- c) air pressure increases.
- d) altitude increases.

Answer ;

- d) altitude increases.**
-

0054. After a cabin pressure loss in approximately 35.000 ft the TUC (Time of Useful Consciousness) will be approximately ;

- a) 10-15 seconds.
- b) 30-40 seconds.
- c) 3-4 minutes.
- d) 5 minutes or more.

Answer ;

- b) 30-40 seconds.**
-

0055. After a decompression to 43.000 ft the TUC (Time of Useful Consciousness) will be approximately ;

- a) 45-60 seconds.
- b) 5-15 seconds.
- c) 60-90 seconds.
- d) 30-45 seconds.

Answer ;

- b) 5-15 seconds.**
-

0056. After a rapid decompression at 35.000 feet , the time of useful consciousness is about ;

- a) 5 minutes.
- b) 30 to 60 seconds.
- c) 15 seconds or less.
- d) 10 minutes.

Answer ;

- b) 30 to 60 seconds.**
-

0057. Active errors/failures are committed at ;

- a) the system/system interface and have a delayed effect.
- b) the system/system interface and have an immediate effect.
- c) the human/system interface and have a delayed effect.
- d) the human/system interface and have an immediate effect.

Answer ;

- d) the human/system interface and have an immediate effect.**
-

0058. After SCUBA diving (more than 30 feet of depth) you have to wait a period of time before flying again. This period is at least ;

- a) 6 hours.
- b) 24 hours.
- c) 48 hours.
- d) 12 hours.

Answer ;

b) 24 hours.

0059. Air at an altitude of 18.000 feet contains , approximately ;

- a) 21% oxygen.
- b) 10% oxygen.
- c) 15% oxygen.
- d) 5% oxygen

Answer ;

a) 21% oxygen.

0060. Alcohol degrades ;

- a) Slow wave sleep.
- b) Stage 2 sleep.
- c) Paradoxical sleep.
- d) Stage 3 and 4 sleep.

Answer ;

c) Paradoxical sleep.

0061. Alcohol , even when taken in minor quantities ;

- a) can make the brain cells to be more susceptible to hypoxia.
- b) may improve the mental functions , so that the symptoms of hypoxia are much better to be identified.
- c) will have no effect at all.
- d) will stimulate the brain , making the pilot resistant to hypoxia.

Answer ;

a) can make the brain cells to be more susceptible to hypoxia.

0062. Alcohol , when taken simultaneously with drugs , may ;

- a) compensate for side effects of drugs.
- b) increase the rate of alcohol elimination from the blood.
- c) show undesired effects only during night flights.
- d) intensify the effects of the drugs.

Answer ;

d) intensify the effects of the drugs.

0063. Although the anticipation of possible events is a good attitude for pilots to acquire , it can sometimes lead to hazardous situations. With this statement in mind , select the response below which could lead to such a hazard ;

- a) anticipating the sequence of items on a check list.
- b) anticipating that the flight will take longer time than planned.
- c) mishearing the contents of a reply from an air traffic controller when a non-standard procedure was given but a standard procedure was anticipated.
- d) anticipating that the weather may deteriorate.

Answer ;

c) mishearing the contents of a reply from an air traffic controller when a non-standard procedure was given but a standard procedure was anticipated.

0064. Although we have a field of vision of more than 180° it is important during flight to use the scanning technique , because ;

- a) it is tiring to look continually in the same direction.
- b) the reduction in the field of vision with decreasing altitude is due to a lack of vitamin A.
- c) only in the peripheral area of the retina resolution is good enough to see an object clearly.
- d) only in the foveal area resolution is good enough to see an object clearly.

Answer ;

d) only in the foveal area resolution is good enough to see an object clearly.

0065. Altitude-hypoxia , when breathing ambient air , should not usually occur (indifferent phase) ;
- a) between 3000 m and 5000 m.
 - b) below 3000 m.
 - c) up to 5000 m.
 - d) between 5000 m and 7000 m.

Answer ;

b) below 3000 m.

0066. Among the external factors that may contribute to an error , cockpit noise and the restriction of the field of view due to windscreen design correspond to ;

- a) organisational factors.
- b) ergonomic factors.
- c) psychobiological factors.
- d) social factors.

Answer ;

b) ergonomic factors.

0067. Among the factors that increase tolerance to long-duration g forces are ;

- a) anti-g suits and correct use of pilot's harness.
- b) correct use of pilot's harness and tensing of the abdominal muscles.
- c) bending forward or supine body position and tensing of the abdominal muscles.
- d) tensing the leg muscles and correct use of pilot's harness.

Answer ;

c) bending forward or supine body position and tensing of the abdominal muscles.

0068. Among the factors which affect night vision are ;

- a) age , cabin altitudes above 5.000 ft , smoking and lack of vitamin E.
- b) age , cabin altitudes above 5.000 ft , smoking and lack of vitamin C.
- c) age , cabin altitudes above 5.000 ft , smoking and alcohol.
- d) age , cabin altitudes above 5.000 ft , smoking and lack of vitamin B.

Answer ;

c) age , cabin altitudes above 5.000 ft , smoking and alcohol.

0069. Among the factors which affect visual acuity are ;

- a) Smoking , colour blindness and angular distance from the fovea.
- b) Colour blindness , alcohol and amount of light available.
- c) Hypoxia , age and angular distance from the fovea.
- d) Colour blindness , visibility and angular distance from the fovea.

Answer ;

c) Hypoxia , age and angular distance from the fovea.

0070. Among the factors which can cause illusions while taxiing are ;

- a) visibility and distance.
- b) distance from the edge of the taxiway and cockpit lighting.
- c) relative movement and cockpit height above the ground.
- d) distance from the edge of the taxiway and taxi lighting.

Answer ;

c) relative movement and cockpit height above the ground.

0071. Among the functions below , which is the most sensitive to hypoxia ?

- a) Hearing.
- b) Motor coordination.
- c) Speech.
- d) Night vision.

Answer ;

d) Night vision.

0072. Among the measures that a pilot should take when in an area where gastro-intestinal upsets are endemic is ;

- a) avoidance of cooked fruits.
- b) avoidance of ice in cold drinks.
- c) avoidance of cooked meats.
- d) avoidance of curries or gassy foods.

Answer ;

b) avoidance of ice in cold drinks.

0073. Among the physiological responses to stress are ;
- a) increased heart rate , apprehension and sweating.
 - b) dry mouth , anxiety and sweating.
 - c) anxiety , fatigue and fear.
 - d) sweating , dry mouth and breathing difficulties.

Answer ;

d) sweating , dry mouth and breathing difficulties.

0074. Among the symptoms of hypoglycaemia are ;
- a) double vision and puffiness around the eyes.
 - b) severe eye and head aches.
 - c) headache and lack of concentration.
 - d) difficulty in focussing on near-by objects.

Hypoglycaemia (Hipoglisemi) :
Kanda Glukoz seviyesinin düşük olması

Answer ;

c) headache and lack of concentration.

0075. An authoritative leader of a group will ;

- a) always be a good listener.
- b) tend to become over-loaded in times of high stress or in an emergency.
- c) be adaptable to different types of groups in any situation.
- d) tend never to interfere with the workings of the group and finds delegation easy.

Answer ;

b) tend to become over-loaded in times of high stress or in an emergency.

0076. An efficient flight deck (synergetic cockpit) will be observed when ;

- a) the plan of action is defined by the Captain because of his experience level.
- b) the Captain delegates the decision making process to other crew members.
- c) decisions are taken by the Captain with the help and participation of the other crew members.
- d) decisions do not need to be discussed because of a common synergy between the crew members.

Answer ;

c) decisions are taken by the Captain with the help and participation of the other crew members.

0077. An example of conflict between status and role is ;

- a) an ATC Controller interviewing an inexperienced Captain.
- b) a senior Captain acting as co-pilot to a junior Captain of an aircraft.
- c) a security officer temporarily standing in for a member of the cabin crew
- d) a Manager interviewing an experienced Captain of an aircraft.

Answer ;

b) a senior Captain acting as co-pilot to a junior Captain of an aircraft.

0078. An experienced pilot ;

- a) prepares thoroughly for a flight and is able to anticipate the majority of possible problems.
- b) seeks complex solutions to problems , thus using his capacities up to 100%.
- c) functions exclusively in a reactive manner , as it is not resource-consuming.
- d) avoids using automated systems as his experience allows him to perform all tasks manually.

Answer ;

a) prepares thoroughly for a flight and is able to anticipate the majority of possible problems.

0079. An identical situation can be experienced by one pilot as exciting in a positive sense and by another pilot as threatening. In both cases ;

- a) the arousal level of both pilots will be raised.
- b) the pilot feeling threatened , will be much more relaxed , than the pilot looking forward to what may happen.
- c) both pilots will experience the same amount of stress.
- d) both pilots will lose their motor-coordination.

Answer ;

a) the arousal level of both pilots will be raised.

0080. An illusion of obtaining greater height above ground can occur when ;

- a) accelerating at low altitude.
- b) flying over high terrain in low visibility.
- c) suddenly flying over small trees after prolonged flying over tall trees.
- d) decelerating at low altitude.

Answer ;

c) suddenly flying over small trees after prolonged flying over tall trees.

0081. An important aspect of Situational Awareness is ;
- a) when the pilot perceives the situation both inside and outside the aircraft.
 - b) when the pilot perceives the instant and immediate situation both inside and outside the cockpit.
 - c) when a pilot's perception equals reality.
 - d) when a pilot is sure that the situation both inside and outside the aircraft is correct.

Answer ;

c) when a pilot's perception equals reality.

0082. An increase in the amount of carbon dioxide in the blood leads to ;
- a) a decrease of acidity in the blood.
 - b) an increased respiratory rate.
 - c) an improving resistance to hypoxia.
 - d) a reduction of red blood cells.

Answer ;

b) an increased respiratory rate.

0083. An under-confident and self-effacing co-pilot is promoted to Captain. It is not uncommon for such a person have a behavioural pattern which is ;

- a) confident and self-assuring because of his/her higher status.
- b) easy to adapt to the role of Captain.
- c) easy-going but having a reliance on others.
- d) aggressive if challenged by another member of the crew.

Answer ;

d) aggressive if challenged by another member of the crew.

0084. An unnoticed way-point error entered in an aircraft data base is an example of ;
- a) Active failure/error.
 - b) Pro-active failure/error.
 - c) Latent failure/error.
 - d) Re-active failure/error.

Answer ;

c) Latent failure/error.

0085. Analysis of accidents involving the human factor in aviation shows that ;
- a) failure of the human factor is always connected with technical breakdowns.
 - b) only pilot training will make it possible to improve the situation.
 - c) there is hardly ever a single cause responsible.
 - d) only front-line operators are involved.

Answer ;

c) there is hardly ever a single cause responsible.

0086. Angular accelerations are perceived by ;
- a) the cochlea.
 - b) the otholiths.
 - c) the receptors in the skin and the joints.
 - d) the semi circular canals.

Answer ;

d) the semi circular canals.

0087. Anxiety and fear can cause ;
- a) spatial disorientation.
 - b) hypoxia.
 - c) hypoglycaemia.
 - d) hyperventilation.

Answer ;

d) hyperventilation.

Hyperventilation (Hiperventilasyon) :
Heyecan , korku , sıkıntı ve diğer bazı etkenler nedeniyle solunumun hızlanması sonucu oluşan ve kandaki karbon dioksitin azalmasına sebep olan durumdur. Baş dönmesi , zihin karışıklığı , el ve ayaklarda spazm , uyuşma ve baygınlık başlıca belirtileri oluşturur.

0088. Anxiety can affect ;

1. Judgement .
 2. Attention .
 3. Memory.
 4. Concentration.
- a) 1 , 2 and only are correct.
b) 1 only is correct.
c) All are correct.
d) 1 and 2 only are correct.

Answer ;

c) All are correct.

0089. Any prolonged exposure to noise in excess of 90 db can result in ;

- a) presbycusis (effects of aging).
b) noise induced hearing loss.
c) conductive hearing loss.
d) a ruptured ear drum.

Answer ;

b) noise induced hearing loss.

0090. Approximately of all communication is achieved by factors other than words (metacommunications).

- a) 80%.
b) 10%.
c) 50%.
d) 30%.

Answer ;

a) 80%.

0091. Approximately how long will a blood/alcohol level of 60 mgm/100 ml take to return to normal ?

- a) 12 hours.
b) 20 hours.
c) 1 hour.
d) 4 hours.

Answer ;

d) 4 hours.

0092. As a cause of accidents , the human factor ;

- a) is cited in approximately 70 - 80 % of aviation accidents.
b) plays a negligible role in commercial aviation accidents. It is much more important in general aviation.
c) which is cited in current statistics, applies to the flight crew and ATC only.
d) has increased considerably since 1980 - the percentage of accident in which this factor has been involved has more than tripled since this date.

Answer ;

a) is cited in approximately 70 - 80 % of aviation accidents.

0093. As a result of automation in cockpits ;

- a) the need for communication between crew members has been decreased.
b) communication and coordination have clearly improved in man-man and man-machine relations.
c) communication and coordination call for an even greater effort on the part of the crew members.
d) it is easier for the captain to monitor the work of the first officer and vice versa.

Answer ;

c) communication and coordination call for an even greater effort on the part of the crew members.

0094. Astigmatism is caused by ;

- a) a lack of accommodation.
b) a lack of empty field.
c) a mis-shapened cornea.
d) a lack of vitamin A.

Answer ;

c) a mis-shapened cornea.

0095. At rest the cardiac output (the quantity of blood the heart pumps in one minute) of an adult is approximately ;
- a) 75 litres/min
 - b) 5 litres/min
 - c) 45 litres/min
 - d) 450 ml/min

Answer ;

b) 5 litres/min

-
0096. At what altitude (breathing 100% oxygen without pressure) could symptoms of hypoxia be expected ?

- a) Approximately 10 - 12 000 ft.
- b) 22 000 ft.
- c) Approximately 38 - 40 000 ft.
- d) Approximately 35 000 ft.

Answer ;

c) Approximately 38 - 40 000 ft.

-
0097. At which altitude is it necessary to breathe 100% oxygen plus pressure after a rapid decompression ?

- a) Approximately 20 000 ft.
- b) Approximately 40 000 ft.
- c) Approximately 14 000 ft.
- d) Approximately 50 000 ft.

Answer ;

b) Approximately 40 000 ft.

-
0098. At which altitude (within the “indifferent zone”) may a degradation of night vision occur ?

- a) Up to 5.000 m
- b) 5.000 m - 7.000 m
- c) From approximately 1.600 m
- d) 3.000 m - 5.000 m

Answer ;

c) From approximately 1.600 m

-
0099. Attitudes are defined as ;

- a) the conditions necessary for carrying out an activity.
- b) tendencies to respond to people , things or events in a particular manner.
- c) a synonym for behaviour.
- d) the genetic predispositions for thinking and acting.

Answer ;

b) tendencies to respond to people , things or events in a particular manner.

-
0100. Autokinesis can give the pilot the impression that ;

- a) the aircraft is climbing.
- b) a star is another aircraft.
- c) lights are further away than in fact they are.
- d) lights from ships are stars.

Autokinesis (Otokinesis) :
Sabit bir ışık kaynağının hareket ediyor gibi görünmesi şeklinde olan görsel ilüzyondur.

Answer ;

b) a star is another aircraft.

-
0101. Autokinesis is ;

- a) the change in diameter of the pupil , when looking in the dark.
- b) the automatic adjustment of the crystalline lens to objects situated at different distances.
- c) the phenomenon of spinning lights after the abuse of alcohol.
- d) the apparent movement of a static single light when stared at for a relatively long period of time in the dark.

Answer ;

d) the apparent movement of a static single light when stared at for a relatively long period of time in the dark.

-
0102. Autokinetic illusion is ;

- a) poor interpretation of the surrounding world.
- b) an illusion in which a stationary point of light , if stared at for several seconds in the dark , may - without a frame of reference - appear to move.
- c) the sensation during a radial acceleration of seeing a fixed reference point moving into the opposite direction of the acceleration.
- d) a conflict between the visual system and bodily sensations.

Answer ;

b) an illusion in which a stationary point of light , if stared at for several seconds in the dark , may - without a frame of reference - appear to move.

0103. Barotrauma caused by gas accumulation in the stomach and intestines can lead to ;

- a) barotitis.
- b) barosinusitis.
- c) decompression sickness.
- d) pressure pain or flatulence.

Answer ;

d) pressure pain or flatulence.

0104. Barotrauma is caused by an increase or decrease in the volume of the gases contained in the body's cavities. Which , in practice , are the most frequent locations of such problems ?

- a) Inner ear , facial sinuses and stomach.
- b) Facial sinuses , outer ear and stomach.
- c) Facial sinuses , middle ear and dental cavities.
- d) Dental cavities , outer ear and eyes.

Answer ;

c) Facial sinuses , middle ear and dental cavities

0105. Barotrauma of the middle ear is usually accompanied by ;

- a) pain in the joints.
- b) dizziness.
- c) noises in the ear.
- d) a reduction in hearing ability and the feeling of increasing pressure.

Answer ;

d) a reduction in hearing ability and the feeling of increasing pressure

0106. Barotrauma of the middle ear most likely will occur ;

- a) during a long high altitude flight.
- b) when descending rapidly.
- c) when climbing.
- d) in sudden steep turns.

Answer ;

b) when descending rapidly.

0107. Barotrauma of the sinuses of the nose ;

- a) is caused by differences in pressure between the sinus cavities and the outside air pressure.
- b) only arises in flying and not in diving.
- c) is only caused by colds and their effects.
- d) is an irritation of the sinuses caused by the over-use of nasal sprays.

Answer ;

a) is caused by differences in pressure between the sinus cavities and the outside air pressure.

0108. Before take-off , a briefing ;

- a) must be done systematically in order to refresh the memory of the crew members and to co-ordinate actions.
- b) is not necessary when departing from home base because this is routine.
- c) is not necessary if the crew members know each other.
- d) is not required unless both crew members agree it is essential.

Answer ;

a) must be done systematically in order to refresh the memory of the crew members and to co-ordinate actions.

0109. Below 70.000 ft , what gas makes up the major part of the atmosphere ?

- a) Oxygen.
- b) Nitrogen.
- c) Ozone.
- d) Carbon dioxide.

Answer ;

b) Nitrogen.

0110. Between which components , with reference to the SHELL Concept , covers pilot misinterpretation of the old three-point altimeter ?

- a) Liveware - Software.
- b) Liveware - Liveware.
- c) Liveware - Environment.
- d) Liveware - Hardware.

Answer ;

d) Liveware - Hardware.

-
0111. Blood-pressure depends on ;
1. the cardiac output.
 2. the resistance of the capillaries.

Which of the following lists all the correct answers ?

- a) 1 and 2 are both false.
- b) 1 and 2 are correct.
- c) 1 is false 2 is correct.
- d) 1 is correct 2 is false.

Answer ;

- b) 1 and 2 are correct.
-

0112. Blood pressure is normally measured in ;

- a) Hectopascals.
- b) Millibars.
- c) inHg.
- d) mmHg.

Answer ;

- d) mmHg.
-

0113. Boyle's law is directly applicable in case of ;

- a) hyperventilation with increasing altitude.
- b) the occurrence of decompression sickness at high altitude.
- c) the occurrence of hypoxia with increasing altitude.
- d) the expansion of trapped gasses in the human body with increasing altitude.

Answer ;

- d) the expansion of trapped gasses in the human body with increasing altitude.
-

0114. Breathing 100% oxygen at 40.000 ft is equivalent to breathing ambient air at ;

- a) 8 000 ft.
- b) 14 000 ft.
- c) 18 000 ft.
- d) 10 000 ft.

Answer ;

- d) 10 000 ft.
-

0115. Breathing 100% oxygen will elevate the pilot's physiological safe altitude to approximately ;

- a) 10 000 ft.
- b) 45 000 ft.
- c) 22 000 ft.
- d) 40 000 ft.

Answer ;

- d) 40 000 ft.
-

0116. Breathing pure oxygen (without pressure) will be sufficient up to an altitude of ;

- a) 60000 ft.
- b) 45000 ft.
- c) 40000 ft.
- d) 80000 ft.

Answer ;

- c) 40000 ft.
-

0117. By "long duration acceleration" , we mean that it lasts more than ;

- a) 5 minutes.
- b) 1 second.
- c) 1 minute.
- d) 20 seconds.

Answer ;

- b) 1 second.
-

0118. By what action could the pressure gradient between the middle and the outer ear be balanced ?

- a) By breathing out.
- b) By swallowing.
- c) By coughing.
- d) By breathing in.

Answer ;

b) By swallowing.

0119. Caffeine may cause an increase in cardiac rate , restlessness/nervousness , insomnia , anxiety and intestinal irritability. Excessive consumption is considered to be in excess of ;

- a) 250 mg/day.
- b) 40-65 mg/day.
- c) 3 mg/day.
- d) 150 mg/day.

Answer ;

a) 250 mg/day.

0120. Can you suffer from aerodontalgia in the descent ?

- a) No.
- b) Only above 10.000 ft.
- c) Yes.
- d) Only in pressurised aircraft.

Answer ;

a) No.

0121. Carbon dioxide ;

- a) is approximately 13% of ambient air at Flight Level 180.
- b) is a normal product of internal respiration.
- c) is increased in the blood and the brain during hyperventilation.
- d) may reach toxic levels in the flight deck due to incomplete fuel combustion.

Answer ;

b) is a normal product of internal respiration.

0122. Carbon monoxide , a product of incomplete combustion , is toxic because ;

- a) it prevents the absorption of food from the digestive tract.
- b) it disturbs gaseous diffusion at the alveoli capillary membrane.
- c) it prevents the excretion of catabolites in the kidneys.
- d) it competes with oxygen in its union with haemoglobin.

Answer ;

d) it competes with oxygen in its union with haemoglobin.

0123. Carbon monoxide (CO) poisoning in flight ;

- a) can be cured by breathing into a plastic bag to retain the carbon monoxide.
- b) is a complication when hyperventilating and requires its own special and individual treatment.
- c) is usually harmless because oxygen is more easily attached to haemoglobin than carbon monoxide to a magnitude of 200 times.
- d) presents an extremely dangerous situation as the blood may not be able carry sufficient amounts of oxygen to vital cells and tissues of the body.

Answer ;

d) presents an extremely dangerous situation as the blood may not be able carry sufficient amounts of oxygen to vital cells and tissues of the body.

0124. Carbon monoxide (CO) poisoning leads to hypoxia because ;

- a) accumulation of CO in blood leads to hyperventilation and thus to hypoxia.
- b) the increasing amounts of CO in the lung alveoles gradually reverses oxygen diffusion.
- c) CO in the blood displaces oxygen from the blood corpuscles impairing oxygen transport.
- d) CO is far less easily attached (200 times) to haemoglobin than oxygen.

Answer ;

c) CO in the blood displaces oxygen from the blood corpuscles impairing oxygen transport.

0125. Carbon monoxide in the human body can lead to ;

1. loss of muscular power.
 2. headache.
 3. impaired judgement.
 4. pain in the joints.
 5. loss of consciousness.
- a) 1 , 2 , 3 , 4 are correct.
b) 1 , 2 , 3 and 5 are correct.
c) 1 , 2 and 4 are correct.
d) 2 and 3 are correct , 1 is false.

Answer ;

b) 1 , 2 , 3 and 5 are correct.

0126. Carbon monoxide is always present in the exhaust gases of engines. If a pilot is exposed to carbon monoxide , which of the following responses is correct ?

- a) A short exposure to relatively high concentrations of carbon monoxide can seriously affect a pilot's ability to operate an aircraft.
b) Carbon monoxide can only affect pilots if they are exposed to it for a long period of time.
c) When exposed to carbon monoxide for a long period of time , the body will adapt to it and no adverse physical effects are experienced
d) Carbon monoxide is easily recognised by odour and taste.

Answer ;

a) A short exposure to relatively high concentrations of carbon monoxide can seriously affect a pilot's ability to operate an aircraft.

0127. Carbon monoxide is particularly dangerous because ;

1. its initial symptoms are not alarming.
 2. it is colourless.
 3. it is odourless.
 4. it is highly toxic.
 5. its effects are cumulative.
- a) 2 , 3 , 4 and 5 only.
b) all of the above.
c) 1 , 2 , 3 and 5 only.
d) 2 , 3 and 4 only.

Answer ;

b) all of the above.

0128. Carbon monoxide poisoning ;

- a) is more likely to occur in aeroplanes with twin-engines because of high engine efficiency.
b) only occurs in jet-driven aeroplanes.
c) occurs only above 15 degrees OAT.
d) is more likely to occur in aeroplanes where the cabin heating is supplied by passing cabin air over the exhaust manifold(s).

Answer ;

d) is more likely to occur in aeroplanes where the cabin heating is supplied by passing cabin air over the exhaust manifold(s).

0129. Carbon monoxide poisoning can be treated by ;

- a) increasing the amount of nitrogen being physically dissolved in the blood.
b) breathing into a paper bag.
c) increasing the amount of oxygen being physically dissolved in the blood.
d) decreasing the amount of oxygen being combined with the hemoglobin in the blood.

Answer ;

c) increasing the amount of oxygen being physically dissolved in the blood.

0130. Cataract is caused by ;

- a) a mis-shaped cornea.
b) a clouding of the lens.
c) a lack of accommodation at the cornea.
d) lack of mobility of the cornea.

Answer ;

b) a clouding of the lens.

0131. Changes in blood-pressure are measured by ;

- a) adrenal glands.
- b) pacemakers.
- c) arteriols.
- d) pressoreceptors.

Answer ;

d) [pressoreceptors.](#)

0132. Cholera can be transmitted through ;

- a) droplets in the air cause by the breath of an infected person.
- b) food or water which has been contaminated.
- c) insect bites.
- d) bacteria in the form of spores via a puncture in the skin.

Answer ;

b) [food or water which has been contaminated.](#)

0133. Cigarette smoking has particular significance to the flyer , because there are long-term and short-term harmful effects. From cigarette smoking the pilot can get ;

- a) a mild carbon monoxide poisoning decreasing the pilot's tolerance to hypoxia.
- b) a suppressed desire to eat and drink.
- c) a mild carbon monoxide poisoning increasing the pilot's tolerance to hypoxia.
- d) a mild carbon dioxide poisoning increasing the pilot's tolerance to hypoxia.

Answer ;

a) [a mild carbon monoxide poisoning decreasing the pilot's tolerance to hypoxia.](#)

0134. CO (carbon monoxide) present in the smoke of cigarettes can lead to ;

- 1. reduction of time of useful consciousness.
 - 2. hypoxia at a lower altitude than normal.
- a) 1 and 2 are both false.
 - b) 1 is correct , 2 is false.
 - c) 1 and 2 are both correct.
 - d) 1 is false , 2 is correct.

Answer ;

c) [1 and 2 are both correct.](#)

0135. Coaction is a mode of coordination which recommends ;

- a) sustained cooperation on actions and the formulation of commitments concerning flight situations.
- b) the application of procedural knowledge in the conduct of specific actions.
- c) working in parallel to achieve one common objective.
- d) working parallel to achieve individual objectives.

Answer ;

c) [working in parallel to achieve one common objective.](#)

0136. Cognitive and physical rehearsal of actions during training ;

- a) is more effective than practical training.
- b) leads to an increased error rate.
- c) is most important for the acquisition of complex perceptual motor skills.
- d) is most important for self-control.

Answer ;

c) [is most important for the acquisition of complex perceptual motor skills.](#)

0137. Cognitive Coping is associated with ;

- a) eliminating the source of the stressor.
- b) a deliberate removal of the stressor.
- c) ignoring or rationalising the stress factor.
- d) taking medicine to relieve the effects of stress.

Answer ;

c) [ignoring or rationalising the stress factor.](#)

0138. Cognitive evaluation which leads to stress is based on ;
- a) the evaluation of the capabilities of the individual and the time available.
 - b) the evaluation of the situation and the evaluation of capabilities to cope with it.
 - c) the evaluation of the situation and the state of fatigue of the individual.
 - d) the capabilities of the individual and the solutions provided by the environment.

Answer ;

- b) the evaluation of the situation and the evaluation of capabilities to cope with it.
-

0139. Cohesion is a major advantage in times of ;

- a) Success.
- b) Personal advancement.
- c) Difficulty.
- d) Achievement.

Answer ;

- c) Difficulty.
-

0140. Colour Blindness ;

- a) may be treated by laser surgery.
- b) may be subtle and only detected using specialised tests.
- c) affects acuity.
- d) is more common in first born males.

Answer ;

- b) may be subtle and only detected using specialised tests.
-

0141. Communication efficiency is most sensitive to ;

- a) packaging of sentences.
- b) workload and interruptions.
- c) packaging of words.
- d) packaging of phrases.

Answer ;

- b) workload and interruptions.
-

0142. Concerning circadian rhythm disruption (jet lag) , adjustment to destination time ;

- 1. Takes longer when travelling west rather than travelling east.
- 2. Takes longer when travelling east rather than travelling west.
- 3. Varies little between individuals.
- 4. Varies substantially among individuals.

Which of the following lists all the correct statements ?

- a) 2 , 3.
- b) 2 , 4.
- c) 1 , 3.
- d) 1 , 4.

Answer ;

- b) 2 , 4.
-

0143. Concerning the effects of drugs and pilot's performance ;

- a) only the primary effect has to be considered ; side effects are negligible.
- b) the side effects only have to be considered.
- c) medication has no influence on pilot's performance.
- d) the primary and the side effects have to be considered.

Answer ;

- d) the primary and the side effects have to be considered.
-

0144. Concerning the light sensitive cells of the eye ;

- a) cones are centrally located in the lens.
- b) the cones detect colour vision and are highly sensitive to hypoxia.
- c) rods are centrally located and increase visual acuity.
- d) blood flow to the rods and cones is maintained even during high g forces.

Answer ;

- b) the cones detect colour vision and are highly sensitive to hypoxia.
-

0145. Concerning the relationship between performance and stress , which of the following statements is correct ?
- a) Domestic stress will not affect the pilot's performance because he is able to leave this type of stress on the ground.
 - b) A student will learn faster and better under severe stress.
 - c) A well trained pilot is able to eliminate any kind of stress completely when he is scheduled to fly.
 - d) A moderate level of stress may improve performance.

Answer ;

- d) A moderate level of stress may improve performance.
-

0146. Conductive hearing loss can be caused by ;
- 1. damage to the ossicles in the middle ear caused by infection or trauma.
 - 2. a damage of the auditory nerve.
 - 3. an obstruction in the outer ear.
 - 4. a ruptured tympanic membrane.
- a) 1 , 2 and 3 are correct , 4 is false.
 - b) 1 , 2 , 3 and 4 are correct.
 - c) 2 , 3 and 4 are correct , 1 is false.
 - d) 1 , 3 and 4 are correct , 2 is false.

Answer ;

- d) 1 , 3 and 4 are correct , 2 is false.
-

0147. Contrary to a person's personality , attitudes ;
- a) are non-evolutive adaptation procedures regardless of the result of the actions associated with them.
 - b) are the product of personal disposition and past experience with reference to an object or a situation.
 - c) are essentially driving forces behind changes in personality.
 - d) form part of personality and , as a result , cannot be changed in an adult.

Answer ;

- b) are the product of personal disposition and past experience with reference to an object or a situation.
-

0148. Coriolis illusion , causing spatial disorientation is the result of ;
- a) undergoing positive G.
 - b) gazing in the direction of a flashing light.
 - c) simultaneous head movements during aircraft manoeuvres.
 - d) normal deterioration of the semicircular canals with age.

Answer ;

- c) simultaneous head movements during aircraft manoeuvres.
-

0149. CRM and MCC training are designed to improve ;
- a) individual efficiency and contribution.
 - b) individual achievement.
 - c) the quality and a number of individuals performance.
 - d) the quality of crew performance.

Answer ;

- d) the quality of crew performance.
-

0150. CRM (Crew Resource Management) training is ;
- a) is mainly of relevance to pilots with personality disorders or inappropriate attitudes.
 - b) intended solely to alter an individual's personality.
 - c) not intended to change the individual's attitude at all.
 - d) intended to develop effectiveness of crew performance by improving attitudes towards flight safety and human relationship management.

Answer ;

- d) intended to develop effectiveness of crew performance by improving attitudes towards flight safety and human relationship management.
-

0151. Dalton's law explains the occurrence of ;
- a) creeps.
 - b) bends.
 - c) altitude hypoxia.
 - d) decompression sickness.

Answer ;

- c) altitude hypoxia.
-

- 0152.** Decision-making can be influenced by the following factors ;
1. people tend to conform to opinions expressed by a majority within the group they belong to.
 2. people always keep the future decisions in line with those their superiors have made in the past.
 3. people more easily tend to select data which meet their expectations.
 4. people rarely base decisions on their personal preferences but rather on rational information.

Which of the following lists all the correct answers ?

- a) 1 , 3.
- b) 1 , 4.
- c) 2 , 3.
- d) 2 , 4.

Answer ;

- a)** 1 , 3.
-

- 0153.** Decision making in emergency situations requires firstly ;

- a) the whole crew to focus on the problem.
- b) delegation of tasks and crew coordination.
- c) speed of reaction.
- d) informing ATC thoroughly about the situation.

Answer ;

- b)** delegation of tasks and crew coordination.
-

- 0154.** Decision-making is a concept which represents ;

- a) a spontaneous act of seeking the most effective solution in a given situation when faced with a defined problem.
- b) a voluntary and conscious process of selection , from among possible solutions , for a given problem.
- c) an automatic process of selection from among the various solutions to a given problem.
- d) an automated or automation-like act of applying defined procedures.

Answer ;

- b)** a voluntary and conscious process of selection , from among possible solutions , for a given problem.
-

- 0155.** Decision-making results in ;

- a) a choice always based on the experience of the PIC.
- b) a choice between different options to achieve a goal.
- c) a subjective choice between options to achieve a goal.
- d) a totally objective choice between options to achieve a goal proposed by the Pilot in Command.

Answer ;

- b)** a choice between different options to achieve a goal.
-

- 0156.** Decisions made by pilots in a cockpit ;

- a) should always be made as quickly as possible.
- b) are automatic when pilots are experts.
- c) remain valid for a limited time only.
- d) are always irreversible.

Answer ;

- c)** remain valid for a limited time only.
-

- 0157.** Decompression sickness can be prevented by ;

1. avoiding cabin altitudes above 18.000 ft.
 2. maintaining cabin pressure below 8.000 ft when flying at high altitudes.
 3. performing physical exercises before and during the flight.
 4. breathing 100% oxygen for 30 min prior and during the flight.
- a) only 3 is correct.
 - b) 2 and 3 are correct , 4 is false.
 - c) 1 , 2 and 3 are correct.
 - d) 1 , 2 and 4 are correct.

Answer ;

- d)** 1 , 2 and 4 are correct.
-

- 0158.** Decompression sickness/illness is characterized by ;

- a) over-saturated nitrogen gas molecules in the body tissues.
- b) headache and dizziness caused by oxygen bubbles due to the low partial pressure of oxygen.
- c) negligible symptoms that need no attention or therapy.
- d) muscle pains in small muscles such as the finger and toe muscles , which are also known as the “chokes”.

Answer ;

- a)** over-saturated nitrogen gas molecules in the body tissues.
-

0159. Decompression sickness symptoms may develop due to ;
- a) sudden pressure surges in the cabin at altitudes below 18.000 ft.
 - b) cabin pressure loss when flying at higher altitudes (above 18.000 ft).
 - c) emergency descents after a cabin pressure loss.
 - d) fast flights from a high-pressure zone into a low pressure area when flying an unpressurised aeroplane.

Answer ;

- b) cabin pressure loss when flying at higher altitudes (above 18.000 ft).
-

0160. Decompression symptoms are caused by ;
- a) dissolved gases from tissues and fluids of the body.
 - b) release of locked gases from joints.
 - c) low oxygen pressure of inhaled air.
 - d) low carbon dioxide pressure of inhaled air.

Answer ;

- a) dissolved gases from tissues and fluids of the body.
-

0161. Define Hypoxia and explain why living tissues require oxygen.
- a) Hypoxia is the lack of sufficient oxygen to meet the needs of the body tissues which require carbon dioxide for metabolism.
 - b) Hypoxia is the lack of sufficient oxygen to meet the needs of the body tissues which require oxygen for oxidation of carbohydrates from food to produce energy.
 - c) Hypoxia is the lack of sufficient oxygen to meet the needs of the body tissues which require oxygen to fight infection or invading bacteria.
 - d) Hypoxia is the lack of sufficient oxygen to meet the needs of the body tissues which require oxygen to coagulate the blood in case of infection.

Answer ;

- b) Hypoxia is the lack of sufficient oxygen to meet the needs of the body tissues which require oxygen for oxidation of carbohydrates from food to produce energy.
-

0162. Differences of opinion should be regarded as
- a) stressful.
 - b) confrontational.
 - c) having negative influence on the group and are not to be encouraged.
 - d) helpful.

Answer ;

- d) helpful.
-

0163. Discussing private matters in the cockpit ;
- a) is appropriate in any phase of flight.
 - b) should be avoided by all means in the cockpit.
 - c) can improve team spirit.
 - d) decreases the captains role of leadership.

Answer ;

- c) can improve team spirit.
-

0164. Disorientation is more likely to occur when the pilot is ;
- 1. flying in IMC.
 - 2. frequently changing between inside and outside references.
 - 3. flying from IMC into VMC.
 - 4. approaching over still water at night.
- a) 1 , 2 and 4 are correct.
 - b) 1 , 3 and 4 are correct.
 - c) 1 , 2 and 3 are correct.
 - d) 2 , 3 and 4 are correct.

Answer ;

- a) 1 , 2 and 4 are correct.
-

0165. Disruptions of pressure equalization in air-filled cavities of the head (nose , ear etc.) are called ;
- a) barotrauma.
 - b) hypoxia.
 - c) hyperventilation.
 - d) embolism.

Answer ;

- a) barotrauma.
-

- 0166.** Disturbance of the biological clock appears after a ;
1. bad night's sleep.
 2. day flight Amsterdam - New York.
 3. day flight Amsterdam - Johannesburg.
 4. night flight New York - Amsterdam.
- a) 1 and 3 are correct.
b) 1 , 2 , 3 and 4 are correct.
c) 1 , 2 and 3 are correct.
d) 2 and 4 are correct.

Answer ;

d) 2 and 4 are correct.

- 0167.** Divided attention is the ability ;
1. to execute several mental activities at almost the same time (i.e. when switching attention from outside the aircraft to the airspeed indicator on the instrument panel).
 2. to monitor the progress of a motor programme (i.e. flying or taxiing the airplane) on a relatively subconscious level , while making a radio call at the same time (requiring a rather conscious level).
 3. to select information and check if it is relevant to the task in hand. At the same time no other operation can be performed.
 4. to delegate tasks to the copilot while concentrating on the procedures.
- a) 1 and 2 are correct , 3 and 4 are false.
b) 1 , 2 and 3 are correct , 4 is false.
c) Only 3 is false.
d) 1 and 3 are correct , 2 and 4 are false.

Answer ;

a) 1 and 2 are correct , 3 and 4 are false.

- 0168.** Dizziness and tumbling sensations , when making head movements in a tight turn , are symptoms of ;
- a) "Oculogravic illusion".
 - b) "Flicker-vertigo".
 - c) "Nystagmus".
 - d) "Pilot's vertigo".

Answer ;

d) "Pilot's vertigo".

- 0169.** Does noise (above 90 dbs) improve performance ?
- a) Depends upon the awareness of the individual.
 - b) Never.
 - c) Yes.
 - d) Sometimes.

Answer ;

b) Never.

- 0170.** Doing a general briefing in the pre-flight phase the captain should emphasize ;
- a) avoidance inadequate handling of flight controls.
 - b) complete delegation of all duties.
 - c) particular requirements in the field of crew coordination and co-operation.
 - d) departure on schedule.

Answer ;

c) particular requirements in the field of crew coordination and co-operation.

- 0171.** Dry air is a mixture of gases. Their volume percentage is about ;
- a) 18% oxygen , 80% nitrogen , 2% other gases.
 - b) 19% oxygen , 80% nitrogen , 1% other gases.
 - c) 21% oxygen , 78% nitrogen , 1% other gases.
 - d) 25% oxygen , 74% nitrogen , 1% other gases.

Answer ;

c) 21% oxygen , 78% nitrogen , 1% other gases.

0172. During a night flight at 10.000 feet you notice that your visual acuity has decreased. In this case you can increase your acuity by ;

- a) closing one eye.
- b) breathing extra oxygen through the oxygen mask.
- c) dim the instrument lights.
- d) scanning sectors of the field of vision.

Answer ;

b) breathing extra oxygen through the oxygen mask.

0173. During a rapid decompression in a 2 crew aircraft , the other pilot becomes incapacitated , your first action is to ;

- a) don your own oxygen mask.
- b) administer oxygen to the other pilot.
- c) commence an emergency descent.
- d) call for immediate assistance from cabin crew.

Answer ;

a) don your own oxygen mask.

0174. During a steady climb above Flight Level 100 if cabin pressurization fails ;

- a) the cabin will cool rapidly.
- b) the cabin altitude will descend.
- c) the onset of anaemia causes hyperventilation.
- d) the effects of hypoxia may be gradual and difficult to recognise.

Answer ;

d) the effects of hypoxia may be gradual and difficult to recognise.

0175. During flight all crewmembers have one or more of the following symptoms ;

- 1. blue lips.
- 2. mental disturbances.
- 3. tingling sensations in arms and/or legs.
- 4. reduction of peripheral vision.

Which is the possible cause ?

- a) Hypothermia.
- b) Hypoglycaemia.
- c) Glaucoma.
- d) Hypoxia.

Answer ;

d) Hypoxia.

0176. During flight in IMC , the most reliable sense which should be used to overcome illusions is the ;

- a) visual sense , interpreting the attitude indicator.
- b) "Seat-of-the-pants-Sense".
- c) vestibular sense.
- d) visual sense by looking outside.

Answer ;

a) visual sense , interpreting the attitude indicator.

0177. During gas exchange , the partial pressure of carbon dioxide in the alveoli is ;

- a) higher than the pressure of carbon dioxide in the blood.
- b) the same as in the atmospheric air.
- c) lower than the pressure of carbon dioxide in the atmospheric air.
- d) lower than in the blood.

Answer ;

d) lower than in the blood.

0178. During hyperventilation ;

- a) acidity level of the blood is reduced.
- b) alkalinity level of the blood is reduced.
- c) oxygen concentration of the blood is below normal.
- d) nitrogen concentration of the blood is above normal.

Answer ;

a) acidity level of the blood is reduced.

0179. During paradoxical sleep ;

- a) physical restoration and recuperation takes place.
- b) rapid eye movements can be observed.
- c) the rhythm of the heart is very regular.
- d) respiration is very regular.

Answer ;

- b) rapid eye movements can be observed.**
-

0180. During poor weather conditions a pilot should fly with reference to instruments because ;

- a) perception of distance and speed is difficult in an environment of low contrast.
- b) pressure differences can cause the altimeter to give wrong information.
- c) his attention will be distracted automatically under these conditions.
- d) the danger of a “greying out” will make it impossible to determine the height above the terrain.

Answer ;

- a) perception of distance and speed is difficult in an environment of low contrast.**
-

0181. During sustained positive G-forces the order of symptoms you can expect is ;

- a) grey-out , unconsciousness , black-out and tunnel vision.
- b) unconsciousness , black-out , tunnel vision and grey out.
- c) grey-out , tunnel vision , black-out and unconsciousness.
- d) black-out , grey-out , tunnel vision and unconsciousness.

Answer ;

- c) grey-out , tunnel vision , black-out and unconsciousness.**
-

0182. During the conflict resolution process it is important to ;

- 1. realise and to accept the conflict.
 - 2. verbalise mutual expectations.
 - 3. search for common agreements.
 - 4. express one’s own viewpoint
- a) 1 and 4 only are correct.
 - b) 1 , 2 and 3 only are correct.
 - c) to express its own emotions.
 - d) 1 , 2 , 3 and 4 are correct.

Answer ;

- d) 1 , 2 , 3 and 4 are correct.**
-

0183. During the cruise , at night , when the workload is low , it is recommended that ;

- a) the lighting in the cockpit be dimmed in order to reduce visual fatigue.
- b) the lighting in the cockpit be dimmed in order to prevent low vigilance.
- c) the lighting in the cockpit be increased in order to prevent low vigilance.
- d) only those instruments that are vital to flying the aircraft be lit in order to avoid dividing one’s attention.

Answer ;

- c) the lighting in the cockpit be increased in order to prevent low vigilance.**
-

0184. During the cruising phase of a short-haul flight the captain starts to smoke a cigarette in the cockpit. The flying copilot asks him to stop smoking because he is a nonsmoker. The captain tells him ; “This is your problem” , and continues smoking. What should the copilot do ?

- a) He should learn to accept the captain smoking cigarettes in the cockpit.
- b) He should not further discuss this issue but should come back to this conflict during the debriefing.
- c) He should repeat his worries about smoking in the cockpit and should argue with the captain about this problem until the conflict is solved.
- d) He should bring a cabin crew member in to arbitrate.

Answer ;

- b) He should not further discuss this issue but should come back to this conflict during the debriefing.**
-

0185. During the decision making process in the cockpit ;

- a) a detailed analysis can only be made once , at the moment the problem occurs.
- b) the amount of time available has a large influence on the analysis of the situation.
- c) habits and routines play no role.
- d) the crew should always decide on ore than one alternative.

Answer ;

- b) the amount of time available has a large influence on the analysis of the situation.**
-

0186. During the pre-flight phase in the cockpit the captain notices that his copilot on the one hand is rather inexperienced and insecure but on the other hand highly motivated. Which kind of leadership behaviour is most appropriate ?

- a) The captain lets the copilot fly and gives him detailed instructions what to do.
- b) The captain flies the first leg by himself and explains each action to the copilot in order to keep him informed about his decisions.
- c) The captain lets the copilot fly and observes his behaviour without any comments.
- d) The captain lets the copilot fly and encourages him to ask for support.

Answer ;

- c)** The captain lets the copilot fly and observes his behaviour without any comments.
-

0187. Early symptoms of hypoxia could be ;

- 1. euphoria.
 - 2. decreased rate and depth of breathing.
 - 3. lack of concentration.
 - 4. visual disturbances.
- a) 1 , 3 and 4 are correct.
 - b) 1 , 2 and 3 are correct.
 - c) 1 , 2 , 3 and 4 are correct.
 - d) 1 , 2 and 4 are correct.

Answer ;

- a)** 1 , 3 and 4 are correct.
-

0188. Empty field myopia is caused by ;

- a) ozone at altitude.
- b) flying over mountainous terrain.
- c) atmospheric perspective.
- d) lack of distant focal points.

Answer ;

- d)** lack of distant focal points.
-

0189. “Environmental capture” is a term used to describe which of the following statements ?

- 1. The tendency for a skill to be executed in an environment in which it is frequently exercised , even if it is inappropriate to do so.
 - 2. The tendency for a skill acquired in one aircraft type to be executed in a new aircraft type , even if it is inappropriate to do so.
 - 3. The tendency for people to behave in different ways in different social situations.
 - 4. The gaining of environmental skills.
- a) 1 , 2 and 3 are correct.
 - b) 1 and 2 are correct.
 - c) 4 is correct.
 - d) 2 and 3 are correct.

Answer ;

- b)** 1 and 2 are correct.
-

0190. Environmental sources of stress in the cockpit could be ;

- 1. noise and vibration.
- 2. acceleration.
- 3. radiation.
- 4. extreme temperatures.

Which of the following lists all the correct answers ?

- a) 2 and 3.
- b) 1 , 2 and 3.
- c) 1 and 4.
- d) 1 , 2 , 3 and 4.

Answer ;

- d)** 1 , 2 , 3 and 4.
-

0191. Equalization of air pressure between the outer and the middle ear is ;

- a) more important during climb because the lower pressure within the middle ear can damage the tympanic membrane.
- b) impossible during descent because the pressure within the middle ear is higher than in the ambient air.
- c) important to prevent pain and loss of hearing during flight.
- d) difficult to achieve when pressure changes are at a low rate.

Answer ;

- c)** important to prevent pain and loss of hearing during flight.
-

- 0192.** Equalization of pressure is limited between the middle ear and the ambient , when ;
- a) the Eustachian tube is blocked.
 - b) barotrauma exists in the sinuses.
 - c) the nose is pinched.
 - d) you breath through the mouth.

Answer ;

a) the Eustachian tube is blocked.

- 0193.** Ergonomics are associated with ;
- a) cognition.
 - b) the human/workplace interface.
 - c) communications.
 - d) information processing.

Answer ;

b) the human/workplace interface.

- 0194.** Even at normal cabin altitudes (i.e. around 8000 ft) you can get severe abdominal pain and flatulence after eating gas forming foods or fizzy drinks. The correct countermeasure is ;
- a) perform “valsalva manoeuvre”.
 - b) descend to a lower altitude.
 - c) climb to a higher altitude.
 - d) use supplemental oxygen.

Answer ;

b) descend to a lower altitude.

- 0195.** Excessive exposure to noise can damage ;
- a) the sensitive membrane in the cochlea.
 - b) the ossicles.
 - c) the semi circular canals.
 - d) the eardrum.

Answer ;

a) the sensitive membrane in the cochlea.

- 0196.** Excessive intake of aspirin can cause ;
- a) memory impairment.
 - b) gastric bleeding.
 - c) sensory loss.
 - d) fits.

Answer ;

b) gastric bleeding.

- 0197.** Expanded abdominal air following rapid decompression at 30.000 feet ;
- a) will not cause pain if you evacuate the air via the mouth.
 - b) is usually not dangerous because only the partial pressure of the oxygen affects the nervous system.
 - c) may be dangerous causing great pain.
 - d) can be reduced if you take drinks with high contents of carbondioxide before take-off.

Answer ;

c) may be dangerous causing great pain.

- 0198.** Expressed as a simple mathematical statement when considering a crew of two , good synergy is ;
- a) $1 - 1 = > 2$.
 - b) $1 + 1 = > 2$.
 - c) $1 + 1 = < 2$.
 - d) $1 +/- 1 = < 2$.

Answer ;

b) $1 + 1 = > 2$.

- 0199.** Fatigue and stress ;
- a) increase the tolerance to hypoxia.
 - b) lower the tolerance to hypoxia.
 - c) will increase the tolerance to hypoxia when flying below 15.000 feet.
 - d) do not affect hypoxia at all.

Answer ;

b) lower the tolerance to hypoxia.

0200. "Feedback" in communication implies that ;
- a) readback of a message is rarely necessary except in the case of specific ATC procedures.
 - b) a message is measured and corrected against the original meaning.
 - c) the flight crew do not need to talk all the time because the feedback is automatic.
 - d) storing of messages in short time memory is available for later release.

Answer ;

- b) a message is measured and corrected against the original meaning.
-

0201. Fixation or tunnel vision is primarily to be expected when ;

- a) stress and motivation are low.
- b) stress is medium.
- c) stress is high.
- d) stress and motivation are medium.

Answer ;

- c) stress is high.
-

0202. Flickering light when reflected from spinning rotor blades ;

- a) should be avoided , because it may destroy the optic nerve.
- b) can be avoided when the strobe-lights are switched on.
- c) can cause spatial disorientation and/or nausea.
- d) has no effect.

Answer ;

- c) can cause spatial disorientation and/or nausea.
-

0203. Flights immediately after SCUBA-diving (compressed gas mixtures , bottles) (>10 m depth) ;

- a) should be avoided because hypoxia may develop.
- b) are allowed , if 38.000 ft are not exceeded.
- c) are forbidden.
- d) can be performed without any danger.

Answer ;

- c) are forbidden.
-

0204. Flights immediately after SCUBA-diving , using compressed gas , to depths greater than 10 metres ;

- a) are to be avoided because of the possibility of decompression sickness.
- b) are allowed if you fly no higher than 38.000 ft.
- c) should be avoided because hypoxia may develop.
- d) can be performed without any danger.

Answer ;

- a) are to be avoided because of the possibility of decompression sickness.
-

0205. Flying a coordinated level turn will ;

- a) make the blood being pooled in the head.
- b) make the seat-of-the-pants sense feel a decreased pressure along the body's vertical axis.
- c) make the body's pressure receptors feel an increased pressure along the body's vertical axis.
- d) first give the impression of climb , then the impression of descent.

Answer ;

- c) make the body's pressure receptors feel an increased pressure along the body's vertical axis.
-

0206. Flying a non-pressurized light aircraft at 9.000 feet your passenger develops blue lips ;

- a) you supply him with extra oxygen , but do not reduce altitude as the partial pressure of oxygen never results in symptoms below 10.000 feet.
- b) assuming he is hyperventilating you order him to stop breathing while you supply additional oxygen.
- c) you will attempt to calm him/her and occupying the passenger with suitable tasks.
- d) assuming he is suffering from hypoxia you reduce altitude and supply him/her with oxygen.

Answer ;

- d) assuming he is suffering from hypoxia you reduce altitude and supply him/her with oxygen.
-

0207. Flying at pressure altitude of 10.000 ft , a pilot , being a moderate to heavy smoker , has an oxygen content in the blood equal to an altitude ;

- a) of 15.000 ft when breathing 100% oxygen.
- b) lower than 10.000 ft.
- c) of 10.000 ft.
- d) above 10.000 ft.

Answer ;

- d) above 10.000 ft.
-

0208. Flying from Frankfurt to Moscow you will have a lay-over of 4 days. What time measure is relevant for your circadian rhythm on the third day ?

- a) UTC (universal time coordinated).
- b) MEZ (middle european time).
- c) ZT (zonal time).
- d) LT (local time).

Answer ;

d) LT (local time).

0209. Flying immediately after SCUBA diving involves the risk of getting ;

- a) hypoxia.
- b) stress.
- c) decompression sickness without having a decompression.
- d) hyperventilation.

Answer ;

c) decompression sickness without having a decompression.

0210. Flying immediately following a dive with SCUBA diving equipment (> 10 m depth) is forbidden because it ;

- a) has no influence on altitude flights.
- b) prevents any dangers caused by DCS (decompression sickness) when climbing to altitudes not exceeding 30 000 ft.
- c) will always lead to hypoxia.
- d) can cause decompression sickness even when flying at pressure altitudes below 18 000 ft.

Answer ;

d) can cause decompression sickness even when flying at pressure altitudes below 18 000 ft.

0211. Flying immediately following a dive with SCUBA diving equipment to a depth greater than 10 metres is not advisable because ;

- a) can cause decompression sickness even when flying at pressure altitudes below 18.000 ft.
- b) prevents any dangers caused by DCS (decompression sickness) when climbing to altitudes not exceeding 30.000 ft.
- c) has no influence on altitude flights.
- d) will always lead to hypoxia.

Answer ;

a) can cause decompression sickness even when flying at pressure altitudes below 18.000 ft.

0212. Flying while suffering from a cold may expose a pilot to ;

- a) otic and sinus barotrauma.
- b) gastro-intestinal barotrauma.
- c) sinus barotrauma only.
- d) autokinesis.

Answer ;

a) otic and sinus barotrauma.

0213. Flying with a “common cold” ;

- a) will cause infection in other crew members if you are flying in a pressurised aircraft.
- b) increases the risk of hypertension.
- c) is permitted as long as you are on treatment with antibiotics.
- d) may lead o incapacitation due to severe sinus or ear pain.

Answer ;

d) may lead o incapacitation due to severe sinus or ear pain.

0214. Following a rapid decompression at 30.000 feet , the time of useful consciousness would be about ;

- a) 10 to 12 minutes.
- b) 3 to 5 minutes.
- c) 5 to 10 minutes.
- d) between 45 seconds and 1 minute 30 seconds.

Answer ;

d) between 45 seconds and 1 minute 30 seconds.

0215. For a normal and healthy person , personality traits are ;

- a) easy changed by an outside influence.
- b) unstable.
- c) easy changeable.
- d) stable.

Answer ;

d) stable.

0216. Free running circadian rhythms normally have a cycle of approximately ;

- a) 6 hours.
- b) 36 hours.
- c) 8 hours.
- d) 25 hours.

Answer ;

d) 25 hours.

0217. From the list below which answers are correct ? Feedback gives information about ;

- 1. the sender.
 - 2. the sender's intentions.
 - 3. the situation.
- a) 2 and 3 only.
 - b) 1 , 2 and 3.
 - c) 1 and 2 only.
 - d) 3 only.

Answer ;

b) 1 , 2 and 3.

0218. Galactic Radiation is ;

- a) unsteady and reasonably predictable.
- b) steady and reasonably predictable.
- c) steady but unpredictable.
- d) unsteady and unpredictable.

Answer ;

b) steady and reasonably predictable.

0219. General Adaptation Syndrome is characterised by the following phases ;

- 1. alarm.
 - 2. alert phase.
 - 3. resistance phase.
 - 4. exhaustion phase.
 - 5. vigilance phase.
- a) 1 , 2 , 4 , 5.
 - b) 2 , 3 , 4.
 - c) 1 , 3 , 4.
 - d) 2 , 3 , 4 , 5.

Answer ;

c) 1 , 3 , 4.

0220. Generally , the time required for dark adaptation is ;

- a) 10 min.
- b) 30 min.
- c) 10 sec.
- d) 1/10 sec.

Answer ;

b) 30 min.

0221. Glaucoma ;

- a) visual field loss due to glaucoma can be significantly improved by giving oxygen.
- b) glaucoma causes red / green colour blindness.
- c) low pressure in the eye ball leads to decreased blood flow to the retina which can cause visual loss.
- d) is a condition detected by pressure testing the eye ball.

Answer ;

d) is a condition detected by pressure testing the eye ball.

0222. Glaucoma ;

1. can lead to total blindness.
 2. can lead to undetected reduction of the visual field.
 3. reduces visual acuity in its final stage.
- a) 1 and 3 are correct , 2 is false.
b) 2 and 3 are correct , 1 is false.
c) 1 is correct , 2 and 3 are false.
d) 1 , 2 and 3 are correct.

Answer ;

d) 1 , 2 and 3 are correct.

0223. Glaucoma is ;

- a) high intra-ocular pressure.
b) disturbed night vision.
c) disturbed colour vision.
d) disturbed adaptation.

Glaucoma (Głokom) :
Göz tansiyonu adıyla da bilinen glockom , göz küresinin iç basıncının artması sonucu göz sinirinin hasarına ve görme kaybına yol açan bir hastalıktır.

Answer ;

a) high intra-ocular pressure.

0224. Glaucoma is due to ;

- a) increase in pressure of the liquid within the eye.
b) excess light on the eyeball.
c) drop in pressure of the liquid around the eye.
d) damage to the eyeball due to high altitude.

Answer ;

a) increase in pressure of the liquid within the eye.

0225. "Greyout" occurs in case of ;

- a) + gX.
b) + gY.
c) - gZ.
d) + gZ.

Answer ;

d) + gZ.

0226. Group norms ;

- a) regulate the interaction and the behaviour between the members of a group.
b) have no influence on the working behaviour of the group.
c) diminish the morale of the group.
d) are always fixed in a written way and can be looked up.

Answer ;

a) regulate the interaction and the behaviour between the members of a group.

0227. Habits and routine can influence decision-making in a way that ;

- a) professional pilots will never question established procedures.
b) shows a tendency to select the most familiar solution first and foremost , sometimes to the detriment of achieving the best possible result.
c) one always selects a choice in accordance with the company's usual practices.
d) habit is the product of experience and should be used as a primary consideration in all decisions.

Answer ;

b) shows a tendency to select the most familiar solution first and foremost , sometimes to the detriment of achieving the best possible result.

0228. Haemoglobin has an affinity for carbon monoxide oftimes over oxygen ;

- a) 500 - 1.000.
b) 50 - 75.
c) 5 - 10.
d) 210 - 250.

Answer ;

d) 210 - 250.

0229. Haemoglobin is ;
- a) dissolved in the plasma.
 - b) in the red blood cells.
 - c) in the platelets.
 - d) in the white blood cells.

Answer ;

b) in the red blood cells.

0230. Having a cold or an infection of the upper respiratory tract you ;
- a) may accept to fly an ambulance flight because “Sinus squeeze” is rare and represents a minor danger to aviation.
 - b) must be careful when flying because the infection may cause hyperglycemia.
 - c) should not fly because the congestion of the frontal sinuses may cause great pain which can seriously affect your ability to control yourself and the aircraft.
 - d) should not fly because the infection may cause hypoxia due to congestion in the nose.

Answer ;

c) should not fly because the congestion of the frontal sinuses may cause great pain which can seriously affect your ability to control yourself and the aircraft.

0231. Having a serious cold it is better not to fly , due to the extra risk of ;

- 1. flatulence.
 - 2. pain in the ear during descent.
 - 3. vertigo.
 - 4. pain in the nasal sinuses.
- a) 1 and 2 are correct.
 - b) 2 , 3 and 4 are correct.
 - c) 1 , 2 and 4 are correct.
 - d) 1 , 3 and 4 are correct.

Answer ;

b) 2 , 3 and 4 are correct.

0232. Having a serious cold , you are going to fly. What can you expect ?

- a) bends.
- b) pain in the sinuses.
- c) hypoxia.
- d) chokes.

Answer ;

b) pain in the sinuses.

0233. Having interrupted the Captain for a sound reason , the co-pilot should ;

- a) remind him/her of his/her last action before the interruption.
- b) repeat the interruption twice so that the Captain’s attention is focussed on the new information.
- c) ensure there is no ATC traffic on the radio that may be causing a possible distraction before continuing.
- d) never interrupt his/her Captain at a critical stage of flight.

Answer ;

a) remind him/her of his/her last action before the interruption.

0234. Having made an important decision , a Commander of an aircraft should ;

- a) always try to make time to explain the reasons for the decision even if it is after landing.
- b) explain the reasons for the decision only if asked.
- c) as the leader of the crew , never need explain the reasons behind the decision.
- d) explain the reasons for the decision only if time permits in the air.

Answer ;

a) always try to make time to explain the reasons for the decision even if it is after landing.

0235. Healthy people are usually capable of compensating for a lack of oxygen up to ;

- a) 20.000 feet.
- b) 10.000 – 12.000 feet.
- c) 15.000 feet.
- d) 25.000 feet.

Answer ;

b) 10.000 – 12.000 feet

0236. Hearing Tests are carried out during pilot medicals to ;

- a) detect early presbyopia.
- b) detect early signs of deafness.
- c) detect vestibular dysfunction.
- d) detect reversible NIHL.

Answer ;

b) detect early signs of deafness.

0237. Heart attack (myocardial infarction) ;

- a) is always preceded by chest pains.
- b) is becoming less common because fewer people smoke.
- c) is not treatable.
- d) is the commonest cause of death in men over the age of 40.

Answer ;

d) is the commonest cause of death in men over the age of 40.

0238. Heart attack (myocardial infarction) ;

- a) is a partial blockage of a coronary artery leading to chest pain on exercise or stress.
- b) is a total blockage of a coronary artery leading to the death of a piece of heart muscle.
- c) an individual's risk of this can not be significantly influenced.
- d) is not treatable once symptoms develop.

Answer ;

b) is a total blockage of a coronary artery leading to the death of a piece of heart muscle.

0239. Henry's Law explains the occurrence of ;

- a) decompression sickness.
- b) hyperventilation.
- c) diffusion.
- d) hypoxia.

Answer ;

a) decompression sickness.

0240. Hepatitis "A" is transmitted through ;

- a) droplets in the air cause by the breath of an infected person.
- b) bacteria in the form of spores via a puncture in the skin.
- c) food or water which has been contaminated.
- d) insect bites.

Answer ;

c) food or water which has been contaminated.

0241. High arousal leads to ;

- a) faster and more accurate responses.
- b) faster but less accurate responses.
- c) slower and less accurate responses.
- d) slower but more accurate responses.

Answer ;

b) faster but less accurate responses.

0242. How are oxygen and carbon dioxide transported throughout the body ?

- a) Metabolism.
- b) Circulation.
- c) Diffusion.
- d) Oxidisation.

Answer ;

b) Circulation.

0243. How can a pilot avoid automation complacency ?

- a) Nothing , because it is system-inherent.
- b) Always try to enhance your aviation related knowledge during low workload periods.
- c) Regard the automatic system as additional crew members that needs to be crosschecked as well.
- d) Always fly the whole flight manually to remain in man-machine loop.

Answer ;

c) Regard the automatic system as additional crew members that needs to be crosschecked as well.

0244. How can a pilot increase his tolerance to +Gz ?
- a) Tighten shoulder harness.
 - b) Tightening of the stomach muscles.
 - c) Relax the muscles and lean upper body forward.
 - d) Take an upright seat position.

Answer ;

- b) Tightening of the stomach muscles.
-

0245. How can a pilot overcome vertigo , encountered during a real or simulated instrument flight ?

- 1. Establish and maintain an effective instrument cross-check.
 - 2. Always believe the instruments , never trust your sense of feeling.
 - 3. Ignore arising illusions.
 - 4. Move the head sideways and back and forth to “shake-off” illusions.
- a) Only 4 is correct.
 - b) 1 , 2 , 3 and 4 are correct.
 - c) 1 , 2 and 3 are correct.
 - d) 1 and 2 are correct , 3 and 4 are false.

Answer ;

- c) 1 , 2 and 3 are correct.
-

0246. How can a pilot prevent spatial disorientation in flight ?

- a) Always try to catch outside visual cues.
- b) Rely on good situational awareness believing your natural senses.
- c) Establish and maintain a good instrument cross check.
- d) Rely on the “seat of the pants” sense.

Answer ;

- c) Establish and maintain a good instrument cross check.
-

0247. How can the process of learning be enhanced ?

- a) By punishing the learner for unsuccessful trials.
- b) By increasing the psychological pressure on the student.
- c) By reinforcing errors.
- d) By reinforcing successful performance.

Answer ;

- d) By reinforcing successful performance
-

0248. How can vertigo be prevented in conditions of good visibility ?

- a) By avoiding blinking.
- b) By closing the eyes momentarily.
- c) By looking approximately 5° to one side of an object.
- d) By looking at the horizon.

Answer ;

- d) By looking at the horizon.
-

0249. How do you understand the statement “one cannot not communicate” ?

- a) No meaningful communication can occur unless it is by verbal means.
- b) Being silent as well as inactive are nonverbal behaviour patterns which express a meaning.
- c) You cannot influence your own communication.
- d) Communication is always possible.

Answer ;

- b) Being silent as well as inactive are nonverbal behaviour patterns which express a meaning.
-

0250. How does an increase in altitude affect the haemoglobin oxygen saturation ?

- a) As altitude increases , the haemoglobin oxygen saturation increases.
- b) As altitude increases , the haemoglobin oxygen saturation will remain constant at 57% from 24.000 ft.
- c) As altitude increases , the haemoglobin oxygen saturation decreases.
- d) As altitude increases , the haemoglobin oxygen saturation will remain constant at 97.5% from 10.000 ft.

Answer ;

- c) As altitude increases , the haemoglobin oxygen saturation decreases.
-

0251. How is haze effecting your perception ?
- a) Haze makes the eyes to focus at infinity
 - b) Objects will give better contrast.
 - c) Objects seem to be further away than in reality.
 - d) Objects seem to be closer than in reality.

Answer ;

- c) Objects seem to be further away than in reality.
-

0252. How is oxygen mainly transported in the blood ?
- a) Plasma.
 - b) Haemoglobin in the red blood cells.
 - c) White blood cells.
 - d) Blood fat.

Answer ;

- b) Haemoglobin in the red blood cells.
-

0253. How is yellow fever contracted ?
- a) Excessive use of alcohol.
 - b) A virus transmitted by an infected mosquito.
 - c) Contaminated food or water.
 - d) By contact with the saliva of infected animals.

Answer ;

- b) A virus transmitted by an infected mosquito.
-

0254. How many stages of sleep are there including REM sleep ?
- a) 2 stages.
 - b) 3 stages.
 - c) 5 stages.
 - d) 4 stages.

Answer ;

- c) 5 stages.
-

0255. How may haze effect perception ?
- a) Haze makes the eyes to focus at infinity
 - b) Objects seem to be closer than in reality.
 - c) Objects will give better contrast.
 - d) Objects seem to be further away than in reality.

Answer ;

- d) Objects seem to be further away than in reality.
-

0256. How will hyperventilation cause the acid-base balance of the blood to be affected ?
- a) Hyperventilation causes too much oxygen to be removed from the blood causing the level of carbonic acid to fall.
 - b) Hyperventilation causes too much CO₂ to be removed from the blood causing the level of carbonic acid to rise.
 - c) Hyperventilation causes too much oxygen to be removed from the blood causing the level of carbonic acid to rise.
 - d) Hyperventilation causes too much CO₂ to be removed from the blood causing the level of carbonic acid to fall.

Answer ;

- d) Hyperventilation causes too much CO₂ to be removed from the blood causing the level of carbonic acid to fall.
-

0257. How will the successful completion of a stressful task affect a pilot's performance when a similar situation is experienced in the future ?
- a) The amount of stress experienced will be less.
 - b) Given a similar situation the amount of stress will be the same.
 - c) Any stress will cause a reduction in performance.
 - d) The stress level experienced will be increased.

Answer ;

- a) The amount of stress experienced will be less.
-

0258. How would you describe the leadership style of a captain who primarily is interested in a friendly atmosphere within his crew , who is always constructive and encouraging , who usually compromises in interpersonal conflicts , who trusts in the capabilities of his crew-members , and who leaves the crew freedom for own decisions ?

- a) High task-orientation and high relationship-orientation.
- b) Low task-orientation and low relationship-orientation.
- c) Low task-orientation and high relationship-orientation.
- d) High task-orientation and low relationship-orientation.

Answer ;

c) Low task-orientation and high relationship-orientation.

0259. Human behaviour is determined by ;

- a) biological characteristics only.
- b) the social environment only.
- c) biological characteristics , social environment and cultural influences.
- d) cultural influences only.

Answer ;

c) biological characteristics , social environment and cultural influences.

0260. Human error rates during the performance of a simple and repetitive task can normally be expected to be approximately ;

- a) 1 in 2000.
- b) 1 in 500.
- c) 1 in 200.
- d) 1 in 100.

Answer ;

d) 1 in 100.

0261. Human errors are frequent and may take several forms ;

- a) an error can be described as the mismatch between the pilots intention and the result of his/her actions.
- b) all errors are avoidable through continuous and thorough training.
- c) a violation is an error which is always involuntary.
- d) representational errors in which the pilot has properly identified the situation and is familiar with the procedure.

Answer ;

a) an error can be described as the mismatch between the pilots intention and the result of his/her actions.

0262. Hypertension ;

- a) increases the risk of a heart attack and stroke.
- b) usually leads to loss of licence.
- c) is otherwise known as “white coat syndrome”.
- d) is only treated when symptoms develop.

Answer ;

a) increases the risk of a heart attack and stroke.

Hypertension (Hipertansiyon) :

Kan basıncının yüksek oluşudur. Hipertansiyon kalp hastalıkları için ana bir risk faktörüdür. Eğer tedavi edilmezse beyin dolaşımı , kalp , damar , göz ve böbrek hastalıkları için ciddi hastalık ve ölüm oranlarında artışa sebep olur.

0263. Hyperventilation causes ;

- a) a reduction of carbon dioxide in the blood.
- b) hypochondria.
- c) acidosis.
- d) an excess of carbon dioxide in the blood.

Answer ;

a) a reduction of carbon dioxide in the blood.

0264. Hyperventilation is ;

- a) a decreased lung ventilation.
- b) an increased lung ventilation.
- c) a too high percentage of oxygen in the blood.
- d) a too high percentage of nitrogen in the blood.

Answer ;

b) an increased lung ventilation.

0265. Hyperventilation is ;
- a) a reduction of partial oxygen pressure in the brain.
 - b) an increased heart rate caused by an increasing blood pressure.
 - c) a normal compensatory physiological reaction to a drop in partial oxygen pressure (i.e. when climbing a high mountain).
 - d) an increased heart rate caused by a decreasing blood-pressure.

Answer ;

- c) a normal compensatory physiological reaction to a drop in partial oxygen pressure (i.e. when climbing a high mountain).
-

0266. Hyperventilation is due to an excessive rate of breathing and can produce the following symptoms ;

- a) a state of overconfidence and reduced heart rate.
- b) blue finger-nails and lips.
- c) reduced heart rate and increase in visual acuity.
- d) dizziness , tingling sensation in the fingers and toes , nausea and blurred vision.

Answer ;

- d) dizziness , tingling sensation in the fingers and toes , nausea and blurred vision.
-

0267. Hypoglycaemia can be caused by ;

- a) not eating regularly or fasting.
- b) excessive g forces.
- c) otic barotraumas.
- d) excessive sugar in the blood.

Answer ;

- a) not eating regularly or fasting.
-

0268. Hypoxia can effect night vision ;

- a) at approximately 5000 ft.
- b) and causes the Coriolis Effect.
- c) and causes the autokinetic phenomena.
- d) less than day vision.

Answer ;

- a) at approximately 5000 ft.
-

0269. Hypoxia can be caused by ;

- a) a lack of red blood cells in the blood or decreased ability of the haemoglobin to transport oxygen.
- b) increasing oxygen partial pressure used for the exchange of gases.
- c) a lack of nitrogen in ambient air.
- d) too much carbon dioxide in the blood.

Answer ;

- a) a lack of red blood cells in the blood or decreased ability of the haemoglobin to transport oxygen.
-

0270. Hypoxia can be caused by ;

- 1. low partial pressure of oxygen in the atmosphere when flying at high altitudes without pressurisation and supplemental oxygen.
 - 2. a decreased saturation of oxygen in the blood due to carbon monoxide attached to the haemoglobin.
 - 3. blood pooling in the lower extremities due to inertia (+Gz).
 - 4. malfunction of the body cells to metabolize oxygen.
- a) 1 is false , 2 , 3 and 4 are correct.
 - b) 1 and 2 are correct , 3 and 4 are false.
 - c) 1 , 2 , 3 are correct , 4 is false.
 - d) 1 , 2 , 3 and 4 are correct.

Answer ;

- d) 1 , 2 , 3 and 4 are correct.
-

0271. Hypoxia is ;

- a) a condition of lacking oxygen in the brain causing the circulatory system to compensate by decreasing the heart rate.
- b) often produced during steep turns when pilots turn their heads in a direction opposite to the direction in which the aircraft is turning.
- c) a physical condition caused by a lack of oxygen saturation in the blood while hyperventilating.
- d) a physical condition caused by a lack of oxygen to meet the needs of the body tissues , leading to mental and muscular disturbances , causing impaired thinking , poor judgement and slow reactions.

Answer ;

- d) a physical condition caused by a lack of oxygen to meet the needs of the body tissues , leading to mental and muscular disturbances , causing impaired thinking , poor judgement and slow reactions.

-
0272. Hypoxia is a situation in which the cells ;
- a) have a shortage of oxygen.
 - b) are saturated with nitrogen.
 - c) have a shortage of carbon dioxide.
 - d) are saturated with oxygen.

Answer ;

- a) have a shortage of oxygen.
-

0273. Hypoxia is caused by ;
- a) an increased number of red blood cells.
 - b) a higher affinity of the red blood cells (haemoglobin) to oxygen.
 - c) reduced partial oxygen pressure in the lung.
 - d) reduced partial pressure of nitrogen in the lung.

Answer ;

- c) reduced partial oxygen pressure in the lung.
-

0274. Hypoxia will effect night vision ;
- a) at approximately 5000 ft.
 - b) and causes hyperventilation.
 - c) and causes the autokinetic phenomena.
 - d) less than day vision.

Answer ;

- a) at approximately 5000 ft.
-

0275. Hypoxic hypoxia may be caused by ;
- 1. climbing to a high altitude without using additional oxygen.
 - 2. malfunction in the oxygen supply system.
 - 3. loss of cabin pressurization at high altitude.

Which of the following lists all the correct answers ?

- a) 2 and 3.
- b) 1 and 3.
- c) 1 , 2 and 3.
- d) 1 and 2.

Answer ;

- c) 1 , 2 and 3.
-

0276. If a crew is having a stop-over for less than 24 hrs before returning to their homebase , what coping strategy should be used for jet lag ?

- a) Only have meals when it is daylight and sleep when it is dark , to ensure sufficient sleep credit.
- b) Maintain eating and sleeping cycle based on home time.
- c) Change eating and sleeping cycle to stop-over (destination) time.
- d) Eat light meals only and have regular naps instead of proper sleep.

Answer ;

- b) Maintain eating and sleeping cycle based on home time.
-

0277. If a stop-over is more than 24 hours , the correct action is to ;
- a) No adjustment is necessary.
 - b) Stay on home time and , on arrival , move to the new time.
 - c) Stay on home time.
 - d) Move to the new time as soon as possible.

Answer ;

- d) Move to the new time as soon as possible.
-

0278. If coping with a stress situation is impossible , one will remain in the state of ;
- a) adaptation.
 - b) hypoxia.
 - c) distress.
 - d) eustress.

Answer ;

- c) distress.
-

0279. If during flight a pilot is in a mental condition of “optimum arousal” he/she will be ;
- a) in a confused mental state.
 - b) prepared best to cope with a difficult task.
 - c) approaching a condition of complacency or fatigue.
 - d) unprepared to handle a difficult situation.

Answer ;

- b) prepared best to cope with a difficult task.
-

0280. If one error is allowed to effect a whole system , the system is described as ;

- a) Error-prone.
- b) Vulnerable.
- c) Error tolerant.
- d) Corrupted.

Answer ;

- b) Vulnerable
-

0281. If somebody starts breathing faster and deeper without physiological need ;

- a) the acid-base balance of the blood will not change.
- b) the blood pressure in the brain will rise significantly.
- c) the blood turns more alkaline.
- d) the blood turns more acid.

Answer ;

- c) the blood turns more alkaline.
-

0282. If someone hyperventilates due to stress his/her blood will become ;

- a) more acid.
- b) more alkaline.
- c) less saturated with oxygen.
- d) more saturated with carbon dioxide.

Answer ;

- b) more alkaline.
-

0283. If the co-pilot continuously feels unfairly treated by the Captain , he/she should ;

- a) point out the problem , concentrate on his/her duties and clarify the matter at a more appropriate time.
- b) speak up and point out the possible consequences if the unfair behaviour persists.
- c) internally retire and think positive.
- d) freeze the communication and thus avoid immediate confrontation.

Answer ;

- a) point out the problem , concentrate on his/her duties and clarify the matter at a more appropriate time.
-

0284. If warned of an imminent thunderstorm ;

- a) cockpit lights should be turned up.
- b) cockpit lights should be dimmed.
- c) no specific action is advised with regards to the adjustment of the brightness of cockpit lights.
- d) cockpit lights should be turned off.

Answer ;

- a) cockpit lights should be turned up.
-

0285. If you are disorientated during night flying you must ;

- a) check your rate of breathing - do not breathe too fast.
- b) descend.
- c) look outside.
- d) rely on your instruments.

Answer ;

- d) rely on your instruments.
-

0286. If you are subjected to an illusion during night flying you should ;

- a) use your oxygen mask.
- b) dim the cockpit lighting.
- c) continue on instruments.
- d) scan the surroundings.

Answer ;

- c) continue on instruments.
-

0287. Illuminated anti-collision lights in IMC ;
- a) can cause colour-illusions.
 - b) can cause disorientation.
 - c) will effect the pilots binocular vision.
 - d) will improve the pilots depth perception.

Answer ;

b) can cause disorientation.

0288. Illusions that pilots experience in conditions of fog or mist are that ;
- a) objects appear closer than they really are and lead to shallow approaches.
 - b) objects appear closer than they really are and lead to steep approaches.
 - c) objects appear further away than they really are and can lead to steep approaches.
 - d) objects appear further away than they really are and lead to shallow approaches.

Answer ;

c) objects appear further away than they really are and can lead to steep approaches.

0289. In 1972 , a psychologist named Edwards presented a concept of the interface between flight crew and other parts of the airspace environment. What is this concept called and the names of those elements involved ?

- a) The SHELL Concept. Software , hardware , economy and liveware.
- b) The SHELL Concept. Software , hardware , environment , and liveware.
- c) The SHELL Concept. Software , harmony , environment , and location.
- d) The SHELL Concept. Software , hardware , environment and location.

Answer ;

b) The SHELL Concept. Software , hardware , environment , and liveware.

0290. In a complex task high levels of arousal ;
- a) lead to better decision-making.
 - b) improve performance.
 - c) reduce failures.
 - d) narrow the span of attention.

Answer ;

d) narrow the span of attention.

0291. In an abnormal situation the pilot has an apparently correct explanation for the problem. The chance that he/she now ignores or devalues other relevant information , not fitting into his/her mental picture is ;

- a) very rare.
- b) not applicable with old and experienced pilots.
- c) not usual.
- d) not possible since correct training should have eliminated this problem.

Answer ;

c) not usual.

0292. In case of in-flight stress , one should ;
- a) always carry out a breathing exercise.
 - b) only trust in oneself , being sure to know the own limits.
 - c) demonstrate aggressiveness to stimulate the crew.
 - d) use all available resources of the crew.

Answer ;

d) use all available resources of the crew.

0293. In civil air transport , linear accelerations (Gx) ;
- 1. do not exist.
 - 2. have slight physiological consequences.
 - 3. may , in the case of pull-out , lead to loss of consciousness.
 - 4. may cause sensory illusions on the pitch axis.
- a) 3.
 - b) 2 , 4.
 - c) 1.
 - d) 3 , 4.

Answer ;

b) 2 , 4.

0294. In decision-making , the selection of a solution depends on ;
1. objective and subjective criteria.
 2. the objective to be achieved.
 3. the risks associated with each solution.
 4. the personality of the decision-maker.
- a) 4.
b) 1 , 2 , 3 , 4.
c) 1 , 2 , 4.
d) 1 , 3.

Answer ;
b) 1 , 2 , 3 , 4.

0295. In-flight incapacitation of a pilot is most frequently caused by ;
- a) chronic pulmonary disorders.
 - b) behavioural disorders.
 - c) cardio-vascular disorders.
 - d) acute gastrointestinal disorders.

Answer ;
d) acute gastrointestinal disorders.

0296. In his/her behaviour , the “ideal professional pilot” is ;
- a) both “person” and “goal” oriented.
 - b) neither “person” nor “goal” oriented.
 - c) “goal” rather than “person” oriented.
 - d) “person” rather than “goal” oriented.

Answer ;
a) both “person” and “goal” oriented.

0297. In order to avoid hypoglycaemia ;
- a) peanuts are recommended because of their high energy value.
 - b) a pilot should eat regularly and ensure he/she has a balanced diet.
 - c) a pilot should never eat sugar or sweets.
 - d) a pilot should not take snacks between meals.

Answer ;
b) a pilot should eat regularly and ensure he/she has a balanced diet.

Hypoglycaemia (Hipoglisemi) :
Kanda Glukoz seviyesinin düşük olması.

0298. In order to completely resynchronise with local time after zone crossing , circadian rhythms require ;
- a) about one week per 2.5 hours of time shift.
 - b) more time when flying from east to west.
 - c) about one day per 2.5 hours of time shift.
 - d) less time when flying from east to west.

Answer ;
d) less time when flying from east to west.

0299. In order to get rid of excess nitrogen following scuba diving , subsequent flights should be delayed ;
- a) 24 hours.
 - b) 36 hours after any scuba diving.
 - c) 3 hours after non decompression diving.
 - d) 48 hours after a continuous ascent in the water has been made.

Answer ;
a) 24 hours.

0300. In order to limit stress when flying , a pilot should ;
- a) maintain his competence by practicing his professional skills and learning from past experiences.
 - b) forget about bad past experiences.
 - c) drop activities outside work so as to focus on his work better.
 - d) avoid anticipating events during a flight to manage his workload.

Answer ;
a) maintain his competence by practicing his professional skills and learning from past experiences.

- 0301.** In order to make communication effective , it is necessary to ;
1. avoid the synchronization of verbal and non-verbal channels.
 2. send information in line with the receiver's decoding abilities.
 3. always concentrate on the informational aspects of the message only.
 4. avoid increasing the number of communication channels , in order to simplify communication.

The correct statement(s) is (are) ;

- a) 1 , 2 and 3 are correct.
- b) 2 and 4 are correct.
- c) only 2 is correct.
- d) 3 and 4 are correct.

Answer ;

- c) only 2 is correct.**
-

- 0302.** In order to make sound decisions it is important to ;
- a) understand why and how we make decisions.
 - b) keep a low level of stress by excluding non-verbal communication.
 - c) avoid situational awareness.
 - d) accept that the Captain , because of his/her authority , tends to be correct.

Answer ;

- a) understand why and how we make decisions.**
-

- 0303.** In order to overcome an overload of work during the flight , it is necessary to ;
1. know how to use one's own reserve of resources in order to ease the burden on the crew.
 2. divide up tasks among the crew.
 3. ensure that the long-term memory is used as much as possible as short term memory will add to the stress.
 4. drop certain tasks and stick to high-level priorities.

The correct statement(s) is (are) ;

- a) 3 and 4 are correct.
- b) 1 , 2 and 3 are correct.
- c) 1 , 2 and 4 are correct.
- d) 1 and 3 are correct.

Answer ;

- c) 1 , 2 and 4 are correct.**
-

- 0304.** In order to perceive colour vision , it is necessary ;
1. for there to be considerable amount of light (ambient luminosity).
 2. at night to look at the point to be observed at an angle of 15°.
 3. to allow the eye a period of time to get used to the light.
 4. to avoid white light.

- a) 1 , 2 , 3.
- b) 1.
- c) 2 , 4.
- d) 3.

Answer ;

- b) 1.**
-

- 0305.** In order to reduce the risk of coronary artery disease , exercise should ;
- a) be avoided since raising the heart rate shortens the life of the heart.
 - b) triple the resting heart rate for 20 minutes , once a week.
 - c) double the resting heart rate for at least an hour , five times a week.
 - d) double the resting heart rate for at least 20 minutes , three times a week.

Answer ;

- d) double the resting heart rate for at least 20 minutes , three times a week.**
-

- 0306.** In problem-solving , what determines the transition from rule-based activities to a knowledge-based activity ?
- a) The unsuitability of the automated actions.
 - b) Knowledge of rules which apply to the problem posed.
 - c) The unsuitability of the known rules for the problem posed.
 - d) Attentional capture.

Answer ;

- c) The unsuitability of the known rules for the problem posed.**
-

0307. In terms of decision-making , the intention to become integrated into the team , to be recognised as the leader or to avoid conflicts may lead to ;

- a) the improvement of internal risk assessment capabilities.
- b) a fast and decisive option.
- c) the attempt to agree on decisions made by other crew members.
- d) an authoritarian approach thus demonstrating ones own ability to lead.

Answer ;

- c) the attempt to agree on decisions made by other crew members.
-

0308. In the absence of external reference points , the sensation that the vehicle in which you sitting is moving when it is in fact the vehicle directly alongside which is moving is called ;

- a) illusion of relative movement.
- b) autokinetic illusion.
- c) cognitive illusion.
- d) somato-gravic illusion.

Answer ;

- a) illusion of relative movement.
-

0309. In the Alarm stage of response to stress what does the body do ?

- a) Attempts to repair any damage caused by the stress.
- b) Recognises danger and slows the body down via the autonomic system.
- c) Attempts to maintain the body's arousal state to the continuing stress.
- d) Recognises the stressor and prepares the body for action.

Answer ;

- d) Recognises the stressor and prepares the body for action.
-

0310. In the alveoli gas exchange takes place (external respiration). Which gas will diffuse from the blood into the lungs ?

- a) Carbon monoxide.
- b) Carbon dioxide.
- c) Ambient air.
- d) Oxygen.

Answer ;

- b) Carbon dioxide.
-

0311. In the case of changes to circadian rhythms , the readjustment to a new time zone ;

- a) is most rapid when flying eastbound.
- b) is the same in both westbound and eastbound flights.
- c) occurs immediately , as circadian rhythms do not change depending on direction of flying.
- d) is most rapid when flying westbound.

Answer ;

- d) is most rapid when flying westbound.
-

0312. In the case of fire in flight most fatalities are caused by ;

- a) Cardiac arrest caused by stress and fear.
- b) Inhalation of hydrogen fumes.
- c) Burns.
- d) Suffocation from fumes caused by aircraft furnishing and wiring.

Answer ;

- d) Suffocation from fumes caused by aircraft furnishing and wiring.
-

0313. In the decision-making process , confirmation bias results in ;

- a) a tendency to look for approval by other crew-members , or by the ATC.
- b) choosing familiar solutions , even if they are not the best ones.
- c) overestimating the frequency of some events.
- d) a tendency to look for information which confirms the validity of the decision.

Answer ;

- d) a tendency to look for information which confirms the validity of the decision.
-

0314. In the event of fuel spillage onto the body it is advisable that the affected area is ;

- a) bandaged and medication is applied to avoid blistering.
- b) initially washed with soap and water.
- c) immediately exposed to air to allow the fuel to evaporate as quickly as possible.
- d) initially washed with copious water and without the use of soap.

Answer ;

- d) initially washed with copious water and without the use of soap.
-

0315. In the event of rapid decompression the first action for the flight deck crew is ;

- a) carry out check for structural damage.
- b) transmit mayday call.
- c) don oxygen masks and ensure oxygen flow.
- d) descent to the higher of 10.000 ft or MSA.

Answer ;

c) don oxygen masks and ensure oxygen flow.

0316. In the short-term-memory , information is stored for approximately ;

- a) 1 hour.
- b) 5 minutes.
- c) a couple of days.
- d) 20 seconds.

Answer ;

d) 20 seconds.

0317. In tropical zones , it is recommended that ;

- a) one eats fruit without peeling them , to ensure a supply of vitamins.
- b) one uses ice cubes in cold drinks.
- c) cold drinks be taken from sealed containers.
- d) one eats raw vegetables.

Answer ;

c) cold drinks be taken from sealed containers.

0318. Incapacitation is most dangerous when it is ;

- a) insidious.
- b) obvious.
- c) sudden.
- d) intense.

Answer ;

a) insidious.

0319. Incorrect perception ;

- a) will only affect one member of the crew at a time.
- b) can be easily recognized.
- c) will only affect flight deck crew.
- d) can be highly persuasive.

Answer ;

d) can be highly persuasive.

0320. In-flight incapacitation of a pilot is most frequently caused by ;

- a) cardio-vascular disorders.
- b) behavioural disorder.
- c) acute gastrointestinal disorders.
- d) chronic pulmonary disorders.

Answer ;

c) acute gastrointestinal disorders.

0321. Informal roles within a crew ;

- a) will always impair the captain's influence.
- b) evolve as a result of the interactions that take place among crew members.
- c) characterize inefficient crews.
- d) are explicitly set out by the crew.

Answer ;

b) evolve as a result of the interactions that take place among crew members.

0322. Information received from the senses is lost ;

- a) only if you suffer from brain damage or disease.
- b) within 10-20 seconds unless it is rehearsed and deliberately placed in the Long Term Memory.
- c) if you choose to store the information in one of the sensory memories (e.g. Iconic or Echoic memories).
- d) if an additional action takes place (e.g. pulling a circuit breaker or extinguishing a warning light).

Answer ;

b) within 10-20 seconds unless it is rehearsed and deliberately placed in the Long Term Memory.

0323. Information stays in the short-term memory ;

- a) less than 1 second.
- b) from 5 to 10 minutes.
- c) around 24 hours.
- d) about 20 seconds.

Answer ;

d) about 20 seconds.

0324. Inhaling carbon monoxide can be extremely dangerous during flying. Which of the following statements is correct ?

- a) Carbon monoxide increases the oxygen saturation in the blood.
- b) With increasing altitude the negative effects of carbon monoxide poisoning can be compensated for.
- c) Small amounts of carbon monoxide are harmless.
- d) Carbon monoxide is odourless and colourless.

Answer ;

d) Carbon monoxide is odourless and colourless.

0325. Inhaling carbon monoxide can be extremely dangerous during flying. Which of the following statements is correct ?

- a) Carbon monoxide increases the oxygen saturation in the blood.
- b) Carbon monoxide is odourless and colourless.
- c) Carbon monoxide has no affinity to haemoglobin.
- d) With increasing altitude the negative effects of carbon monoxide poisoning can be compensated for.

Answer ;

b) Carbon monoxide is odourless and colourless.

0326. Internal conflict within oneself is termed as ;

- a) Intermediate-personal conflict.
- b) Intra-personal conflict.
- c) Inter-personal conflict.
- d) Intro-personal conflict.

Answer ;

b) Intra-personal conflict.

0327. Internal respiration ;

- a) is the exchange of carbon dioxide and oxygen in the lung.
- b) is the process of inflating the lungs during mouth-to-mouth resuscitation.
- c) is a metabolic process that takes place inside the cells during which oxygen is used and carbon dioxide produced.
- d) is the term for respiration inside the pressure cabin.

Answer ;

c) is a metabolic process that takes place inside the cells during which oxygen is used and carbon dioxide produced.

0328. Is it possible to suffer from Aerodontalgia in the descent ?

- a) Only in pressurized aircraft.
- b) Yes.
- c) Only above 10.000 ft AMSL.
- d) No.

Answer ;

d) No.

Aerodontalgia (Aerodontalji) ;
Hava basıncındaki ani değişimin sebep olduğu diş ağrısı.

0329. Is it possible to survive depressurized flight at 40.000 feet for more than 10 minutes ?

- a) Yes - because the partial pressure of oxygen in the blood is independent of that in the lungs.
- b) No - because the partial pressure of oxygen in the lung alveoles at 40.000 feet is lower than in the blood , causing oxygen to flow backwards from blood to lungs depleting the brain of its oxygen.
- c) Yes - provided 100% oxygen is immediately available from masks.
- d) No - because the brain can never receive sufficient oxygen at 40.000 feet even with an oxygen mask.

Answer ;

c) Yes - provided 100% oxygen is immediately available from masks.

0330. It is desirable to standardize as many patterns of behaviour (operating procedures) as possible in commercial aviation mainly because ;

- a) this reduces the amount of training required.
- b) it makes the flight deck easier to design.
- c) this lowers the ability requirement in pilot selection.
- d) such behaviour reduces errors even under adverse circumstances.

Answer ;

- d) such behaviour reduces errors even under adverse circumstances.
-

0331. It is generally considered that the most serious non-professional stressor is ;

- a) death of a spouse or partner.
- b) personal injury or illness.
- c) loss of a job.
- d) marital separation.

Answer ;

- a) death of a spouse or partner.
-

0332. It is inadvisable to fly when suffering from a cold. The reason for this is ;

- a) swollen tissue in the Eustachian tube will cause permanent hearing loss.
- b) pain and damage to the eardrum can result , particularly during fast descents.
- c) swollen tissue in the inner ear will prevent the air from ventilating through the tympanic membrane.
- d) gentle descents at high altitude can result in damage to the ear drum.

Answer ;

- b) pain and damage to the eardrum can result , particularly during fast descents.
-

0333. It is inadvisable to fly when suffering from a cold. The reason for this is ;

- a) the tissue around the nasal end of the Eustachian tube is likely to be swollen thus causing difficulty in equalising the pressure within the middle ear and the nasal/throat area. Pain and damage to the eardrum can result , particularly during fast descents.
- b) because it will seriously affect peripheral vision.
- c) swollen tissue in the inner ear will increase the rate of metabolic production resulting in hyperventilation
- d) although the change in air pressure during a climb at lower altitudes is very small , it increases rapidly at high altitudes. If the tissue in the Eustachian tube of the ear is swollen , gentle descents at high altitude would result in damage to the ear drum.

Answer ;

- a) the tissue around the nasal end of the Eustachian tube is likely to be swollen thus causing difficulty in equalising the pressure within the middle ear and the nasal/throat area. Pain and damage to the eardrum can result , particularly during fast descents.
-

0334. Judgement is based upon ;

- a) the development of skills through constant practice of flight manoeuvres.
- b) a process involving a pilot's attitude to take and to evaluate risks by assessing the situation and making decisions based upon knowledge , skill and experience.
- c) the ability to interpret the flight instruments.
- d) a decision-making process involving the 5 physical senses and their use to manually operate the aircraft controls.

Answer ;

- b) a process involving a pilot's attitude to take and to evaluate risks by assessing the situation and making decisions based upon knowledge , skill and experience.
-

0335. Less experienced pilots differ from experienced pilots in the following way ;

- a) flight planning performance decreases with age , and experience is unable to mask this deficiency.
- b) task for task , an expert's workload is greater than a novice's one.
- c) experienced pilots are less routine-minded than young pilots because they know that routine causes mistakes.
- d) inexperienced pilots refer to information more than experts when carrying out the same task.

Answer ;

- d) inexperienced pilots refer to information more than experts when carrying out the same task.
-

0336. Linear acceleration when flying straight and level in IMC may give the illusion of ;

- a) spinning.
- b) descending.
- c) yawing.
- d) climbing.

Answer ;

- d) climbing.
-

0337. List the constituent parts of the blood and state their function.

- a) Red cells carry oxygen , white cells are involved in immunity and platelets coagulate blood.
- b) Red cells carry oxygen , white cells coagulate blood and platelets are involved in immunity.
- c) Red cells are involved in immunity , white cells carry oxygen and platelets coagulate blood.
- d) Red cells coagulate blood , white cells are involved in immunity , and platelets carry oxygen.

Answer ;

- a) Red cells carry oxygen , white cells are involved in immunity and platelets coagulate blood.

0338. Loss of elasticity of the crystalline lens due to ageing occurs in ;

- a) hypermetropia.
- b) presbyopia.
- c) astigmatism.
- d) myopia.

Presbyopia (Presbiyopi) ;
Yaşlılık nedeni ile gözdeki lensin esnekliğini kaybetmesi sonucu yakını iyi görememe.

Answer ;

- b) presbyopia.

0339. Man possesses a system for maintaining his internal equilibrium in the face of variations brought about by external stimulations. This internal equilibrium is called ;

- a) Isothermy.
- b) Metastasis.
- c) Heterostasis.
- d) Homeostasis.

Homeostasis (Homeostazi) ;
Tüm vücut fonksiyonlarının , normal seviyelerde , dengeli olmasıdır ve vücutta gerçekleşen değişikliklere karşı var olan dengenin korunmaya çalışılmasıdır

Answer ;

- d) Homeostasis.

0340. Mark the two most important attributes for a positive leadership style ;

- 1. dominant behaviour.
 - 2. excellent role-behaviour.
 - 3. mastery of communication skills.
 - 4. “Laissez-faire” behaviour.
- a) 2 and 3.
 - b) 1 and 3.
 - c) 1 and 4.
 - d) 2 and 4.

Answer ;

- a) 2 and 3.

0341. Medical conditions such as high blood pressure , coronary problems and diabetes are associated with ;

- a) hypoxia.
- b) cholera.
- c) anorexia nervosa.
- d) obesity.

Obesity (Obezite) ;
Vücutta özellikle de karın bölgesinde aşırı yağ birikimi sonucu oluşan şişmanlık.

Answer ;

- d) obesity.

0342. Mental rehearsal is helpful to improve flying skills ;

- a) only for instructor pilots.
- b) only for student pilots.
- c) only at a certain level of flying experience.
- d) at all levels of flying proficiency.

Answer ;

- d) at all levels of flying proficiency.

0343. Metacommunication is defined as ;

- a) having an assessment conversation.
- b) those tools , other than the actual words , which compliment those words in order to communicate.
- c) active listening.
- d) balancing the own ideas and interests with those of the receiver.

Answer ;

- b) those tools , other than the actual words , which compliment those words in order to communicate.

0344. Mode Error is associated with ;

- a) Hardware.
- b) Check lists.
- c) Vertigo.
- d) Automation.

Answer ;

d) Automation.

0345. Motivation , crew-climate and “Confirmation Bias” are factors that ;

- a) have no influence on teamwork.
- b) can influence the situational awareness.
- c) are easy to control.
- d) depend only on personality and are unchangeable.

Answer ;

b) can influence the situational awareness.

0346. Motivation is an important attribute which contributes to flight safety. Which of the following statements is correct with regards to motivation ?

- a) A high degree of motivation makes it possible to make up for insufficient knowledge in complete safety.
- b) A high degree of motivation lowers the level of vigilance.
- c) Excessive motivation leads to stress which adversely affects performance.
- d) Motivation reduces the intensity of sensory illusions.

Answer ;

c) Excessive motivation leads to stress which adversely affects performance.

0347. Motor programmes are ;

- a) stored routines that enable patterns of behaviour to be executed without continuous conscious control.
- b) rules that enable us to deal with novel situations.
- c) rules that enable us to deal with preconceived situations.
- d) stored routines that enable patterns of behaviour to be executed only under continuous conscious control.

Answer ;

a) stored routines that enable patterns of behaviour to be executed without continuous conscious control.

0348. Name the commonest problem of the Long Term Memory and how would you counteract it ?

- a) Facts can be stored in the episodic memory instead of the semantic memory.Careful attention in learning will counteract this error.
- b) Unless an item is recalled regularly , it becomes more and more difficult to retrieve from the LTM store.
- c) Recent events can become muddled in the LTM.Careful attention to detail associated with “memory bench marking” will go a long way towards avoiding this tendency.
- d) Any item can be recalled from the LTM with ease as long as it has been stored correctly.

Answer ;

b) Unless an item is recalled regularly , it becomes more and more difficult to retrieve from the LTM store.

0349. Night vision ;

- a) is insensitive to short duration light sources.
- b) requires up to 30 minutes to reach its best performance.
- c) requires 5 minutes to reach its best performance.
- d) is insensitive to lightning flashes in storms.

Answer ;

b) requires up to 30 minutes to reach its best performance.

0350. Night vision is significantly reduced (i.e. a loss of more than 25%) at 15.000 ft in which stage of hypoxia ?

- a) Anaemic phase.
- b) Clinical manifestation phase.
- c) Compensatory.
- d) Indifferent.

Answer ;

c) Compensatory.

0351. Noise Induced Hearing Loss ;

- a) causes Eustationary Tube dysfunction.
- b) is a condition resulting in permanent hearing loss of selected frequencies.
- c) is also known as presbycusis and is associated with pressure damage to the middle ear.
- d) is not a permanent hearing loss , the nerve cells frequently recover.

Answer ;

b) is a condition resulting in permanent hearing loss of selected frequencies.

-
0352. Noise induced hearing loss is influenced by ;
- a) the suddenness of onset of a noise.
 - b) the duration and intensity of a noise.
 - c) the intensity of the noise but not its duration.
 - d) the duration of a noise but not its intensity.

Answer ;

- b) the duration and intensity of a noise.
-

0353. Noise induced hearing loss (NIHL) is caused by ;
- a) damage to the sensitive membrane in the cochlea due to overexposure to noise
 - b) reduced mobility of the ossicles
 - c) pressure differences on both sides of the eardrum
 - d) a blocked Eustachian tube

Answer ;

- a) damage to the sensitive membrane in the cochlea due to overexposure to noise
-

0354. Non-verbal communication ;
- a) should be avoided by all means in the cockpit.
 - b) is always used intentionally.
 - c) supports verbal communication.
 - d) is of no meaning in the cockpit.

Answer ;

- c) supports verbal communication.
-

0355. Non-verbal communication ;
- a) is always clearly interpretable.
 - b) is cross-culturally understood.
 - c) can serve as a substitute for oral speech.
 - d) is used only if verbal clarity is not possible.

Answer ;

- c) can serve as a substitute for oral speech.
-

0356. “Non-verbal communication” ;
- a) is unimportant , inherited and cannot be taught.
 - b) means “a sterile cockpit” below 10.000 feet.
 - c) rarely causes accidents.
 - d) constitutes approximately 70% of human communication.

Answer ;

- d) constitutes approximately 70% of human communication.
-

0357. Of the following alternatives , which effects are due to positive acceleration (+Gz) ?
- 1. Decrease in heart rate.
 - 2. Pooling of blood into lower parts of the body.
 - 3. Drop in blood pressure above heart-level.
 - 4. Downward displacement or deformation of soft or mobile organs.

- a) 1.
- b) 2 , 3 , 4.
- c) 1 , 3 , 4.
- d) 1 , 2 , 3.

Answer ;

- b) 2 , 3 , 4.
-

0358. Of the following statements regarding stress , which is correct ?
- a) Stress and anxiety are the same in every sense.
 - b) Stress and fatigue are synonymous.
 - c) Stress may be positive , fatigue is always negative.
 - d) Stress always has a negative effect on performance.

Answer ;

- c) Stress may be positive , fatigue is always negative.
-

0359. Of the following statements , which apply to coordinated cooperation ?

1. It allows for synergy in the actions between the captain and the co-pilot.
2. It represents the simultaneous execution of a single action by the various members of the crew.
3. Communication here results in synchronised actions and the natural and easy distribution of responsibilities.
4. Communication is centred around the outside world.

The correct statement(s) is(are) ;

- a) 1 , 2 and 4.
- b) 1 and 3.
- c) 1 and 4.
- d) 2 and 3.

Answer ;

b) 1 and 3.

0360. On approach , by day , in snowy conditions , without reference to glideslope information ;

- a) there is a risk of you entering a spin.
- b) there is a risk of you underestimating your speed.
- c) estimation of height and distance is not a problem for a well-trained pilot.
- d) judgement of height is extremely difficult.

Answer ;

d) judgement of height is extremely difficult.

0361. On ascent the gases in the digestive tract will ;

- a) shrink.
- b) be absorbed by tissues and blood.
- c) stay the same.
- d) expand.

Answer ;

d) expand.

0362. On average most human adults require approximately how many hours sleep per night ?

- a) 6 hrs
- b) 10 hrs
- c) 4 hrs
- d) 8 hrs

Answer ;

d) 8 hrs

0363. On experiencing a vestibular illusion in straight and level flight , it is recommended that ;

- a) you close your eyes for a moment so that the oculo-vestibular conflict disappears.
- b) you avoid head movements and rely on your instruments.
- c) you rely on your sensations only , as humans are much more reliable than flight instruments.
- d) you tilt your head to the side opposite to the turn to nullify the stimulus causing the illusion.

Answer ;

b) you avoid head movements and rely on your instruments.

0364. Once a pilot has constructed a mental model he/she tends to ;

- a) give undue weight to information that contradicts the model.
- b) alter that model unnecessarily frequently.
- c) give undue weight to information that confirms the model.
- d) give equal weight to contradicting and confirming information.

Answer ;

c) give undue weight to information that confirms the model.

0365. Once a pilot has developed a certain way of thinking about a problem he will probably ;

- a) find it difficult to stick to his/her interpretation of the data.
- b) find it difficult to get out of that way of thinking and difficult to try a different interpretation of the data.
- c) find it impossible to get out of that way of thinking , whatever happens.
- d) find it easy to interpret the data in different ways.

Answer ;

b) find it difficult to get out of that way of thinking and difficult to try a different interpretation of the data.

0366. Once we have constructed a mental model we tend ;
- a) to give undue weight to information that contradicts the model.
 - b) to alter that model unnecessarily frequently.
 - c) to give undue weight to information that confirms the model.
 - d) to give equal weight to contradicting and confirming information.

Answer ;

- c) to give undue weight to information that confirms the model.
-

0367. One negative aspect of the highly automated cockpit results in ;
- a) complacency among the crewmembers.
 - b) pilots disregarding the automatic equipment.
 - c) less experienced crews because of more transparent system details.
 - d) constantly high crew overload with regard to the monitoring tasks.

Answer ;

- a) complacency among the crewmembers.
-

0368. One of the dangers of extreme cold is that ;
- a) sleepiness occurs associated with a feeling of contentment or apathy.
 - b) sleepiness occurs very slowly with a feeling of alertness.
 - c) sleepiness occurs very quickly with a feeling of agitation.
 - d) it degrades the fatty tissues of the body.

Answer ;

- a) sleepiness occurs associated with a feeling of contentment or apathy.
-

0369. One of the earliest and most significant features of Hypothermia (extreme cold) is ;
- a) aggression and anxiety.
 - b) aggression.
 - c) anxiety.
 - d) apathy.

Answer ;

- d) apathy.
-

0370. One of the first effects to be noticed on gradual exposure to high positive radial accelerations is ;
- a) red-vision.
 - b) black-out.
 - c) grey-out.
 - d) loss of consciousness.

Answer ;

- c) grey-out.
-

0371. One of the major protective measures against illusions is ;
- a) regular eye tests.
 - b) wearing of good quality sun glasses.
 - c) comprehensive briefing and de-briefing.
 - d) wearing of polaroid sun glasses.

Answer ;

- c) comprehensive briefing and de-briefing.
-

0372. One of the most dangerous symptoms of hypoxia concerning flight safety is ;
- a) impaired judgement.
 - b) reduced coordination of limb movements , causing the pilot to spin.
 - c) hyperventilation , causing emotional stress.
 - d) cyanosis , reducing then pilots ability to hear.

Answer ;

- a) impaired judgement.
-

0373. One of the substances present in the smoke of cigarettes can make it significantly more difficult for the red blood cells to transport oxygen and as a consequence contributes to hypoxia. Which substance are we referring to ?

- a) Carbon monoxide.
- b) Tar.
- c) Carbon dioxide.
- d) Carbonic anhydride.

Answer ;

- a) Carbon monoxide.
-

0374. One of the waste products of the metabolic process in the cell is ;

- a) carbon dioxide.
- b) sugar.
- c) fat.
- d) protein.

Answer ;

a) carbon dioxide.

0375. Organisational factors which affect or may have some influence on human error are ;

- 1. malfunction of technical systems.
- 2. fuel-saving policies.
- 3. rostering.
- 4. weather phenomena.

Which of the following lists all the correct answers ?

- a) 3 and 4.
- b) 1 and 4.
- c) 2 and 3.
- d) 1 and 2.

Answer ;

c) 2 and 3.

0376. Orientation in flight is accomplished by ;

- 1. eyes.
 - 2. utricle and saccule.
 - 3. semicircular canals.
 - 4. Seat-of-the-pants-Sense.
- a) 2, 3 and 4 are correct, 1 is false.
 - b) only 1 and 4 are correct.
 - c) 1, 2, 3 and 4 are correct.
 - d) 2, 3 and 4 are false, only 1 is correct.

Answer ;

c) 1, 2, 3 and 4 are correct.

0377. Otic Barotrauma is worse during the descent than climb because ;

- a) during the climb the humidity in the middle ear is greater than the cabin.
- b) the air pressure in the inner ear increases more slowly than the cabin.
- c) the air in the middle ear is at a lower pressure than the cabin, this closes the eustachian tube so that the pressure cannot be equalised.
- d) during the descent the cabin altitude increases faster than the eustachian can open.

Answer ;

c) the air in the middle ear is at a lower pressure than the cabin, this closes the eustachian tube so that the pressure cannot be equalised.

0378. Our body takes its energy from ;

- 1. minerals.
 - 2. protein.
 - 3. carbohydrates.
 - 4. vitamins.
- a) 1, 2, 3, 4.
 - b) 1, 3.
 - c) 1, 4.
 - d) 2, 3.

Answer ;

d) 2, 3.

0379. Our mental model of the world is based ;

- a) on both our past experiences and the sensory information we receive.
- b) entirely on past experiences.
- c) entirely on the sensory information we receive.
- d) on both our past experiences and our motor programmes.

Answer ;

a) on both our past experiences and the sensory information we receive.

0380. Out of the list of possible measures to counteract hyperventilation , the most effective measure is ;
- a) speak soothingly and get the person to breathe slowly.
 - b) avoid strenuous flight manoeuvres.
 - c) breathe into a plastic or paper bag.
 - d) hold breath.

Answer ;

c) breathe into a plastic or paper bag.

0381. Oxygen , combined with haemoglobin in blood is transported by ;
- a) white blood cells.
 - b) platelets.
 - c) blood plasma.
 - d) red blood cells.

Answer ;

d) red blood cells.

0382. Oxygen transport to the tissues ;
- a) is not affected by carbon monoxide.
 - b) will be maintained even during severe hypoxia.
 - c) depends on haemoglobin level and blood flow.
 - d) does not depend on haemoglobin levels as oxygen is mainly dissolved in the plasma.

Answer ;

c) depends on haemoglobin level and blood flow.

0383. Ozone in the air of a pressurized cabin can be eliminated by ;
- a) ozone-converters.
 - b) climbing to altitudes above 45.000 ft.
 - c) avoiding flights along the equator.
 - d) spraying detergents.

Answer ;

a) ozone-converters.

0384. Pain in the Joints (“bends”) is a symptom of ;
- a) barotrauma.
 - b) air-sickness.
 - c) decompression sickness.
 - d) hypoxia.

Answer ;

c) decompression sickness.

0385. Pain in the middle ear during descent may be eased by ;
- a) levelling off and possibly climbing.
 - b) increasing the rate of descent.
 - c) using an oxygen mask.
 - d) blocking the effected ear with the palm of your hand.

Answer ;

a) levelling off and possibly climbing.

0386. Perception is based upon ;
- a) solely upon technical ability.
 - b) information received as well as past experience and knowledge.
 - c) past experience and knowledge only.
 - d) information received only.

Answer ;

b) information received as well as past experience and knowledge.

0387. Perceptual illusions are ;
- a) never caused by the brain’s ability to distort normal sense inputs.
 - b) abnormal and indicate weakness in one or more of the senses.
 - c) normal and can be prevented by trusting instrument read-out.
 - d) usually not serious except in bad weather.

Answer ;

c) normal and can be prevented by trusting instrument read-out.

0388. Personality is based on ;

1. Heredity .
 2. Upbringing.
 3. Experience.
 4. Childhood.
- a) None of the listed answers is correct.
b) 1 , 2 and 4 only are correct.
c) 1 , 2 , 3 and 4 are correct.
d) 2 , 3 and 4 only are correct.

Answer ;

c) 1 , 2 , 3 and 4 are correct.

0389. Phobic states or obsessional disorders will ;

- a) may require successful treatment before flying may be permitted.
b) require immediate termination of all flying duties.
c) result in the loss of a Class I medical classification after successful treatment.
d) always result in permanent loss of a flying licence.

Answer ;

a) may require successful treatment before flying may be permitted.

0390. Physical exercise in high temperatures (tropical climates) may ;

- a) increase a pilot's resistance to decompression sickness.
b) cause the pilot to collapse , due to an excess of fluid in the body.
c) cause dysentery.
d) cause painful muscle and abdominal cramps.

Answer ;

d) cause painful muscle and abdominal cramps.

0391. Physical Stress takes place when ;

- a) overload of the neurones in the nervous system.
b) the brain can no longer cope due to either quantitative or qualitative overload.
c) outside conditions put a strain upon the homeostatic mechanisms of the body.
d) the perceived demand is greater than the perceived ability.

Answer ;

c) outside conditions put a strain upon the homeostatic mechanisms of the body.

0392. Physiological problems due to increasing altitude are caused by ;

- a) disorientation.
b) increased atmospheric pressure.
c) accelerations.
d) decreased atmospheric pressure.

Answer ;

d) decreased atmospheric pressure.

0393. Physiological stressors include ;

- a) noise , temperature (low or high) , humidity , sleep deprivation.
b) temperature , hunger , thirst , divorce.
c) heat , humidity , fatigue , administrative problems.
d) noise , hunger , conflicts , a death.

Answer ;

a) noise , temperature (low or high) , humidity , sleep deprivation.

0394. Pilot stress reactions ;

- a) are related to an internationally recognized list of stressors where the top-ten items should be avoided by every means.
b) seem to be the same for most pilots.
c) differ from pilot to pilot , depending on how a person manages the particular stressors.
d) do not change with the environment or different situations but mainly with the characters themselves.

Answer ;

c) differ from pilot to pilot , depending on how a person manages the particular stressors.

0395. Pilots are more easily inclined to take greater risks when ;
- a) they are not constrained by time.
 - b) making decisions independently of others.
 - c) they are part of a group of pilots and they feel that they are being observed and admired (e.g. air shows).
 - d) making a flight over unfamiliar territory.

Answer ;

- c) they are part of a group of pilots and they feel that they are being observed and admired (e.g. air shows).

0396. Pilots should not fly immediately following giving a donation of blood ;
- a) because there is a small increased risk of low blood pressure due to loss of blood volume.
 - b) the haemoglobin level is too low.
 - c) pilots should not give blood because of the risk of infection.
 - d) there is a risk of bleeding from the needle site during flight.

Answer ;

- a) because there is a small increased risk of low blood pressure due to loss of blood volume.

0397. Pilots should not take antihistamines because they may cause ;
- 1. Drowsiness and dizziness.
 - 2. Hypoglycaemia.
 - 3. Hyperventilation.
 - 4. Flatulence.

Which of the following lists all the correct answers ?

- a) only 1.
- b) 2 , 3 and 4.
- c) 3 and 4.
- d) 1 and 3.

Answer ;

- a) only 1.

Antihistamine (Antihistamin) :
Anti-alerjik ilaçlara verilen isimdir.
Drowsiness :
Uyuşukluk , uykulu olma hali.
Dizziness :
Baş dönmesi , sersemleme.

0398. "Pilot's vertigo" ;
- a) is the sensation to keep a rotation after completing a turn.
 - b) announces the beginning of airsickness.
 - c) is the condition of dizziness and/or tumbling sensation caused by contradictory impulses to the central nervous system (CNS).
 - d) is the sensation of climbing caused by a strong linear acceleration.

Answer ;

- c) is the condition of dizziness and/or tumbling sensation caused by contradictory impulses to the central nervous system (CNS).

0399. Planning ;
- a) in the cockpit typically results in plans that are always easy to modify when things are not as anticipated.
 - b) allows crew members to anticipate potential risky situations and decide on possible responses.
 - c) is dangerous in the cockpit , as it interrupts flight crew creativity.
 - d) is unnecessary in the cockpit , as crew members are so highly trained , they will always know what to do in unusual situations.

Answer ;

- b) allows crew members to anticipate potential risky situations and decide on possible responses.

0400. Positive g will cause the blood-pressure in the brain to ;
- a) decrease.
 - b) first increase , then decrease.
 - c) remain constant.
 - d) increase.

Answer ;

- a) decrease.

0401. Positive linear acceleration when flying in IMC may cause a false sensation of ;
- a) pitching up.
 - b) pitching down.
 - c) vertigo.
 - d) apparent sideward movement of objects in the field of vision.

Answer ;

- a) pitching up.

0402. Presbycusis causes initial loss of ;

- a) conductive hearing.
- b) low and high tones equally.
- c) high tones.
- d) low tones.

Presbycusis :
Yaşlılığa bağlı olarak işitme gücünün azalması.

Answer ;

c) high tones.

0403. Presbycusis results in ;

- a) reduction in the perception of all tones equally.
- b) reduction in the perception of low tones first.
- c) will not affect a pilot's hearing if he/she is wearing ear-plugs all the time.
- d) reduction in the perception of high tones first.

Answer ;

d) reduction in the perception of high tones first.

0404. Presbyopia ;

- a) is common over the age of 50.
- b) is caused by long-termed exposure to stimuli over 90 dB.
- c) is partial visual loss due to pressure changes in the eye.
- d) surgical replacement of the lens the usual treatment and is compatible with flying.

Answer ;

a) is common over the age of 50.

0405. Presbyopia causes a ;

- a) decrease of accommodation.
- b) lack of vitamin A.
- c) lack of empty field.
- d) mis-shapened cornea.

Answer ;

a) decrease of accommodation.

0406. Presbyopia is ;

- a) high intraocular pressure.
- b) long sightedness linked with age.
- c) short sightedness.
- d) myopia.

Answer ;

b) long sightedness linked with age.

0407. Presbyopia results in ;

- a) a lack of empty field.
- b) a decrease of accommodation.
- c) a mis-shapened cornea.
- d) a lack of vitamin A.

Answer ;

b) a decrease of accommodation.

0408. Preventing hypoxia at altitude is achieved by ;

- a) hyperventilating to increase the intake of oxygen , trading oxygen for carbon dioxide.
- b) reducing the contents of carbon dioxide by filtering the cabin air.
- c) pressurizing the air inside the aircraft to maintain oxygen saturation in blood of 30%.
- d) pressurizing the air inside the aircraft to achieve oxygen saturation in blood of more than 90%.

Answer ;

d) pressurizing the air inside the aircraft to achieve oxygen saturation in blood of more than 90%.

- 0409.** Professional languages have certain characteristics , for example ;
1. They use a limited vocabulary.
 2. They are rich and adapted to the context , which sometimes lead to ambiguities.
 3. Their grammar is rather complicated and complex.
 4. Context provides meaning , therefore reduces the risk of ambiguities.

The correct statement(s) is (are) ;

- a) 1 and 4 are correct.
- b) 2 and 3 are correct.
- c) only 4 is correct.
- d) 1 and 3 are correct.

Answer ;

- a) 1 and 4 are correct.**
-

- 0410.** Raising the sensory threshold of a sensory organ means ;

- a) a lesser sensitivity.
- b) a greater selectivity.
- c) a greater sensitivity.
- d) a lesser selectivity.

Answer ;

- a) a lesser sensitivity.**
-

- 0411.** Records of radiation are normally kept for flights above ;

- a) 55.000 ft.
- b) 40.000 ft.
- c) 49.000 ft.
- d) 45.000 ft.

Answer ;

- c) 49.000 ft.**
-

- 0412.** Resonance of the body parts can result from ;

- a) acceleration along the longitudinal body axis.
- b) vibrations from 1 to 100 Hz.
- c) angular velocity.
- d) vibrations from 16 Hz to 18 kHz.

Answer ;

- b) vibrations from 1 to 100 Hz.**
-

- 0413.** Rising the perceptual threshold of a sensory organ means ;

- a) a lesser sensitivity.
- b) a greater selectivity.
- c) a greater sensitivity.
- d) a lesser selectivity.

Answer ;

- a) a lesser sensitivity.**
-

- 0414.** Rods (scotopic visual cells) allow for ;

- a) red vision , both during the day and at night.
- b) good , virtually instantaneous night-vision (scotopic vision).
- c) good night-vision after adaptation to darkness (30 min).
- d) precise vision of contours and colours.

Answer ;

- c) good night-vision after adaptation to darkness (30 min).**
-

- 0415.** Safety is often improved by applying the principles of CRM , e.g. ;

- a) expression of doubts or of a different opinion.
- b) abstention from any suggestion which might be untimely.
- c) the avoidance of any conflict in order to preserve the crew's synergy.
- d) unquestioned obedience to all the Captain's decisions

Answer ;

- a) expression of doubts or of a different opinion.**
-

- 0416.** Saturation of oxygen in the blood at sea level is approximately 98%. This saturation decreases with ;
1. decreasing air pressure.
 2. carbon monoxide poisoning.
 3. increasing altitude.
 4. increasing air pressure.
- a) 1 , 2 and 4 are correct , 3 is false.
b) 1 , 3 and 4 are correct , 2 is false.
c) 1 , 2 and 3 are correct , 4 is false.
d) 2 , 3 and 4 are correct , 1 is false.

Answer ;

c) 1 , 2 and 3 are correct , 4 is false.

- 0417.** Scanning at night should be performed by ;
- a) scanning with one eye open.
 - b) concentrated fixation on an object (image must fall on the fovea).
 - c) slight eye movements to the side of the object.
 - d) avoiding food containing Vitamin A.

Answer ;

c) slight eye movements to the side of the object.

- 0418.** Select from the following list the advantages of teamwork ;
1. Workload is lessened.
 2. Conformity to group norm.
 3. Flight safety enhanced.
 4. Risky Shift.
 5. Work stress is reduced.
 6. Improved decision-making.
- a) 1 , 3 , 4 & 5.
b) 1 , 3 , 5 & 6.
c) 2 , 3 , 4 , 5 & 6.
d) 2 , 3 , 5 & 6.

Answer ;

b) 1 , 3 , 5 & 6.

- 0419.** Select from the following list the environmental stressors ;
1. Heat.
 2. Humidity.
 3. Divorce.
 4. Lack of rest.
 5. Loss of job.
 6. Noise.
- a) 1 , 2 , 4 & 6.
b) 1 , 2 , 3 & 6.
c) 1 , 2 & 6.
d) 1 , 2 , 3 , 5 & 6.

Answer ;

c) 1 , 2 & 6.

- 0420.** Sensory input to the “Seat-of-the-Pants” sense is given by ;
- a) subcutaneous pressure receptors and muscle activity sensors.
 - b) acceleration of the stomach (nausea).
 - c) blood rushing into legs.
 - d) pressure of the heart on the diaphragm.

Answer ;

a) subcutaneous pressure receptors and muscle activity sensors.

- 0421.** Signs of stress include ;
- a) faster , deep inhalation , stabbing pain around the heart.
 - b) perspiration , dry mouth , dilated pupils , fast breathing.
 - c) rising of the blood pressure , pupils narrowing , stabbing pain around the heart.
 - d) lowering of the blood pressure.

Answer ;

b) perspiration , dry mouth , dilated pupils , fast breathing.

0422. Situations particularly vulnerable to “reversion to an earlier behaviour pattern” are ;

1. when concentration on a particular task is relaxed.
 2. when situations are characterised by medium workload.
 3. when situations are characterised by stress.
- a) 2 and 3.
b) 1 and 2.
c) 1 and 3.
d) 3.

Answer ;

c) 1 and 3.

0423. Sleeplessness or the disruption of sleeping patterns ;

1. can lead to symptoms of drowsiness , irritability and lack of concentration.
 2. will make an individual more prone to make errors.
- a) 1 is correct , 2 is not correct.
b) 1 and 2 are both correct.
c) 1 is not correct , 2 is correct.
d) 1 and 2 are both not correct.

Answer ;

b) 1 and 2 are both correct.

0424. Smoking 3 cigarettes in 1 hour at sea level will ;

- a) cause a lower degree of hypoxic tolerance.
b) increase your G-tolerance considerably by increasing blood pressure.
c) not affect night vision when flying at sea level.
d) prevent serious hyperventilation due to sedation effect of nicotine.

Answer ;

a) cause a lower degree of hypoxic tolerance.

0425. Smoking cigarettes reduces the capability of the blood to carry oxygen because ;

- a) haemoglobin has a greater affinity for carbon monoxide than it has for oxygen
b) the smoke of one cigarette can cause an obstruction in the respiratory tract
c) carbon monoxide increases the partial pressure of oxygen in the alveoli
d) carbon monoxide in the smoke of cigarettes assists diffusion of oxygen in the alveoli

Answer ;

a) haemoglobin has a greater affinity for carbon monoxide than it has for oxygen

0426. Some hours after a rapid decompression at FL 300 you experience pain in the joints. Which of following answers is correct ?

- a) This phenomenon is treated by breathing 100% nitrogen.
b) You should ask for medical advice (flight surgeon) since this is a symptom of decompression sickness.
c) This phenomenon is treated by physiotherapy.
d) This symptom indicates decompression sickness and will disappear when you take some exercise.

Answer ;

b) You should ask for medical advice (flight surgeon) since this is a symptom of decompression sickness.

0427. Someone who has anaemia has ;

- a) not enough functional haemoglobin.
b) not enough platelets.
c) not enough plasma.
d) not enough white blood cells.

Answer ;

a) not enough functional haemoglobin.

0428. Sound is measured in ;

- a) millicverts.
b) mmHg.
c) milliamps.
d) dB.

Answer ;

d) dB.

0429. Spatial disorientation will be most likely to occur during flight ;

- a) when flying in light rain below the ceiling.
- b) if the brain receives conflicting information and the pilot does not believe the instruments.
- c) when flying in and out of clouds and the pilot maintains good instrument cross check.
- d) when flying in bright sunlight above a cloud layer.

Answer ;

- b) if the brain receives conflicting information and the pilot does not believe the instruments.
-

0430. State the conditions which cause the “black hole effect” and the danger to flight safety.

- a) The “black hole effect” can be caused by flying over featureless terrain on the approach to an airfield which can create the illusion that the aircraft is at a higher altitude than it is , leading to a high approach being flown.
- b) The “black hole effect” can be caused by flying under instruments with poor cockpit lighting and can lead to disorientation.
- c) The “black hole effect” can be caused by flying over water at night on the approach to an airfield which can create the illusion that the aircraft is at higher altitude than it is , leading to a low approach being flown.
- d) The “black hole effect” can be caused by flying over water at night on the approach to an airfield which can create the illusion that the aircraft is at higher altitude than it is , leading to a high approach being flown.

Answer ;

- c) The “black hole effect” can be caused by flying over water at night on the approach to an airfield which can create the illusion that the aircraft is at higher altitude than it is , leading to a low approach being flown.
-

0431. “Stereotypes” are preconceptions or prejudices which can lead us to ;

- a) act in the same manner in all situations and thus assuring stability.
- b) develop better teamwork by standardizing procedures.
- c) mis-judge individuals even if we have contact with them.
- d) communicate non-verbally with a stranger.

Answer ;

- c) mis-judge individuals even if we have contact with them.
-

0432. Stimuli must be of a certain strength for the receptors to pick them up. This is called ;

- a) Sensory strength.
- b) Sensory filter.
- c) Sensory threshold.
- d) Sensory volume.

Answer ;

- c) Sensory threshold.
-

0433. Stress can be defined as ;

- a) a low level of pressure acting on the individual’s mental system which inhibits completion of even the simplest tasks.
- b) a feeling of well-being enabling difficult tasks to be completed more easily.
- c) a physical limit acting upon the individual’s physical systems inhibiting completion of any task.
- d) the unresolved pressure , strain or force acting upon an individual’s physical or mental systems and the human response.

Answer ;

- d) the unresolved pressure , strain or force acting upon an individual’s physical or mental systems and the human response.
-

0434. Stress is ;

- a) a psychosomatic disease that one can learn to control.
- b) a response by man to his problems , which automatically leads to a reduction in his performance.
- c) a phenomenon which is specific to modern man.
- d) a mechanism by which an individual can respond to situations which he/she may have to face.

Answer ;

- d) a mechanism by which an individual can respond to situations which he/she may have to face.
-

0435. Stress is a frequent aspect of the pilot's job. Under which of the following circumstances does it occur ?
1. Stress occurs whenever the pilot must revise his plan of action and does not immediately have a solution.
 2. Stress occurs with inexperienced pilots when the situational demands exceed their individual capabilities.
 3. Stress occurs if a pilot is convinced that he will not be able to find a solution for the problem he/she faces.
- a) 1 and 2 are correct , 3 is false.
b) 1 is correct , 2 and 3 are false.
c) 1 , 2 and 3 are correct.
d) Only 1 is false.

Answer ;

c) 1 , 2 and 3 are correct.

0436. Stress management programmes usually involve ;
- a) the prevention and/or the removal of stress.
 - b) the use of psychoactive drugs.
 - c) only the removal of stress.
 - d) only the prevention of stress.

Answer ;

a) the prevention and/or the removal of stress.

0437. Stress may be defined as ;
- a) a human reaction which one must manage to eliminate.
 - b) a normal phenomenon which enables an individual to adapt to encountered situations.
 - c) a poorly controlled emotion which leads to a reduction in capabilities.
 - d) a psychological phenomenon which only affects fragile personalities.

Answer ;

b) a normal phenomenon which enables an individual to adapt to encountered situations.

0438. Stress may cause Regression which can lead to ;
- a) experimentation and trying out of new ideas.
 - b) a more efficient use of mental capacity.
 - c) an improvement in performance.
 - d) correct actions being forgotten and substituted for procedures learnt in the past.

Answer ;

d) correct actions being forgotten and substituted for procedures learnt in the past.

0439. Stress occurs ;
1. only in a situation of imminent danger.
 2. only when faced with real , existing and actual situation.
 3. sometimes via imagination , the anticipation of a situation or its outcome.
 4. because of the similarity with a formerly experienced stressful situation.

The correct statement(s) is(are) ;

- a) 2 , 3.
- b) 1 , 2.
- c) 1 , 2 , 4.
- d) 3 , 4.

Answer ;

d) 3 , 4.

0440. Stress occurs ;
1. only in a situation of imminent danger.
 2. only when faced with real , existing and palpable phenomenon.
 3. sometimes via imagination , the anticipation of a situation or its outcome.
 4. because of the similarity with a formerly experienced stressful situation.

The correct statement(s) is(are) ;

- a) 2 , 3.
- b) 1 , 2 , 4.
- c) 3 , 4.
- d) 1 , 2.

Answer ;

c) 3 , 4.

0441. Stress will effect ;

1. attention.
 2. concentration.
 3. memory.
 4. judgment.
- a) 1 and 3 are correct.
b) 1 , 2 , 3 and 4 are correct.
c) 1 and 2 are correct.
d) 2 , 3 and 4 are correct.

Answer ;

b) 1 , 2 , 3 and 4 are correct.

0442. Subcutaneous pressure receptors are stimulated by ;

- a) the pressure created on the corresponding body parts when sitting , standing or lying down.
b) the condition of the body itself.
c) a touch on the skin indicating the true vertical.
d) environmental stressors.

Answer ;

a) the pressure created on the corresponding body parts when sitting , standing or lying down.

0443. Symptoms caused by gas bubbles in the lungs , following a decompression are called ;

- a) leans.
b) creeps.
c) bends.
d) chokes.

Answer ;

d) chokes.

0444. Symptoms of decompression sickness ;

- a) are flatulence and pain in the middle ear.
b) are bends , chokes , creeps and neurological symptoms.
c) are only relevant when diving.
d) can only develop at altitudes of more than 40.000 ft.

Answer ;

b) are bends , chokes , creeps and neurological symptoms.

0445. Tattooing and immunization with the use of improperly sterilized needles or syringes may expose the receiver to which of the following ?

- a) Hepatitis B or C
b) Diabetes
c) Hypertension
d) Osteoarthritis

Answer ;

a) Hepatitis B or C

0446. Tetanus is transmitted through ;

- a) insect bites.
b) food or water which has been contaminated.
c) bacteria in the form of spores via a puncture in the skin.
d) droplets in the air cause by the breath of an infected person.

Answer ;

c) bacteria in the form of spores via a puncture in the skin.

0447. The 5 human senses tend to adapt.

- a) This statement is true.
b) This statement is untrue. Only sight adapts (i.e. night flying).
c) This statement is untrue. Touch is the exception.
d) This statement is untrue. None adapt.

Answer ;

a) This statement is true.

0448. The ability of the human eye to read alphanumeric information is ;
- a) governed by peripheral vision over an area of approximately 20 degrees of angle.
 - b) limited to the foveal area of the retina.
 - c) limited to daytime using the rod cells.
 - d) almost equally shared by the entire retina.

Answer ;

b) limited to the foveal area of the retina.

0449. The ability of the lens to change its shape is called ;

- a) accommodation.
- b) adaptation.
- c) depth perception.
- d) binocular vision.

Answer ;

a) accommodation.

0450. The ability to monitor information which could indicate the development of a critical situation ;

- a) makes no sense because the human information processing system is limited anyway.
- b) is necessary to maintain good situational awareness.
- c) is dangerous , because it distracts attention from flying the aircraft.
- d) is responsible for the development of inadequate mental models of the real world.

Answer ;

b) is necessary to maintain good situational awareness.

0451. The ability to withstand g forces is reduced by a number of factors , among these factors are ;

- a) low blood sugar , obesity and hypoxia.
- b) smoking , obesity and vibrations.
- c) physical fitness , heat and hypotension.
- d) hypoxia , fatigue and vibrations.

Answer ;

a) low blood sugar , obesity and hypoxia.

0452. The acquisition of a motor programme (skill) will mean that ;

- a) the more behaviour is automated , the more it requires attention and the less it frees resources.
- b) the less behaviour is automated , the less it requires attention and the more it frees resources.
- c) the more behaviour is automated , the more it requires attention and the more it frees resources.
- d) the more behaviour is automated , the less it requires conscious attention and thus the more it frees mental resources.

Answer ;

d) the more behaviour is automated , the less it requires conscious attention and thus the more it frees mental resources.

0453. The acquisition of a skill comprises three stages (Anderson model) ;

- a) cognitive , associative and knowledge.
- b) associative , autonomous and expert.
- c) automatic , cognitive and knowledge.
- d) cognitive , associative and automatic.

Answer ;

d) cognitive , associative and automatic.

0454. The amount of light which strikes the retina is controlled by ;

- a) the cornea.
- b) the ciliary body.
- c) the lens.
- d) the pupil.

Answer ;

d) the pupil.

0455. The approximate time required for complete adaptation of the eye when moving from darkness to light is ;

- a) 10 minutes.
- b) 10 seconds.
- c) 7 minutes.
- d) 30 minutes.

Answer ;

b) 10 seconds.

0456. The assessment of risk in a particular situation will be based on ;
- a) subjective perception and evaluation of situational factors.
 - b) the emergency checklist only.
 - c) external factors only.
 - d) situational factors only.

Answer ;

- a) subjective perception and evaluation of situational factors.
-

0457. The atmosphere contains the following gases ;
- a) 78% nitrogen , 21% oxygen , 0.03% carbon dioxide , rest ; rare gases.
 - b) 78% nitrogen , 21% oxygen , 1% carbon monoxide , rest ; rare gases.
 - c) 78% helium , 21% oxygen , 1% carbon monoxide , rest ; rare gases.
 - d) 78% helium , 21% oxygen , 0.03% carbon dioxide , rest ; rare gases.

Answer ;

- a) 78% nitrogen , 21% oxygen , 0.03% carbon dioxide , rest ; rare gases.
-

0458. The atmospheric pressure at 18.000 feet altitude is half the atmospheric pressure at sea level. In accordance with this statement ;

- a) the partial oxygen pressure at that altitude will be doubled.
- b) the partial oxygen pressure at that altitude will also drop to 1/2 of the pressure of oxygen at sea level.
- c) the oxygen saturation of the blood at that altitude will drop by 50 % too.
- d) the oxygen percentage of the air at that altitude will drop by one half also.

Answer ;

- b) the partial oxygen pressure at that altitude will also drop to 1/2 of the pressure of oxygen at sea level.
-

0459. The autocratic leadership style is synonymous with ;

- a) an authoritarian style.
- b) a “laissez-faire” style.
- c) an anarchic style.
- d) a democratic style.

Answer ;

- a) an authoritarian style.
-

0460. The Autonomic Nervous System (ANS) regulates ;

- a) reflexes , body temperature and reasoning.
- b) reflexes only.
- c) sweating , arterial pressure , body temperature and the General Adaption Syndrome.
- d) motor programmes and reflexes

Answer ;

- c) sweating , arterial pressure , body temperature and the General Adaption Syndrome.
-

0461. The available cognitive resources of the human brain ;

- a) are limited and make it impossible to perform two attentional tasks at the same time.
- b) are limited but make it possible to easily perform several tasks at the same time.
- c) are virtually unlimited.
- d) allow for twin-tasks operation without any loss of effectiveness.

Answer ;

- a) are limited and make it impossible to perform two attentional tasks at the same time.
-

0462. The average pulse of a healthy adult at rest is about ;

- a) 60 to 80 beats/min.
- b) 30 to 50 beats/min.
- c) 90 to 100 beats/min.
- d) 110 to 150 beats/min.

Answer ;

- a) 60 to 80 beats/min.
-

0463. The barometric pressure has dropped to 1/2 of the pressure at sea level at ;

- a) 10 000 feet.
- b) 25 000 feet.
- c) 18 000 feet.
- d) 30 000 feet.

Answer ;

- c) 18 000 feet.
-

0464. The behaviour of a leader depends on ;
- a) his/her ability to prevent discussions among the crew members in order to avoid conflict.
 - b) only his/her ability to dominate and be assertive.
 - c) the authority gradient only.
 - d) the situation , the goals and composition of the group.

Answer ;

- d) the situation , the goals and composition of the group.
-

0465. The behavioural effects of stress may include ;
- 1. manifestation of aggressiveness.
 - 2. a willingness to improve communication.
 - 3. a willingness for group cohesion.
 - 4. a tendency to withdrawal.

The combination of correct statements is ;

- a) 1 and 4 are correct.
- b) 3 and 4 are correct.
- c) 2 , 3 and 4 are correct.
- d) 1 , 2 and 3 are correct.

Answer ;

- a) 1 and 4 are correct.
-

0466. “The Bends” as a symptom of decompression sickness consists of ;

- a) CNS-disturbances.
- b) loss of peripheral vision.
- c) pain in the joints.
- d) pain in the thorax and a cough.

Answer ;

- c) pain in the joints.
-

0467. The best method to avoid Confirmation Bias is to ;

- a) deliberately confirm that the receiver has understood the information that has been passed to him/her.
- b) double-check that you have not fitted the ground over which you are passing to your map.
- c) search for information that will falsify the hypothesis.
- d) precise and careful pre-flight briefings.

Answer ;

- c) search for information that will falsify the hypothesis.
-

0468. The biological reaction to stress is identical regardless of the cause of stress. This mechanism occurs in three phases and is referred to , by Selye , as the “General Adaptation Syndrome”. The sequence is ;

- a) alarm phase - resistance phase - exhaustion phase.
- b) exhaustion phase - resistance phase - adaptation phase.
- c) alarm phase - denial phase - exhaustion phase.
- d) resistance phase - exhaustion phase - recovery phase.

Answer ;

- a) alarm phase - resistance phase - exhaustion phase.
-

0469. The “Black hole” phenomenon occurs during approaches at night and over water , jungle or desert. When the pilot is lacking visual cues other than those of the aerodrome there is an illusion of ;

- a) being too close , landing long.
- b) being too high and too far away , dropping low and landing short.
- c) being too low , flying a steeper approach than normal.
- d) climbing.

Answer ;

- b) being too high and too far away , dropping low and landing short.
-

0470. The blood in the pulmonary artery is ;

- a) lacking in oxygen and rich in carbon dioxide.
- b) rich in oxygen and lacking in carbon dioxide.
- c) rich in both oxygen and carbon dioxide.
- d) lacking in both oxygen and carbon dioxide.

Answer ;

- a) lacking in oxygen and rich in carbon dioxide.
-

0471. The blood-pressure depends on ;
1. the work of the heart.
 2. the peripheral resistance.
 3. the elasticity of the arterial walls.
 4. the blood volume and viscosity.
- a) 1 , 2 , 3 and 4 are correct.
b) 1 , 2 and 3 are correct , 4 is false.
c) 1 , 3 and 4 are correct , 2 is false.
d) 2 , 3 and 4 are correct , 1 is false.

Answer ;

- a) 1 , 2 , 3 and 4 are correct.
-

0472. The body loses water via ;
1. the skin and the lungs.
 2. the kidneys.
- Which of the following lists all the correct answers ?
- a) 1 is not correct and 2 is correct.
b) both are false.
c) 1 is correct and 2 is not correct.
d) 1 and 2 are correct.

Answer ;

- d) 1 and 2 are correct.
-

0473. The body obtains its energy from ;
- a) minerals and vitamins.
b) carbohydrates , proteins and fats.
c) protein and vitamins.
d) minerals , carbohydrates and vitamins.

Answer ;

- b) carbohydrates , proteins and fats.
-

0474. The “Break Point” is that point after which , if stress continues to rise ;
- a) performance is degraded.
b) performance is enhanced.
c) performance is initially degraded and is then enhanced.
d) performance peaks and remains at this level.

Answer ;

- a) performance is degraded.
-

0475. The cabin pressure in airline operation is ;
- a) normally between 2 000 to 3 000 feet.
b) normally between 6 000 to 8 000 feet.
c) normally between 4 000 to 5 000 feet.
d) always equivalent to sea level.

Answer ;

- b) normally between 6 000 to 8 000 feet.
-

0476. The carcinogen (a substance with the ability to produce modifications in cells which develop a cancer) in cigarettes is ;
- a) tar.
b) nicotine.
c) lead.
d) carbon monoxide.

Answer ;

- a) tar.
-

0477. The causes of human error can be abundant and complex. Which of the following factors could contribute to human error ?
1. Personality.
 2. Motivation and attitudes.
 3. Emotional state.
 4. External environmental factors.

Which of the following lists all the correct answers ?

- a) 1 , 2 , 3 and 4.
- b) 2 and 3.
- c) 2 , 3 and 4.
- d) 1 , 2 and 3.

Answer ;

- a) 1 , 2 , 3 and 4.
-

0478. The changes in atmospheric gas pressure with altitude are ;

- a) rises with altitude.
- b) decreases in a linear manner with altitude.
- c) non-linear , with a higher rate of change at lower levels.
- d) decreases more slowly at lower altitudes compared with at higher levels.

Answer ;

- c) non-linear , with a higher rate of change at lower levels.
-

0479. The chemical substance responsible for addiction to tobacco is ;

- a) the combination of nicotine , tar and carbon monoxide.
- b) tar.
- c) carbon monoxide.
- d) nicotine.

Answer ;

- d) nicotine.
-

0480. The choice of the moment you select flaps depending on situation and conditions of the landing is ;

- a) skill and/or rule based behaviour.
- b) knowledge based behaviour.
- c) pressure based behaviour.
- d) automated behaviour.

Answer ;

- a) skill and/or rule based behaviour.
-

0481. The choice of words and their packaging become more important in the cockpit of an aircraft because ;

- a) loss of body language.
- b) loss of hypertext.
- c) loss of perception.
- d) loss of situational awareness.

Answer ;

- a) loss of body language.
-

0482. The circulation of blood ;

1. transports oxygen to the body cells.
2. withdraws waste products from the cells.
3. conveys nutrients to the cells.

Which of the following lists all the correct answers ?

- a) 2 and 3.
- b) 1 , 2 and 3.
- c) 1 and 2.
- d) 1 and 3.

Answer ;

- b) 1 , 2 and 3.
-

0483. The circulatory system , amongst other things , allows for ;
1. transportation of oxygen and carbon dioxide.
 2. transportation of information by chemical substances.

Which of the following lists all the correct statements ?

- a) 1 and 2 are correct.
- b) 1 is false and 2 is correct.
- c) both are false.
- d) 1 is correct and 2 is false.

Answer ;

- a) 1 and 2 are correct.
-

0484. The “cocktail party effect” is ;
- a) the tendency to believe information that reinforces our mental model of the world.
 - b) the tendency not to perceive relevant information.
 - c) the ability to drink too much at social gathering.
 - d) the ability to pick up relevant information unintentionally.

Answer ;

- d) the ability to pick up relevant information unintentionally.
-

0485. The “Cocktail Party Effect” is associated with ;
- a) cohesion.
 - b) adaption.
 - c) attention mechanisms.
 - d) communications.

Answer ;

- c) attention mechanisms.
-

0486. The cognitive effects of stress may include ;
1. excessive haste.
 2. an improvement in memory.
 3. a complete block , action is impossible.
 4. a risk of focusing on a particular aspect.
 5. ease of decision-making.
 6. an increase in the rate of mistakes.

The combination which brings together all correct statements is ;

- a) 1 , 3 , 4 , 6.
- b) 2 , 3 , 5 , 6.
- c) 1 , 2 , 5.
- d) 3 , 4 , 5.

Answer ;

- a) 1 , 3 , 4 , 6.
-

0487. The confirmation bias of decision making is ;
- a) a tendency not to seek for information which confirms a judgement.
 - b) a tendency to ignore that information which indicates that a hypothesis or decision is poor.
 - c) a tendency to agree with the decision made by the group.
 - d) a tendency not to look for information which would reassure oneself about a decision.

Answer ;

- b) a tendency to ignore that information which indicates that a hypothesis or decision is poor.
-

0488. The consumption of medicines or other substances may have consequences on qualification to fly for the following reasons ;
1. The disease requiring a treatment may be cause for disqualification.
 2. Flight conditions may modify the reactions of the body to a treatment.
 3. Drugs may cause adverse side effects impairing flight safety.
 4. The effects of medicine do not necessarily immediately disappear when the treatment is stopped.
- a) Only 2 is false.
 - b) 1 , 2 , 3 and 4 are correct.
 - c) 3 and 4 are false , 1 and 2 are correct.
 - d) 1 , 2 and 3 are correct , 4 is false.

Answer ;

- b) 1 , 2 , 3 and 4 are correct.
-

0489. The Coriolis effect in spatial disorientation occurs as a result of ;
- a) stimulation of the saccule and the utricle (otoliths) of the inner ear.
 - b) simultaneous stimulation of several semicircular canals.
 - c) absence of semicircular canal stimulation.
 - d) on stimulating the cochlea intensely.

Answer ;

b) simultaneous stimulation of several semicircular canals.

0490. The cornea and the crystalline lens of the eye ;
- a) cause the convergence of light rays onto the retina.
 - b) allow for the regulation of the amount of light admitted into the eye.
 - c) permit the reception and conversion of visual stimuli to images interpreted by the brain.
 - d) keep the retina clean and healthy.

Answer ;

a) cause the convergence of light rays onto the retina.

0491. The damaging effects of noise on the pilot depends on ;
- 1. The intensity of the noise.
 - 2. The frequency of the noise.
 - 3. The exposure time to noise.
 - 4. Proper use of the headset in the cockpit.

Which of the following lists all the correct statements ?

- a) 2 , 3 and 4.
- b) 1 and 4.
- c) 1 , 2 , 3 and 4.
- d) 3 only.

Answer ;

c) 1 , 2 , 3 and 4.

0492. The decision making in emergency situations requires firstly ;
- a) informing ATC thoroughly about the situation.
 - b) the whole crew to focus on the problem.
 - c) speed of reaction.
 - d) delegation of tasks and crew coordination.

Answer ;

d) delegation of tasks and crew coordination.

0493. The distribution of primary causes of accidents in the worldwide jet aircraft commercial fleet shows that human error is involved in ;

- a) about 90% of cases.
- b) about 55% of cases.
- c) all cases , one way or another.
- d) about 70% of cases.

Answer ;

d) about 70% of cases.

0494. The dry atmosphere of the flight deck may cause dehydration , which may lead to a reduction in the ability to pay attention. To prevent this , it is appropriate to ;

- a) drink sufficient non-carbonated liquids.
- b) drink tea.
- c) drink plenty of coffee.
- d) drink cool cola drinks.

Answer ;

a) drink sufficient non-carbonated liquids.

0495. The duration of a period of sleep is governed primarily by ;
- a) the amount of time you have been awake.
 - b) the point within your circadian rhythm at which you try to sleep.
 - c) the duration of your previous sleep.
 - d) the number of points you have in your “credit/deficit” system.

Answer ;

b) the point within your circadian rhythm at which you try to sleep.

0496. The effect of experience and repetition on performance ;
- a) is always negative.
 - b) is always beneficial.
 - c) is never negative.
 - d) can both be beneficial and negative.

Answer ;

d) can both be beneficial and negative.

0497. The “Effective Performance Time” or “Time of Useful Consciousness” after a decompression at 35.000 ft is ;
- a) less than 20 seconds.
 - b) approximately 3 minutes.
 - c) between 30 and 60 seconds.
 - d) approximately 5 minutes.

Answer ;

c) between 30 and 60 seconds.

0498. The effects of Galactic radiation ;
- a) is unaffected by altitude.
 - b) decreases with altitude.
 - c) increases with altitude.
 - d) remains steady up to 49.000 ft and thereafter increases.

Answer ;

c) increases with altitude.

0499. The effects of sleep deprivation on performance ;
- 1. increase with altitude.
 - 2. decrease with altitude.
 - 3. increase with higher workload.
 - 4. decrease with higher workload.
- Which of the following lists all the correct statements ?
- a) 1 , 2 and 3 are correct.
 - b) 2 , 3 and 4 are correct.
 - c) 1 , 3 and 4 are correct.
 - d) 1 and 3 are correct.

Answer ;

d) 1 and 3 are correct.

0500. The errors of a motor programme are ;
- a) Action Slip and Confirmation Bias.
 - b) Confirmation error and Error of Commission.
 - c) Action Slip and Environmental Capture (Habituation).
 - d) Habituation and Error of Commission.

Answer ;

c) Action Slip and Environmental Capture (Habituation).

0501. The errors resulting from an illogical indexing system in an operations manual are related to an interface mismatch between ;
- a) Liveware - Hardware.
 - b) Liveware - Software.
 - c) Liveware - Liveware.
 - d) Liveware - Environment.

Answer ;

b) Liveware - Software.

0502. The Eustachian tube connects the ;
- a) middle ear and the inner ear.
 - b) semi circular canals.
 - c) middle ear and the throat.
 - d) auditory duct and the inner ear.

Answer ;

c) middle ear and the throat.

0503. The Eustachian tube is the passage way between the ;
- a) throat and the middle ear.
 - b) nose , throat and inner ear.
 - c) sinuses and the throat.
 - d) nose , throat and the external auditory canal.

Answer ;

- a) throat and the middle ear.
-

0504. The exchange of gases between the alveoli and the blood is due to ;
- a) inspiration.
 - b) changes in atmospheric pressure.
 - c) physical exercise.
 - d) diffusion.

Answer ;

- d) diffusion.
-

0505. The expression “when perception matches reality” describes ;
- a) Situational Awareness.
 - b) Vigilance.
 - c) Attention.
 - d) Alertness.

Answer ;

- a) Situational Awareness.
-

0506. The eye can fully adjust to ;
- a) high levels of illumination in 10 seconds and darkness in 30 minutes.
 - b) high levels of illumination in 30 minutes and darkness in 10 seconds.
 - c) high levels of illumination in 10 minutes and darkness in 30 minutes.
 - d) high levels of illumination in 30 minutes and darkness in 10 minutes.

Answer ;

- a) high levels of illumination in 10 seconds and darkness in 30 minutes.
-

0507. The first cockpit tool that tends to suffer as a result of stress ;
- a) Cockpit entity.
 - b) Cockpit synergy.
 - c) Cockpit cohesion.
 - d) Cockpit communications.

Answer ;

- d) Cockpit communications.
-

0508. The first stage in the information process is ;
- a) selective attention.
 - b) the recognition of information.
 - c) sensory stimulation.
 - d) perception.

Answer ;

- c) sensory stimulation.
-

0509. The following actions are appropriate when faced with symptoms of decompression sickness ;
1. climb to higher level.
 2. descent to the higher of 10000 ft or MSA and land as soon as possible.
 3. breathe 100 % oxygen.
 4. get medical advice about recompression after landing.
- a) 1 and 3 are correct.
 - b) 1 , 2 and 3 are correct.
 - c) 1 and 4 are correct.
 - d) 2 , 3 and 4 are correct.

Answer ;

- d) 2 , 3 and 4 are correct.
-

0510. The following can be observed when the internal body temperature falls below 35°C ;
- a) shivering , will tend to cease , and be followed by the onset of apathy.
 - b) mental disorders , and even coma.
 - c) the appearance of intense shivering.
 - d) profuse sweating.

Answer ;

- a) shivering , will tend to cease , and be followed by the onset of apathy.
-

0511. The following course of action should be taken if gastrointestinal complaints in flight crew occur before to take-off ;
- 1. take the standard medicines and advise the doctor on returning from the flight.
 - 2. assess your own ability to fly , if necessary with the help of a doctor.
 - 3. if in doubt about fitness to fly - do not fly !
 - 4. reduce the cabin temperature , and drink before you are thirsty so as to avoid dehydration.
- a) 1 , 4.
 - b) 1 , 2 , 4.
 - c) 1 , 3.
 - d) 2 , 3.

Answer ;

- d) 2 , 3.
-

0512. The following occurs in man if the internal body temperature increases to 39°C ;
- a) nothing significant happens at this temperature. The first clinical signs only start to appear at 39°C.
 - b) impairment of physical and mental performance.
 - c) apathy.
 - d) considerable dehydration.

Answer ;

- b) impairment of physical and mental performance.
-

0513. The following three conditions will decrease your G-tolerance ;
- a) loss of body fluids , full stomach , 20 degrees backward tilted sitting position.
 - b) cold , 30 degrees backward tilted sitting position , short body (short stature).
 - c) old age , short body (short stature) , an empty stomach.
 - d) obesity , alcohol , sleep deprivation.

Answer ;

- d) obesity , alcohol , sleep deprivation.
-

0514. The fovea ;
- a) is the area responsible for night vision.
 - b) is an area in which cones predominate.
 - c) is an area in which rods predominate.
 - d) is sensitive to very low intensities of light.

Answer ;

- b) is an area in which cones predominate.
-

0515. The fovea is ;
- a) where the optic nerves come together with the pupil.
 - b) the area of best day vision and no night vision at all.
 - c) the area of best day vision and best night vision.
 - d) the area of the blind spot (optic disc).

Answer ;

- b) the area of best day vision and no night vision at all.
-

0516. The function of the Eustachian tube is to equalise the pressure between the ;
- a) middle ear and the external atmosphere.
 - b) sinuses of the nose and the external atmosphere.
 - c) nose , throat and the external atmosphere.
 - d) sinuses.

Answer ;

- a) middle ear and the external atmosphere.
-

0517. The General Adaption Syndrome is associated with the ;

- a) NIHL (Noise Induced Hearing Loss).
- b) CNS (Central Nervous System).
- c) ANS (Autonomic Nervous System).
- d) PNS (Peripheral Nervous System).

Answer ;

c) ANS (Autonomic Nervous System).

0518. The group of tiny bones (the hammer , anvil and stirrup) are situated in ;

- a) the inner ear.
- b) the middle ear.
- c) the maxillary sinus.
- d) the outer ear.

Answer ;

b) the middle ear.

0519. The heart muscle is supplied with blood by ;

- a) the pulmonary veins.
- b) the auricles.
- c) the coronary arteries.
- d) ventricles.

Answer ;

c) the coronary arteries.

0520. The human circadian rhythm is based on a cycle of about ;

- a) 12 hours.
- b) 48 hours.
- c) 1.5 hours.
- d) 25 hours.

Answer ;

d) 25 hours.

0521. The human ear is capable of perceiving vibrations between the frequencies ;

- a) 20.000 - 40.000 Hz.
- b) 0 - 16 Hz.
- c) 30 - 15.000 dB.
- d) 20 - 20.000 Hz.

Answer ;

d) 20 - 20.000 Hz.

0522. The human vestibular system performs spatial orientation by ;

- a) comparing the body's position to gravity over the tympanic membrane.
- b) comparing linear and radial speeds of the body to the attitude of the aircraft.
- c) measuring fluid movements in the tympanic membrane of the middle ear.
- d) measuring linear and radial acceleration within the inner ear.

Answer ;

d) measuring linear and radial acceleration within the inner ear.

0523. The "ideal" cockpit can be termed as ;

- a) Cohesive.
- b) Co-operative.
- c) Synergistic.
- d) Communicative.

Answer ;

c) Synergistic.

0524. The impression of an apparent movement of light when stared at for a relatively long period of time in the dark is called ;

- a) "autokinesis".
- b) "white out".
- c) "oculogyral illusion".
- d) "oculografic illusion".

Answer ;

a) "autokinesis".

0525. The individual's perception of stress depends on ;
- a) the pilot's increasing level of arousal.
 - b) the subjective evaluation of the situation and one's abilities to cope with it.
 - c) the objective evaluation of the situation and one's abilities to cope with it.
 - d) the conditions of the current situation only.

Answer ;

b) the subjective evaluation of the situation and one's abilities to cope with it.

0526. The individual's personality ;
- a) is variable and easily changeable.
 - b) changes with a different emotional mood.
 - c) refers to unique psychological characteristics.
 - d) is the tendency to evaluate things in a positive or negative way.

Answer ;

c) refers to unique psychological characteristics.

0527. The inner ear is able to perceive ;
- 1. angular acceleration.
 - 2. linear acceleration.
 - 3. noise.
- a) 2 and 3 are correct , 1 is false.
 - b) 2 is correct , 1 and 3 are both false.
 - c) 1 and 2 and 3 are correct.
 - d) 1 and 2 are correct , 3 is false.

Answer ;

c) 1 and 2 and 3 are correct.

0528. The intensity of a sound is measured in ;
- a) curies.
 - b) decibels.
 - c) hertz.
 - d) cycles per second.

Answer ;

b) decibels.

0529. The "Leans" or Somatogyral illusion can be caused by ;
- a) going into a turn too quickly.
 - b) bunting the aircraft.
 - c) reducing bank following a prolonged turn.
 - d) a carrier take-off.

Answer ;

c) reducing bank following a prolonged turn.

0530. The level at which a pilot will experience a situation as stressful ;
- a) depends on the level of demand but not on individual interpretation of the situational demands.
 - b) depends on the individual's perception of available abilities in comparison to the perceived demands.
 - c) depends on self-confidence alone.
 - d) does not depend on his capacity to absorb information.

Answer ;

b) depends on the individual's perception of available abilities in comparison to the perceived demands.

0531. The level of automation helps to conserve resources. On the other hand , it may result in ;
- a) decision-making errors.
 - b) routine errors (slips).
 - c) mistakes.
 - d) errors in selecting an appropriate plan of action.

Answer ;

b) routine errors (slips).

0532. The long term memory is influenced by ;
- a) the time of the day , gender , intelligence and blood pressure.
 - b) short term memory only.
 - c) experience , repetition , suggestion , desires and expectations.
 - d) short term memory and intelligence only.

Answer ;

c) experience , repetition , suggestion , desires and expectations.

-
0533. The low level of humidity found in a pressurised aircraft cabin ;
- a) should be compensated for by a regular supply of coffee or tea.
 - b) should be compensated for by breathing in to a paper bag.
 - c) requires a regular supply of water only for those who are in poor health. Healthy people compensate for this naturally.
 - d) should be compensated for by drinking water before one gets thirsty.

Answer ;

- d)** should be compensated for by drinking water before one gets thirsty.
-

0534. The lowest level of Maslow's Model of the hierarchy of Human Needs is ;

- a) safety and security needs.
- b) physiological needs.
- c) belonging and affection needs.
- d) self-esteem needs.

Answer ;

- b)** physiological needs.
-

0535. The main function of the red blood cells is ;

- a) to transport oxygen.
- b) to contribute to the immune response of the organism.
- c) the cellular defence of the organism.
- d) to participate in the process of coagulation of the blood.

Answer ;

- a)** to transport oxygen.
-

0536. The main limitation of the long-term memory is ;

- a) the data storage time.
- b) unless information is accessed from time to time , retrieval of information can be difficult.
- c) if enough information is entered at the same time , saturation may take place.
- d) the quantity of data which may be stored.

Answer ;

- b)** unless information is accessed from time to time , retrieval of information can be difficult.
-

0537. The main purpose of lumbar support is to ;

- a) allow the most comfortable position for the upper spine.
- b) allow the most comfortable position for the spine and shoulder bones.
- c) produce an even pressure of the discs by allowing the lower spine to curve naturally.
- d) allow the most comfortable position for the spine and higher neck bones.

Answer ;

- c)** produce an even pressure of the discs by allowing the lower spine to curve naturally.
-

0538. The main reason why it is inadvisable to fly when suffering from a cold is that ;

- a) swollen tissue in the inner ear will prevent the air from ventilating through the tympanic membrane.
- b) pain and damage to the eardrum can result , particularly during fast descents.
- c) swollen tissue in the Eustachian tube will cause permanent hearing loss.
- d) gentle descents at high altitude can result in damage to the ear drum.

Answer ;

- b)** pain and damage to the eardrum can result , particularly during fast descents.
-

0539. The maintenance of man's internal equilibrium is called ;

- a) Homeostasis.
- b) Homeothermy.
- c) Heterostasis.
- d) Poikilothermy.

Answer ;

- a)** Homeostasis.
-

0540. The maximum number of unrelated items that can be stored in working memory is ;

- a) about 7 items.
- b) about 30 items.
- c) very limited - only 3 items.
- d) unlimited.

Answer ;

- a)** about 7 items.
-

0541. The mechanism of accommodation is caused by ;
- a) the diameter of the pupil.
 - b) the functioning of the muscles of the eye.
 - c) the elasticity of the optic nerves.
 - d) the functioning of the ciliary muscle around the lens.

Answer ;
d) the functioning of the ciliary muscle around the lens.

0542. The metabolism of alcohol ;
- a) is accelerated by consuming alcohol with a meal.
 - b) can be influenced by easily obtained medication.
 - c) is a question of time.
 - d) can be accelerated even more by coffee.

Answer ;
c) is a question of time.

0543. The most dangerous characteristic of perception is , that it ;
- a) will mainly occur under conditions of relaxation.
 - b) will only occur under conditions of stress.
 - c) can easily be changed.
 - d) is frequently extremely resistant to correction.

Answer ;
d) is frequently extremely resistant to correction.

0544. The most dangerous symptoms of hypoxia at altitude are ;
- a) euphoria and impairment of judgement.
 - b) sensation of heat and blurred vision.
 - c) breathlessness and reduced night vision.
 - d) hyperventilation.

Answer ;
a) euphoria and impairment of judgement.

0545. The most dangerous type of incapacitation in flight is one that ;
- a) is not severe in its effect.
 - b) is temporary.
 - c) develops slowly and gradually (insidious).
 - d) occurs very quickly.

Answer ;
c) develops slowly and gradually (insidious).

0546. The most significant item of technical equipment introduced in the 1980s & 1990s which has contributed to a reduction in accidents is ;

- a) EFIS.
- b) AFDS.
- c) ATIS.
- d) GPWS.

Answer ;
d) GPWS.

0547. The noise induced damage to a pilot depends on ;
- 1. The intensity of the noise.
 - 2. The frequency of the noise.
 - 3. The duration of the noise.
 - 4. Proper usage of the headset.

- a) 3.
- b) 1 , 2 , 3 , 4.
- c) 1 , 4.
- d) 2 , 3 , 4.

Answer ;
b) 1 , 2 , 3 , 4.

0548. The normal arterial blood-pressure of a healthy adult is (systolic/diastolic) ;

- a) 180/120 mmHg.
- b) 120/80 mmHg.
- c) 80/20 mmHg.
- d) 220/180 mmHg.

Answer ;

b) 120/80 mmHg.

0549. The normal rate of breathing of an adult at rest is about ;

- a) 72 cycles per minute.
- b) 32 cycles per minute.
- c) 4 cycles per minute.
- d) 16 cycles per minute.

Answer ;

d) 16 cycles per minute.

0550. The normal rate of breathing when at rest is ;

- a) 60 to 100 cycles a minute.
- b) 32 to 40 cycles a minute.
- c) 25 to 30 cycles a minute.
- d) 12 to 20 cycles a minute.

Answer ;

d) 12 to 20 cycles a minute.

0551. The normal sleep cycle is approximately every minutes ;

- a) 90
- b) 60
- c) 120
- d) 180

Answer ;

a) 90

0552. The organ which metabolises alcohol from the body is the ;

- a) pancreas.
- b) liver.
- c) spleen.
- d) kidneys.

Answer ;

b) liver.

0553. The ossicles (the malleus , incus and stapes) are ;

- a) located in the middle ear.
- b) located in the inner ear.
- c) are not located in the ear.
- d) located in the outer ear.

Answer ;

a) located in the middle ear.

0554. The otoliths in the inner ear are sensitive to ;

- a) angular speed.
- b) linear acceleration and gravity.
- c) constant speed only.
- d) angular acceleration.

Answer ;

b) linear acceleration and gravity.

0555. The ozone-layer is situated in the ;

- a) stratosphere.
- b) troposphere.
- c) ionosphere.
- d) thermosphere.

Answer ;

a) stratosphere.

0556. The partial pressure of carbon dioxide in the alveoli is ;
- a) lower than in the blood.
 - b) higher than the pressure of carbon dioxide in the blood.
 - c) the same as in the atmospheric air.
 - d) lower than the pressure of carbon dioxide in the atmospheric air.

Answer ;

- a) lower than in the blood.
-

0557. The partial pressure of oxygen in inspired air ;
- a) is independent of the proportion of oxygen in the ambient air.
 - b) remains constant at all altitudes.
 - c) is a highly significant factor in ensuring that oxygen diffuses from the lungs to the blood.
 - d) can not be varied when using a oxygen mask.

Answer ;

- c) is a highly significant factor in ensuring that oxygen diffuses from the lungs to the blood.
-

0558. The part(s) of the eye responsible for night vision ;
- a) are the cones.
 - b) are the rods.
 - c) are rods and cones.
 - d) is the cornea.

Answer ;

- b) are the rods.
-

0559. The percentage of oxygen in cabin air ;
- a) varies with cabin altitude.
 - b) is proportional to the cabin pressure.
 - c) is the same as at sea level.
 - d) will reduce during a rapid decompression.

Answer ;

- c) is the same as at sea level.
-

0560. The percentage of oxygen in the air at an altitude of approximately 34.000 ft is ;
- a) 5%.
 - b) 42%.
 - c) 10.5%.
 - d) 21%.

Answer ;

- d) 21%.
-

0561. The percentage of oxygen in the troposphere in dry air ;
- a) is dependent of the partial pressure which is constant above sea level.
 - b) increases with longitude.
 - c) is variable because oxygen replaces water vapour.
 - d) is independent of altitude.

Answer ;

- d) is independent of altitude.
-

0562. The Peripheral Nervous System passes information from the ;
- a) brain to the sensory stores through sensory nerves.
 - b) brain to all parts of the body through sensory and autonomic nerves.
 - c) brain to all parts of the body through sensory nerves.
 - d) sensory inputs to the CNS through sensory and motor nerves.

Answer ;

- d) sensory inputs to the CNS through sensory and motor nerves.
-

0563. The peripheral vision is important for ;
- a) detecting moving objects.
 - b) visual acuity.
 - c) binocular vision.
 - d) colour vision.

Answer ;

- a) detecting moving objects.
-

0564. The phases of General Adaption Syndrome are ;
- a) Alarm , Resistance and Exhaustion.
 - b) Alarm , Resistance and Recovery.
 - c) Emergency , Medial and Recovery.
 - d) Emergency , Resistance and Exhaustion.

Answer ;

- a) Alarm , Resistance and Exhaustion.
-

0565. The phases of learning a Motor Programme are ;
- a) 1st , 2nd and 3rd phases.
 - b) Induction , Simulation and Practice phases.
 - c) Cognitive , Associative and Automatic phases.
 - d) Initial , Interim and Final phases.

Answer ;

- c) Cognitive , Associative and Automatic phases.
-

0566. The photosensitive cells being responsible for night vision are called ;
- a) the fovea.
 - b) the cones and the rods.
 - c) the cones.
 - d) the rods.

Answer ;

- d) the rods.
-

0567. The physiological effects of accelerations to the human body depend on ;
- 1. the duration of the G-forces.
 - 2. the onset rate of the G-forces.
 - 3. the magnitude of the G-forces.
 - 4. the direction of the G-forces..
- a) 1 , 2 , 3 and 4 are correct.
 - b) 1 , 2 , 3 are correct , 4 is false.
 - c) 1 and 4 are correct , 3 is false.
 - d) 2 , 3 and 4 are correct , 1 is false.

Answer ;

- a) 1 , 2 , 3 and 4 are correct.
-

0568. The physiological rhythms of a pilot in a new time zone will resynchronise to this new time zone at a rate of about ;
- a) 3 - 3.5 hours a day.
 - b) 2 - 2.5 hours a day.
 - c) 4 - 4.5 hours a day.
 - d) 1 - 1.5 hours a day.

Answer ;

- d) 1 - 1.5 hours a day.
-

0569. The pressoreceptors are located in ;
- a) the lungs.
 - b) the intestines.
 - c) the carotid and aortic arterial vessels.
 - d) the heart.

Answer ;

- c) the carotid and aortic arterial vessels.
-

0570. The primary factor in controlling the rate and depth of breathing is the ;
- a) partial pressure of nitrogen.
 - b) total air pressure in the blood.
 - c) pressure of carbon dioxide in the blood.
 - d) partial pressure of oxygen in the blood.

Answer ;

- c) pressure of carbon dioxide in the blood.
-

0571. The primary symptom of Decompression sickness/illness is ;

- a) the Bends.
- b) the Chokes.
- c) red coloured cheeks and lips.
- d) neurological damages to the CNS.

Answer ;

a) the Bends.

0572. The procedure to be followed in the event of decompression when flying above 10.000 ft must ;

- a) allow for the rapid supply of oxygen in order to prevent hypoxia.
- b) allow for a rapid descent independent from sufficient supply of oxygen in order to prevent disorders due to hypoxia.
- c) make it possible to eliminate the risk of fogging due to the sudden pressure changes.
- d) make it possible to prevent hyperventilation owing to the inhalation of 100 % oxygen.

Answer ;

a) allow for the rapid supply of oxygen in order to prevent hypoxia.

0573. The process of responding to a sender by confirming the reception of a message is called ;

- a) feedback.
- b) redundancy.
- c) transference.
- d) synchronization.

Answer ;

a) feedback.

0574. The proprioceptive senses (Seat of-the-Pants-Sense) ;

- a) can be used , if trained , to avoid spatial disorientation in IMC.
- b) may give incorrect information , when outside visual reference is lost.
- c) is a natural human instinct , always indicating the correct attitude.
- d) can neither be used for motor coordination in IMC and VMC.

Answer ;

b) may give incorrect information , when outside visual reference is lost.

0575. The proprioceptive senses (seat-of-the-pants sense) are important for motor coordination. They ;

- a) are completely unreliable for orientation when flying in IMC.
- b) allow the pilot to determine the absolute vertical at flight condition.
- c) are important senses for flight training in IMC.
- d) indicate the difference between gravity and G-forces.

Answer ;

a) are completely unreliable for orientation when flying in IMC.

0576. The proprioceptors do not orient an individual to his/her surroundings , but informs him/her of ;

- a) the relative motion and relative position of his body parts.
- b) the condition in the body itself.
- c) our surroundings.
- d) a touch on the skin.

Answer ;

a) the relative motion and relative position of his body parts.

0577. The quality of learning ;

- a) depends on long-term memory capacity.
- b) is dependent on age and professional role.
- c) is independent of the level of motivation.
- d) is promoted by feedback on the value of one's own performance.

Answer ;

d) is promoted by feedback on the value of one's own performance.

0578. The rate and depth of breathing is primarily controlled by ;

- a) the total atmospheric pressure.
- b) the amount of nitrogen in the blood.
- c) the amount of carbon monoxide in the blood.
- d) the amount of carbon dioxide in the blood.

Answer ;

d) the amount of carbon dioxide in the blood.

0579. The rate and depth of breathing is primarily regulated by the concentration of ;

- a) oxygen in the cells.
- b) nitrogen in the air.
- c) carbon dioxide in the blood.
- d) water vapour in the alveoli.

Answer ;

c) carbon dioxide in the blood.

0580. The re-adjustment of the biological rhythms after a time shift is normally more difficult ;

- a) with flights towards the East.
- b) with flights towards the West.
- c) with flights towards the South.
- d) with flights towards the North.

Answer ;

a) with flights towards the East.

0581. The relationship between arousal and flying performance is ;

- a) approximately the form of an inverted U.
- b) approximately linear increasing.
- c) approximately sinusoidal.
- d) approximately exponential.

Answer ;

a) approximately the form of an inverted U.

0582. The relevance of check procedures during flight becomes even more important when ;

- a) flying an unfamiliar type of aircraft and experiencing mental pressure.
- b) flying an aircraft which you have flown recently.
- c) flying an aircraft which you have flown many times before.
- d) conducting a longer flight than you would normally perform.

Answer ;

a) flying an unfamiliar type of aircraft and experiencing mental pressure.

0583. The requirement of good sunglasses is to ;

- a) absorb enough visible light to eliminate glare without decreasing visual acuity.
- b) increase the time for dark adaptation.
- c) fit to the pilots individual taste.
- d) eliminate distortion in aircraft windshields.

Answer ;

a) absorb enough visible light to eliminate glare without decreasing visual acuity.

0584. The resistance phase of stress ;

- a) is characterised by the secretion of adrenalin , increased heart rate and blood pressure.
- b) allows fats to be transformed into sugars , thereby prolonging the mobilization of energy in the body.
- c) may cause a person to die by stopping the defence mechanisms of the body.
- d) is very short duration and is unlikely to give a pilot the opportunity to resolve a problem.

Answer ;

b) allows fats to be transformed into sugars , thereby prolonging the mobilization of energy in the body.

0585. The respiratory process consists mainly of ;

- a) the transportation of carbon dioxide to the cells and elimination of oxygen.
- b) the transportation of oxygen to the cells and the elimination of nitrogen.
- c) the diffusion of oxygen through the respiratory membranes into the blood , transportation to the cells , diffusion into the cells and elimination of carbon dioxide from the body.
- d) the transportation of oxygen to the cells and the elimination of carbon monoxide.

Answer ;

c) the diffusion of oxygen through the respiratory membranes into the blood , transportation to the cells , diffusion into the cells and elimination of carbon dioxide from the body.

0586. The retina allows for colour perception as a result of the ;

- a) rods located in its peripheral zone.
- b) cones located in its central part.
- c) crystalline lens.
- d) rods located in its central part.

Answer ;

b) cones located in its central part.

0587. The retina of the eye ;
- a) is the muscle , changing the size of the crystalline lens.
 - b) is the light-sensitive inner lining of the eye containing the photoreceptors essential for vision.
 - c) filters the UV-light.
 - d) only regulates the light that falls into the eye.

Answer ;

- b) is the light-sensitive inner lining of the eye containing the photoreceptors essential for vision.
-

0588. The risk of a barotrauma of the middle ear is more likely to occur ;

- a) with colds and fast climbs.
- b) with colds and slow ascents.
- c) after a decompression.
- d) with colds and rapid descents.

Answer ;

- d) with colds and rapid descents.
-

0589. The risk of spatial disorientation increases when ;

- a) information from the vestibular organ in the inner ear are ignored.
- b) there is contradictory information between the instruments and the vestibular organs.
- c) the pilot is buckled tightly to his/her seat and cannot sense the attitude changes of the aircraft by the Seat-of-the-Pants-Sense.
- d) the pilot is performing an effective instrument cross-check and is ignoring illusions.

Answer ;

- b) there is contradictory information between the instruments and the vestibular organs.
-

0590. The “Seat-of-the-Pants-Sense” ;

- a) can be used as a reference for determining attitude when operating in visual and instrument meteorological conditions.
- b) is a natural human instinct which will always indicate the correct body position in space.
- c) can give false inputs to body orientation when visual reference is lost.
- d) can be used , if trained , to avoid disorientation in space.

Answer ;

- c) can give false inputs to body orientation when visual reference is lost.
-

0591. The Seat-of-the-Pants Sense involves receptors in the ;

- a) skin only.
- b) muscles , tendons and joints sensitive to the position and movement of body parts.
- c) utricle and saccule.
- d) semicircular canals.

Answer ;

- b) muscles , tendons and joints sensitive to the position and movement of body parts.
-

0592. The semicircular canals form part of the ;

- a) ear drum.
- b) external ear.
- c) inner ear.
- d) middle ear.

Answer ;

- c) inner ear.
-

0593. The semicircular canals of the inner ear monitor ;

- a) movements with constant speeds.
- b) relative speed and linear accelerations.
- c) gravity.
- d) angular accelerations.

Answer ;

- d) angular accelerations.
-

0594. The severity of hypoxia depends on the ;
1. rate of decompression.
 2. physical fitness.
 3. flight level.
 4. individual tolerance.
- a) 2 , 3 and 4 are correct , 1 is false.
b) 1 , 2 , 3 and 4 are correct.
c) 1 and 3 are correct , 2 and 4 are false.
d) 1 , 2 and 3 are correct , 4 is false.

Answer ;

b) 1 , 2 , 3 and 4 are correct.

0595. The Skill-based , Rule-based , Knowledge-based Model (Jens Rasmussen) is associated with ;
- a) behaviour.
 - b) risk.
 - c) personality.
 - d) motor programmes.

Answer ;

a) behaviour.

0596. The sleep cycles repeat during the course of a night's sleep ;
1. Each succeeding cycle contains a greater amount of REM-sleep.
 2. Frequent interruption of the REM-sleep may be harmful.
- a) 1 and 2 are both correct.
b) 1 is false , 2 is correct.
c) 1 and 2 are both false.
d) 1 is correct , 2 is false.

Answer ;

a) 1 and 2 are both correct.

0597. The sleep pattern is closely associated with ;
- a) glucose level.
 - b) heart rate.
 - c) body temperature.
 - d) blood pressure.

Answer ;

c) body temperature.

0598. The so-called Coriolis effect (a conflict in information processing in the brain) in spatial disorientation occurs ;
- a) on stimulating several semicircular canals simultaneously.
 - b) on stimulating the cochlea intensely.
 - c) when no semicircular canal is stimulated.
 - d) on stimulating the saccule and the utricle of the inner ear.

Answer ;

a) on stimulating several semicircular canals simultaneously.

0599. The so-called "Seat-of-the-Pants" sense is ;
- a) useful for instrument and contact flight.
 - b) not suitable for spatial orientation when outside visual references are lost.
 - c) the only sense a pilot can rely on , when flying in IMC.
 - d) only to be used by experienced pilots with the permission to fly in IMC.

Answer ;

b) not suitable for spatial orientation when outside visual references are lost.

0600. The Somatogravic illusion gives the pilot a false impression of ;
- a) descending and turning to the right.
 - b) climbing or descending.
 - c) climbing and turning to the right.
 - d) descending and turning to the left.

Answer ;

b) climbing or descending.

0601. The symptoms caused by gas bubbles under the skin following a decompression are called ;

- a) creeps.
- b) chokes.
- c) leans.
- d) bends.

Answer ;

a) creeps.

0602. The symptoms of hypoxia include ;

- a) visual disturbances , lack of concentration , euphoria.
- b) nausea and barotitis.
- c) dull headache and bends.
- d) dizziness , hypothermia.

Answer ;

a) visual disturbances , lack of concentration , euphoria.

0603. The system which controls breathing , digestion , heart rate , etc. , over which there is no voluntary control , is ;

- a) the critical nervous system.
- b) non-existent , as all these functions are under conscious control.
- c) the local nervous system.
- d) the autonomic nervous system.

Answer ;

d) the autonomic nervous system.

0604. The team spirit of a cockpit-crew depends most likely on both pilots ;

- a) wearing the same uniform.
- b) flying together very often for a long period.
- c) respecting each other and striving for the same goals.
- d) having the same political and ideological attitude.

Answer ;

c) respecting each other and striving for the same goals.

0605. The team spirit of a cockpit-crew most likely depends on both pilots ;

- a) respecting each other and striving for the same goals.
- b) wearing the same uniform.
- c) flying together very often for a long period.
- d) having the same political and ideological attitude.

Answer ;

a) respecting each other and striving for the same goals.

0606. The thin walls of capillaries are permeable for ;

- a) gases.
- b) red blood cells.
- c) protein.
- d) platelets.

Answer ;

a) gases.

0607. The three types of Authority Gradient Cockpits are ;

- a) Authoritative , Followership and Synergistic.
- b) Authoritative , Laisser-Faire and Synergistic.
- c) Autocratic , Followership and Synergistic.
- d) Autocratic , Laisser-Faire and Synergistic.

Answer ;

d) Autocratic , Laisser-Faire and Synergistic.

0608. The time an eye needs to adapt fully to the dark is about ;

- a) 10 seconds.
- b) 5 minutes.
- c) 10 minutes.
- d) 25 - 30 minutes.

Answer ;

d) 25 - 30 minutes.

- 0609.** The time between inadequate oxygen supply and incapacitation is called TUC (Time of Useful Consciousness). It ;
- a) is the same amount of time for every person.
 - b) varies individually and depends on cabin pressure altitude.
 - c) varies individually and does not depend on altitude.
 - d) is not dependent on physical or psychological pressure.

Answer ;

- b)** varies individually and depends on cabin pressure altitude.
-

- 0610.** The Time of Useful Consciousness may vary according to ;
- 1. physical activity of the subjected crew.
 - 2. the experience of the pilot on the type of aircraft in question.
 - 3. the strength and time of decompression.
 - 4. the time of day.
- a) 1 , 3.
 - b) 3 , 4.
 - c) 4.
 - d) 1 , 2.

Answer ;

- a)** 1 , 3.
-

- 0611.** The Time of Useful Consciousness may vary according to ;
- 1. physical activity of the subjected crew.
 - 2. the experience of the pilot on the type of aircraft in question.
 - 3. the strength and time of decompression.
 - 4. the cabin temperature.
- a) 3 , 4.
 - b) 1 , 3.
 - c) 1 , 2.
 - d) 4.

Answer ;

- b)** 1 , 3.
-

- 0612.** The time required for complete adaptation is ;
- a) for day and night ; 30 min.
 - b) for high levels of illumination 10 sec and for full dark adaptation 30 min.
 - c) for night 10 sec and for day 30 min.
 - d) for high levels of illumination 10 minutes and for low levels of illumination 30 minutes.

Answer ;

- b)** for high levels of illumination 10 sec and for full dark adaptation 30 min.
-

- 0613.** The total gas volume of the lung is the sum of ;
- 1. tidal volume.
 - 2. inspiratory reserve volume.
 - 3. expiratory reserve volume.
 - 4. residual volume.
- Which of the following lists the correct combination ?
- a) 1 and 2.
 - b) 2 and 3.
 - c) 1 , 2 and 3.
 - d) 1 , 2 , 3 and 4.

Answer ;

- d)** 1 , 2 , 3 and 4.
-

- 0614.** The total pressure of a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture. This is ;
- a) Henry's law.
 - b) Dalton's law.
 - c) Graham's law.
 - d) Boyle Mariotte's law.

Answer ;

- b)** Dalton's law.
-

0615. The transfer of carbon dioxide from the blood to the alveoli can be described by ;

- a) Dalton's Law.
- b) the law of diffusion.
- c) Henry's Law.
- d) Boyles Law.

Answer ;

b) the law of diffusion.

0616. The transfer of oxygen from the alveoli to the blood can be described by ;

- a) Boyle's Law.
- b) Dalton's Law.
- c) the law of diffusion.
- d) Henry's Law.

Answer ;

c) the law of diffusion.

0617. The trend in aeroplane hull-loss rate over the last three decades seems to be related to ;

- a) the manufacturer.
- b) the year of manufacture.
- c) the number of engines.
- d) the crew.

Answer ;

d) the crew.

0618. The two types of fatigue are ;

- a) Oppressive and negative.
- b) Short-term and oppressive.
- c) Chronic short-term and acute.
- d) Heavy and light.

Answer ;

c) Chronic short-term and acute.

0619. The type of hypoxia , which occurs at altitude is explained by ;

- a) Graham's law.
- b) Boyle Mariotte's law.
- c) Dalton's law.
- d) Henry's law.

Answer ;

c) Dalton's law.

0620. The unique organisation of characteristics which determine the typical or standard behaviour of an individual is known as ;

- a) learning.
- b) development.
- c) heredity.
- d) personality.

Answer ;

d) personality.

0621. The use of check lists should be carried out in such a way that ;

- a) their execution may be done simultaneously with other actions.
- b) they should only be carried out when time is available.
- c) their execution should be combined with other important tasks.
- d) their execution should not be done simultaneously with other actions.

Answer ;

d) their execution should not be done simultaneously with other actions.

0622. The use of professional languages offers ;

- a) more precise grammar.
- b) quick comprehension and simplified grammar.
- c) better communication even if professionals are not from the same field.
- d) longer attention span.

Answer ;

b) quick comprehension and simplified grammar.

0623. The vestibular apparatus ;
- a) reacts to vibrations of the cochlea.
 - b) reacts to linear/angular acceleration and gravity.
 - c) reacts to pressure changes in the middle ear.
 - d) gives the impression of hearing.

Answer ;

- b) reacts to linear/angular acceleration and gravity.
-

0624. The Vestibular Apparatus consists of the ;
- a) Cochlea and the auditory nerve.
 - b) Eustachian tube and the semi-circular canals.
 - c) Semi-circular canals and the otoliths.
 - d) Eustachian tube and the pinna.

Answer ;

- c) Semi-circular canals and the otoliths.
-

0625. The vestibular system is composed of ;
- 1. two ventricles.
 - 2. a saccule.
 - 3. a utricle.
 - 4. three semicircular channels.
- a) 2 , 3 , 4.
 - b) 2 , 3.
 - c) 1 , 3 , 4.
 - d) 1 , 4.

Answer ;

- a) 2 , 3 , 4.
-

0626. The volume of air exchanged during a normal breathing cycle (tidal volume) is about ;
- a) 500 ml of air.
 - b) 75 ml of air.
 - c) 150 ml of air.
 - d) 350 ml of air.

Answer ;

- a) 500 ml of air.
-

0627. The volume percentage of oxygen in the atmosphere at 30.000 feet remains at 21% , but the partial pressure of oxygen ;
- a) increases by expansion.
 - b) decreases significantly with lower temperatures.
 - c) decreases with decreasing barometric pressure.
 - d) remains constant , independent from altitude.

Answer ;

- c) decreases with decreasing barometric pressure.
-

0628. The volume percentage of oxygen in the atmosphere is 21% which ;
- a) increases with increasing altitude.
 - b) decreases with increasing altitude.
 - c) is dependent on the present air pressure.
 - d) is constant at all altitudes for conventional aircraft.

Answer ;

- d) is constant at all altitudes for conventional aircraft.
-

0629. Thinking on human reliability is changing ;
- a) human errors can be avoided. All it takes is to be vigilant and to extend one's knowledge.
 - b) human errors are now considered as being inevitable.
 - c) it is thought that it will be possible to eliminate errors in the near future.
 - d) the individual view of safety has gradually replaced the systemic view of safety.

Answer ;

- b) human errors are now considered as being inevitable.
-

0630. Through which part of the ear does the equalization of pressure take place , when altitude is changed ?

- a) Eustachian tube.
- b) Cochlea.
- c) Tympanic membrane.
- d) External auditory canal.

Answer ;

- a) Eustachian tube.
-

0631. To avoid incapacity due to gastro-intestinal problems , it is recommended that a simple , easily digested meal before boarding the aircraft should be accompanied by ;

- a) no drinks , as fluids are not important.
- b) non-carbonated water.
- c) carbonated water.
- d) wine or beer.

Answer ;

- b) non-carbonated water.
-

0632. To avoid wrong decisions by the pilot , an aircraft system should at least be able to ;

- a) report the deviation.
- b) report its malfunction.
- c) correct the deviation.
- d) tolerate the deviation.

Answer ;

- b) report its malfunction.
-

0633. To avoid wrong decisions by the pilot , an aircraft system should at the minimum to ;

- a) report the deviation.
- b) correct the deviation.
- c) tolerate the deviation.
- d) report its malfunction.

Answer ;

- d) report its malfunction.
-

0634. To maintain good situational awareness you should ;

- 1. believe only in your own interpretation of the data.
 - 2. gather as much data as possible from every possible source before making inferences.
 - 3. question whether your hypothesis still fits the situation as events progress and try to make time to review the situation.
 - 4. consider ways of testing your situational hypothesis to see whether it is correct.
- a) 2 , 3 and 4 are correct.
 - b) 1 and 3 are correct.
 - c) all answers are correct.
 - d) 1 and 4 are correct.

Answer ;

- a) 2 , 3 and 4 are correct.
-

0635. To optimise one's night-vision performance , it is necessary ;

- 1. to spend some time getting adapted to low levels of illumination.
 - 2. to increase the instrument panel lighting by reducing the cockpit lighting.
 - 3. not to focus on the point to be observed.
 - 4. to avoid blinding sources of light.
- a) 1 , 3 , 4.
 - b) 2.
 - c) 1 , 2 , 4.
 - d) 2 , 3 , 4.

Answer ;

- a) 1 , 3 , 4.
-

0636. To prevent empty field myopia at higher altitudes ;
- a) one should book an appointment with an ophthalmologist as soon as possible.
 - b) one should concentrate only on instruments.
 - c) it is recommended that a pilot uses short sharp scans , or periodically focuses on a distant object such as a wingtip or cloud edge.
 - d) it is recommended that a pilot focuses on an object known to be approximately one metre away.

Answer ;

- c) it is recommended that a pilot uses short sharp scans , or periodically focuses on a distant object such as a wingtip or cloud edge.
-

0637. To prevent gastro-intestinal problems in tropical climates you should ;

- 1. not eat salad or raw vegetables.
- 2. always peel fruit.
- 3. only eat food that has been properly cooked.
- 4. avoid ice-cream.

Which of the following lists all the correct answers ?

- a) 1 , 2 and 3.
- b) 1 and 2.
- c) 2 and 4.
- d) 1 , 2 , 3 and 4.

Answer ;

- d) 1 , 2 , 3 and 4.
-

0638. To prevent vertigo in flight we should ;

- a) keep breathing normally.
- b) breath deeply but control the respiratory frequency.
- c) look towards the sides when we make a turn.
- d) not move the head suddenly while we are turning.

Answer ;

- d) not move the head suddenly while we are turning.
-

0639. To resynchronize a circadian rhythm , it takes more time after ;

- a) north-south flights.
- b) eastbound flights.
- c) westbound flights.
- d) south-north flights.

Answer ;

- b) eastbound flights.
-

0640. Tolerance to decompression sickness is decreased by ;

- 1. SCUBA diving.
 - 2. Obesity.
 - 3. Age.
 - 4. Body height.
- a) 2 and 4 are correct.
 - b) 1 , 2 and 3 are correct.
 - c) only 4 is correct.
 - d) 1 , 3 and 4 are correct.

Answer ;

- b) 1 , 2 and 3 are correct.
-

0641. Trace elements should be obtained through ;

- a) vitamin pills.
- b) eating plenty of meat and vegetables.
- c) a balanced diet.
- d) dairy products and vegetables.

Answer ;

- c) a balanced diet.
-

0642. Trapped intestinal gases can cause severe pain. When is this the case ?

- a) Only in pressurized aircraft when flying at higher flight levels.
- b) During descent as well as during climb , when the cabin pressure altitude exceeds 2.000 ft.
- c) At lower altitudes.
- d) More frequent when flying above 18.000 ft in a non-pressurized aircraft.

Answer ;

- d) More frequent when flying above 18.000 ft in a non-pressurized aircraft.

-
0643. TUC (Time of Useful Consciousness) is ;
- a) the time before becoming unconscious at a sudden pressure loss.
 - b) the time after pressure loss until decompression sickness sets in.
 - c) the length of time during which an individual can act with both mental and physical efficiency , measured from the moment at which he/she is exposed to hypoxia.
 - d) the time between the start of hypoxia and death.

Answer ;

- c) the length of time during which an individual can act with both mental and physical efficiency , measured from the moment at which he/she is exposed to hypoxia.
-

0644. Under normal circumstances , which gas will diffuse from the blood to the alveoli ;
- a) nitrogen.
 - b) carbon dioxide.
 - c) carbon monoxide.
 - d) oxygen.

Answer ;

- b) carbon dioxide.
-

0645. Up to what altitude is blood oxygen saturation reduction unlikely to affect a pilot's judgement ?
- a) 20.000 ft.
 - b) 10.000 ft.
 - c) 25.000 ft.
 - d) 30.000 ft.

Answer ;

- b) 10.000 ft.
-

0646. Using the Body Mass Index (BMI) formula , when will a pilot be classed as obese ?
- a) Over 20 for males and 19 for females.
 - b) Over 10 for males and 9 for females.
 - c) Over 26 for males and 25 for females.
 - d) Over 30 for males and 29 for females.

Answer ;

- d) Over 30 for males and 29 for females.
-

0647. Very high ambition and need for achievement ;
- a) improves the coping process with personal failures.
 - b) always promote teamwork.
 - c) fulfil the requirements of stress resistance.
 - d) disturb the climate of cooperation.

Answer ;

- d) disturb the climate of cooperation.
-

0648. Vision of terrain relief (e.g. hills , valleys , etc.) ;
- a) is based on binocular vision at short distances and the rules of proportion and perspective for objects that are further away.
 - b) is impossible with only one eye.
 - c) is impossible if a pilot wears sunglasses.
 - d) is possible at only at distances over 200 m due to binocular vision.

Answer ;

- a) is based on binocular vision at short distances and the rules of proportion and perspective for objects that are further away.
-

0649. Visual acuity during flight at high altitudes can be affected by ;
- 1. anaemia.
 - 2. smoking in the cockpit.
 - 3. carbon monoxide poisoning.
 - 4. hypoxia.
- a) 1 , 3 and 4 are correct.
 - b) 1 , 2 and 3 are correct.
 - c) 2 , 3 and 4 are correct.
 - d) 1 , 2 , 3 and 4 are correct.

Answer ;

- d) 1 , 2 , 3 and 4 are correct.
-

0650. Visual disturbances can be caused by ;
1. hyperventilation.
 2. hypoxia.
 3. hypertension.
 4. fatigue.
- a) 2 , 3 and 4 are correct.
b) 1 , 2 and 4 are correct.
c) 1 , 2 , 3 and 4 are correct.
d) 1 , 2 and 3 are correct.

Answer ;

b) 1 , 2 and 4 are correct.

0651. Visual perception of depth at close to medium distance is primarily due to ;
- a) the high sensitivity of the retina.
 - b) binocular vision.
 - c) peripheral vision.
 - d) interactions between cones and rods.

Answer ;

b) binocular vision.

0652. Vitamin A and possibly vitamins B and C are chemical factors and essential to good night vision ;
1. Vitamin deficiencies may decrease night vision performance.
 2. An excess intake of vitamin A will improve night vision performance significantly.
 3. Pilots should be carefully concerned to take a balanced diet containing sufficient vitamin A.
 4. Vitamin deficiencies may decrease visual acuity in photopic vision but not in scotopic vision.
- a) 1 and 3 are false , 2 and 4 are correct.
b) 1 , 2 , 3 and 4 are correct.
c) Only 4 is false.
d) 1 and 3 are correct , 2 and 4 are false.

Answer ;

d) 1 and 3 are correct , 2 and 4 are false.

0653. Waiting for take-off on the runway , a pilot is informed by the tower that the tail wind component is 12 kt. According to the manual , the limitation is 10 kt max. The pilot makes a decision to take-off anyway , as the runway is long.
- a) This is neither an error or a violation - the pilot is using his initiative.
 - b) This is an error of reasoning.
 - c) This is a decision error.
 - d) This is a violation.

Answer ;

d) This is a violation.

0654. We can observe the following in relation to a state of hypothermia ;
- a) widespread pain in the joints.
 - b) greater capacity for adaptation than in a hot atmosphere.
 - c) reasoning problems as soon as body temperature falls below 37°C.
 - d) a substantial increase in internal body temperature whereas peripheral temperature at the skin is stable.

Answer ;

c) reasoning problems as soon as body temperature falls below 37°C.

0655. We generate Mental Models on ;
- a) imagination.
 - b) stereopsis.
 - c) 3-D vision.
 - d) past experience and learning.

Answer ;

d) past experience and learning.

0656. What aircraft equipment marked a substantial decrease in hull loss rates in the eighties ?
- a) SSR
 - b) DME
 - c) GPWS
 - d) TCAS

Answer ;

c) GPWS

0657. What are main signs indicating the loss of vigilance ?
1. Decrease in sensory perception.
 2. Increase in selective attention.
 3. Sensation of muscular heaviness.
 4. Decrease in complacency.

Which of the following lists all the correct statements ?

- a) 2 and 4.
- b) 2 and 3.
- c) 1 and 4.
- d) 1 and 3.

Answer ;

- d) 1 and 3.
-

0658. What are the characteristics of the alarm phase of the stress reactions ?

1. increased arousal level as a result of adrenaline secretion.
2. an increase in heart rate , respiration and release of glucose.
3. a decrease in stress resistance.
4. activation of the digestive system.
5. secretion of cortisol to mobilize attention.

The combination of correct statements is ;

- a) 1 , 2 , 3.
- b) 1 , 2.
- c) 2 , 4 , 5.
- d) 1 , 3 , 5.

Answer ;

- a) 1 , 2 , 3.
-

0659. What are the communication qualities of a good briefing ? A good briefing must ;

1. contain as much information and be as comprehensive as possible.
2. be of a standard type so that it can be reused for another flight of the same type.
3. be short and precise.
4. be understandable to the other crew member(s).

The correct statement(s) is (are) ;

- a) 1 and 4 are correct.
- b) 1 , 2 and 4 are correct.
- c) 1 and 2 are correct.
- d) 2 , 3 and 4 are correct.

Answer ;

- d) 2 , 3 and 4 are correct.
-

0660. What are the effects of excessive stress ?

- a) It increases vigilance for a longer period than stress itself , but may focus attention inappropriately.
- b) It activates resources stored in memory.
- c) It reduces vigilance and focuses attention.
- d) It has very little immediate effect on vigilance and attention.

Answer ;

- a) It increases vigilance for a longer period than stress itself , but may focus attention inappropriately.
-

0661. What are the main advantages of the human over the machine ?

- a) Ability to recognize system failures and rectify them quicker than a machine.
- b) Creativity , innovation , and aptitude to deal with novel situations.
- c) Ability to recognize limits on systems being exceeded quicker than a machine and resolve the situation.
- d) Logic and the use of human senses.

Answer ;

- b) Creativity , innovation , and aptitude to deal with novel situations.
-

0662. What are the main advantages that the human possesses over those of a machine in the decision making process ?

- a) The ability to reach the most advantageous option from a number of choices.
- b) The ability to analyse and compare.
- c) Creativity , innovation and adaptability.
- d) The ability to analyse and memorise.

Answer ;

- c) Creativity , innovation and adaptability.
-

- 0663.** What are the main characteristics of active errors ? They ;
1. are detectable only with difficulty by first-line operators.
 2. have rapid and direct consequences on the action in progress.
 3. occur at the human/machine interface.
 4. lie dormant and are undetected at first.
- a) 3 , 4.
b) 1 , 4.
c) 2 , 3.
d) 1 , 2.

Answer ;

c) 2 , 3.

- 0664.** What are the main consequences of latent errors ? They ;
1. remain undetected in the system for a certain length of time.
 2. may only manifest themselves under certain conditions.
 3. are quickly detectable by the front-line operator whose mental schemas on the instantaneous situation filter out formal errors.
 4. lull pilots into a false sense of security.

The correct statement(s) is (are) ;

- a) 1 , 2 and 4.
b) 1 and 3.
c) 2 , 3 and 4.
d) 1 and 2.

Answer ;

a) 1 , 2 and 4.

- 0665.** What are the main effects of a lack of sleep on performance ?

- a) It causes muscular spasms.
b) It reduces concentration and fatigue only with sleep loss greater than 48 hours.
c) It increases fatigue , reduces concentration and increases the risk of sensory illusions.
d) It increases fatigue and concentration difficulties , but facilitates stress management by muscular relaxation.

Answer ;

c) It increases fatigue , reduces concentration and increases the risk of sensory illusions.

- 0666.** What are the main factors which bring about reduced or low vigilance (hypovigilance) ?

1. The monotony of the task.
2. Tiredness and the need for sleep.
3. A lack of stimulation.
4. Excessive stress.

- a) 1 , 3.
b) 2 , 4.
c) 3 , 4.
d) 1 , 2 , 3.

Answer ;

d) 1 , 2 , 3.

- 0667.** What are the main limits of short-term memory ? It is ;

1. very sensitive to interruptions and interference.
2. difficult to access.
3. limited in size.
4. information is lost within approximately 20 seconds unless it is actively or deliberately placed into the long term memory.

- a) 1 , 2 , 3.
b) 2 , 4.
c) 2 , 3.
d) 1 , 3 , 4.

Answer ;

d) 1 , 3 , 4.

- 0668.** What are the main parts of the Central Nervous System and where is vision processed ?

- a) The brain and the brain stem are the main parts of the CNS and vision is processed at the cerebellum.
b) The brain and spinal chord are the main parts of the CNS and vision is processed at the retina.
c) The brain and the brain stem are the main parts of the CNS and vision is processed at the optic nerve centre.
d) The brain and spinal chord are the main parts of the CNS and vision is processed at the cortex.

Answer ;

d) The brain and spinal chord are the main parts of the CNS and vision is processed at the cortex.

0669. What are the main sources of radiation at height ?

- a) Ionisation of the layers in the atmosphere due to the bombardment of Galactic (Cosmic) radiation.
- b) Solar radiation caused by solar flares and Nuclear radiation caused by the earth's natural radiation.
- c) Galactic (Cosmic) caused by galactic particles and Solar radiation caused by solar flares.
- d) Galactic (Cosmic) and Nuclear radiation which are both caused by the earth's natural radiation.

Answer ;

- c) Galactic (Cosmic) caused by galactic particles and Solar radiation caused by solar flares.
-

0670. What are the most frequent results of an self-centred captain on the flight deck ?

- a) In a two-pilot flight deck , the co-pilot is ignored and may react by disengaging , showing delayed responses or aggression.
- b) Performance is very poor as self-centred behaviour leads to an increase of cooperation and efficiency.
- c) High group performance despite the strained relations.
- d) A major risk of authority inversion if the co-pilot is unassertive.

Answer ;

- a) In a two-pilot flight deck , the co-pilot is ignored and may react by disengaging , showing delayed responses or aggression.
-

0671. What are the three phases of General Adaptation Syndrome ?

- a) Alarm , resistance , exhaustion.
- b) Alert , resistance , performance.
- c) Alert , resistance , exhaustion.
- d) Alarm , resistance , performance.

Answer ;

- a) Alarm , resistance , exhaustion.
-

0672. What are the two inputs to a simple model of stress ?

- a) The ease of achieving the task and its usefulness in the scheme of things.
- b) The value of the task to the pilot and what the pilot can get out of it.
- c) The value of the task overall and its usefulness in the scheme of things.
- d) The perceived demand of the task to the pilot and the pilot's perceived ability to complete the task.

Answer ;

- d) The perceived demand of the task to the pilot and the pilot's perceived ability to complete the task.
-

0673. What are the two main categories of fatigue ?

- a) Normal and Severe.
- b) Acute and Chronic.
- c) Normal and Chronic.
- d) Acute and Sympathetic.

Answer ;

- b) Acute and Chronic.
-

0674. What are the two types of attention ?

- a) Cognitive and intuitive.
- b) Intuitive and behavioural.
- c) Divided and behavioural.
- d) Selective and divided.

Answer ;

- d) Selective and divided.
-

0675. What can a pilot do to avoid automation complacency ?

- a) Nothing , because it is system-inherent.
- b) Always fly the whole flight manually to remain in man-machine loop.
- c) Regard the automatic system as additional crew members that needs to be crosschecked as well.
- d) Always try to enhance your aviation related knowledge during low workload periods.

Answer ;

- c) Regard the automatic system as additional crew members that needs to be crosschecked as well.
-

0676. What can a pilot do to avoid "Flicker vertigo" when flying in the clouds ?

- a) Engage the autopilot until breaking the clouds.
- b) Dim the cockpit lights to avoid reflections.
- c) Switch strobe-lights off.
- d) Fly straight and level and avoid head movements.

Answer ;

- c) Switch strobe-lights off.
-

0677. What can be said concerning the following two statements ?

1. Euphoria can be a symptom of hypoxia.
 2. Someone in an euphoric condition is more prone to error.
- a) 1 and 2 are both correct.
 - b) 1 and 2 are both false.
 - c) 1 is false , 2 is correct.
 - d) 1 is correct , 2 is false.

Answer ;

- a) 1 and 2 are both correct.
-

0678. What can cause spatial disorientation ?

- a) Damage to the receptor cells in the eyes.
- b) Flying in clear daylight conditions.
- c) False perception of orientation of the aircraft with respect to spatial references.
- d) Reference to well-defined external visual references , such as the horizon.

Answer ;

- c) False perception of orientation of the aircraft with respect to spatial references.
-

0679. What could be symptoms of hypoxia (when flying without oxygen) above 12.000 ft ?

- a) Headache , thirst , somnolence , collapse.
- b) Euphoria , headache , improvement in judgement , loss of consciousness.
- c) Trembling , increase in body temperature , convulsions , slowing of the rate of breathing.
- d) Headache , fatigue , dizziness , lack of coordination.

Answer ;

- d) Headache , fatigue , dizziness , lack of coordination.
-

0680. What could cause hyperventilation ?

- a) Fear , anxiety and distress.
- b) Abuse of alcohol.
- c) Extreme low rate of breathing.
- d) Fatigue.

Answer ;

- a) Fear , anxiety and distress.
-

0681. What counter-measure can be used against barotrauma of the middle ear ?

- a) Pilots should apply anti-cold remedies prior every flight to prevent barotrauma in the middle ear.
- b) Increase rate of descent.
- c) Stop climbing , start descent.
- d) Close the mouth , pinch the nose tight and blow out thereby increasing the pressure in the mouth and throat.
At the same time try to swallow or move lower jaw.

Answer ;

- d) Close the mouth , pinch the nose tight and blow out thereby increasing the pressure in the mouth and throat.
At the same time try to swallow or move lower jaw.
-

0682. What diseases can be associated with contaminated water ?

- a) Typhoid , Cholera , Dysentery.
- b) Tetanus , Typhoid , Cholera , Dysentery.
- c) Hydrophobia , Cholera , Dysentery.
- d) Tetanus , Typhoid , Cholera.

Answer ;

- a) Typhoid , Cholera , Dysentery.
-

0683. What distinguishes status from role ?

- a) While role defines- via behaviour- the functions that must be performed by individuals , status defines the hierarchical position and its recognition by the group.
- b) While role defines the enjoyment of a hierarchical position and its recognition by the group , status defines - via behaviour - the functions that must be performed by individuals.
- c) Unlike status , role is fixed and is not modified either by the situation in flight or by the interactions of a new crew.
- d) Unlike status , role is fixed and is modified either by the situation in flight or by the interactions of a new crew.

Answer ;

- a) While role defines- via behaviour- the functions that must be performed by individuals , status defines the hierarchical position and its recognition by the group.
-

- 0684.** What do you do , when you are affected by “pilot’s vertigo” ?
1. Establish and maintain an effective instrument cross-check.
 2. Believe the instruments.
 3. Ignore illusions.
 4. Minimize head movements.
- a) 1 , 2 , 3 and 4 are correct.
b) 1 , 2 and 3 are correct , 4 is false.
c) Only 4 is false.
d) 1 and 2 are correct , 3 and 4 are false.

Answer ;

- a) 1 , 2 , 3 and 4 are correct.
-

- 0685.** What does not impair the function of the photosensitive cells ?
- a) Toxic influence (alcohol , nicotine , medication).
 - b) High speed.
 - c) Acceleration.
 - d) Oxygen deficiency.

Answer ;

- b) High speed.
-

- 0686.** What elements establish synergy within the crew ?
- a) Synergy is independent of the natural individual characteristics of the group members (communication , mutual confidence , sharing of tasks , etc.)
 - b) Synergy establishes itself automatically within the crew , right through from briefing to debriefing.
 - c) It is only the captain’s status which allows the establishment of synergy within the crew.
 - d) Synergy must be built up from the start of the mission (briefing) and be maintained until it comes to an end (debriefing).

Answer ;

- d) Synergy must be built up from the start of the mission (briefing) and be maintained until it comes to an end (debriefing).
-

- 0687.** What event can cause a hyperventilation (not required by physical need) ?
1. Pressure breathing.
 2. Anxiety or fear.
 3. Overstress.
 4. Strong pain.
 5. Jogging.
- a) 1 and 5 are both false.
b) 1 , 2 , 3 , 4 and 5 are correct.
c) 1 , 2 , 3 and 4 are correct , 5 is false.
d) Only 2 and 3 are correct.

Answer ;

- c) 1 , 2 , 3 and 4 are correct , 5 is false.
-

- 0688.** What factors can lead to a deterioration in vision ?
1. Hypoxia.
 2. Smoking.
 3. Poor diet , lacking Vitamin A.
 4. Exposure to bright lights.
- Which of the following lists all the correct statements ?
- a) 1 , 3 and 4.
 - b) 1 and 3.
 - c) 1 , 2 , 3 and 4.
 - d) 2 and 4.

Answer ;

- c) 1 , 2 , 3 and 4.
-

- 0689.** What human function is most sensitive to lack of oxygen ?
- a) Hearing.
 - b) Motor co-ordination.
 - c) Touch.
 - d) Night vision.

Answer ;

- d) Night vision.
-

0690. What illusion may occur if an aircraft is flying into fog , snow or haze ?

- a) Objects will appear bigger in size than in reality.
- b) Objects seem to be farther away than in reality.
- c) Objects seem to move slower than in reality.
- d) Objects will appear closer than they really are.

Answer ;

- b)** Objects seem to be farther away than in reality.
-

0691. What is a physiological Stressor ?

- a) An unresolved force acting upon the individual's mental or physical systems.
- b) A means of coping with a stressful situation.
- c) A stimulus that produces a change in any of the homeostatic mechanisms of the body.
- d) The body's non-specific response to demands put upon it.

Answer ;

- c)** A stimulus that produces a change in any of the homeostatic mechanisms of the body.
-

0692. What is a stressor ?

- a) All external stimuli are stressors since they modify the internal equilibrium.
- b) A psychological problem developed in a situation of danger.
- c) An external or internal stimulus which is interpreted by an individual as being stressful.
- d) The adaptation response of the individual to his environment.

Answer ;

- c)** An external or internal stimulus which is interpreted by an individual as being stressful.
-

0693. What is "automation complacency" ?

- a) The circumstance where monitoring and cross-checking is reduced because of the belief in the infallibility of the automatic systems.
- b) No over-ride facility is available on auto-flight selection as there are enough levels of redundancy to cope with any system failure.
- c) Automation is now so reliable that the human input is no longer necessary.
- d) The circumstance where the pilot flies the aircraft manually because he is unable to rely on the automatic systems flying the aircraft safely.

Answer ;

- a)** The circumstance where monitoring and cross-checking is reduced because of the belief in the infallibility of the automatic systems.
-

0694. What is characterized by a "laissez-faire" cockpit ?

- a) The high level of independence granted to each member by the captain quickly leads to tension between the various crew members.
- b) The captain's authority rules all the actions or decisions associated with the situation.
- c) A passive approach by the captain allows decisions , choices and actions by other crew members.
- d) Each member carries out actions and makes choices informing the other members about them.

Answer ;

- c)** A passive approach by the captain allows decisions , choices and actions by other crew members.
-

0695. What is decompression sickness ?

- a) The formation of air bubbles in bodily tissues , with no consequences for people's capabilities.
- b) A condition resulting from the formation of nitrogen bubbles in bodily tissues and fluids after a cabin pressure loss at high altitude.
- c) A disorder which is solely encountered below 18.000 ft.
- d) A frequent disorder in commercial aviation due to the pressurisation curve of modern aircraft.

Answer ;

- b)** A condition resulting from the formation of nitrogen bubbles in bodily tissues and fluids after a cabin pressure loss at high altitude.
-

0696. What is "divided attention" ?

- a) The adverse effect of motivation which leads to one's attention being dispersed.
- b) Ease of concentrating on a particular objective.
- c) Difficulty of concentrating on a particular objective.
- d) The management of several matters of interest dealt with individually one after the other.

Answer ;

- d)** The management of several matters of interest dealt with individually one after the other.
-

0697. What is hypoxia ?

- a) A state characterised by an excessive supply of oxygen which may be due to maladjustment of the mask.
- b) The total absence of oxygen in the blood of the body.
- c) Any condition where the oxygen concentration of the body is below normal limits to meet the needs of he body tissues.
- d) The respiratory symptom associated with altitude decompression sickness.

Answer ;

- c) Any condition where the oxygen concentration of the body is below normal limits to meet the needs of he body tissues.
-

0698. What is hypoxia ?

- a) A state characterised by an excessive supply of oxygen which may be due to maladjustment of the mask.
- b) The respiratory symptom associated with altitude decompression sickness.
- c) Any condition where the oxygen concentration of the body is below normal limits or where the oxygen available to the body cannot be used due to some pathological condition.
- d) The total absence of oxygen in the blood of the body.

Answer ;

- c) Any condition where the oxygen concentration of the body is below normal limits or where the oxygen available to the body cannot be used due to some pathological condition.
-

0699. What is meant by metabolism ?

- a) Information exchange.
- b) The generation and utilisation of energy by the body's cells and tissues.
- c) Transfer of chemical messages.
- d) Exchange of substances between the lung and the blood.

Answer ;

- b) The generation and utilisation of energy by the body's cells and tissues.
-

0700. What is meant by "passive monitoring" in terms of automation complacency ?

- a) A pilot knowing where the aircraft has been , where it is now , and where it is going to.
- b) A pilot's concentration on one particular area of the system instead of monitoring the whole system.
- c) A pilot watching what is going on rather than analyzing and constantly checking.
- d) A pilot's belief in the fallibility of systems and constantly checking and analyzing the situation.

Answer ;

- c) A pilot watching what is going on rather than analyzing and constantly checking.
-

0701. What is meant by presbycusis ?

- a) Gradual loss of hearing with age.
- b) Total loss of hearing in one ear due to use of a headset in the cockpit.
- c) Gradual onset of long-sightedness with age.
- d) Total loss of hearing in both ears due to disease.

Answer ;

- a) Gradual loss of hearing with age.
-

0702. What is meant by the term "complacency" ?

- a) Physiological effects on pilots because of fear of flying.
- b) An agreement between captain and co-pilot due to Crew Resources Management.
- c) Unjustified self-confidence.
- d) To question possible solutions.

Answer ;

- c) Unjustified self-confidence.
-

0703. What is self-concept ?

- a) How you see yourself.
- b) How other people see you.
- c) How you would like to see yourself.
- d) How you would like to be seen by other people.

Answer ;

- a) How you see yourself.
-

0704. What is synergy in a crew ?

- a) The uncoordinated action of the crewmembers towards a common objective.
- b) A behavioural expedient associated with the desynchronisation of the coordinated actions.
- c) The coordinated action of unrelated individual performances in achieving a non-standard task.
- d) The coordinated action of all members towards a common objective , in which collective performance is proving to be more than the sum of the individual performances.

Answer ;

- d) The coordinated action of all members towards a common objective , in which collective performance is proving to be more than the sum of the individual performances.
-

0705. What is the approximate Time of Useful Consciousness for a seated pilot following a rapid decompression at 35.000 ft ?

- a) 3 seconds.
- b) 45 seconds.
- c) 12 seconds.
- d) 5 minutes.

Answer ;

- b) 45 seconds.
-

0706. What is the correct action to counteract vertigo ?

- a) Rely on the information from the semicircular canals , since these will be transmitting correct information.
- b) Believe the instruments.
- c) Concentrate on the vertical speed indicator.
- d) Reduce rate of breathing until all symptoms disappear , then breathe again normally.

Answer ;

- b) Believe the instruments
-

0707. What is the current approach to human error ?

- a) Realisation that humans are fallible and that systems and procedures should be brought into line to prevent system , latent and procedural errors.
- b) Realisation that humans are infallible and that systems/procedures should be designated to minimise human error.
- c) Realisation that humans are fallible and that systems and procedures should be designated to minimise human error.
- d) Realisation that humans are infallible and that systems and procedures should be brought into line to prevent system/procedural errors.

Answer ;

- c) Realisation that humans are fallible and that systems and procedures should be designated to minimise human error.
-

0708. What is the effect of increasing altitude on the total pressure and partial pressure of the main gases in the atmosphere ?

- a) The total pressure will decrease and the partial pressures will decrease.
- b) The total pressure will decrease and the partial pressures will remain constant.
- c) The total pressure will decrease and the partial pressures will increase.
- d) The total pressure remains constant and the partial pressures will decrease.

Answer ;

- a) The total pressure will decrease and the partial pressures will decrease.
-

0709. What is the effect of stress on performance ?

- 1. It always reduces performance.
- 2. Optimum performance is obtained with optimum arousal.
- 3. Excessive stress weakens performance.
- 4. Insufficient stress weakens performance.

The combination of correct statements is ;

- a) 1 , 3 , 4.
- b) 1 , 2 , 4.
- c) 1 , 2 , 3.
- d) 2 , 3 , 4.

Answer ;

- d) 2 , 3 , 4.
-

0710. What is the effect of tiredness on attention ?

- a) It leads to one's attention being shared between different centres of interest.
- b) It increases the ability to manage multiple matters.
- c) It reduces the ability to manage multiple matters.
- d) It has no specific effects on attention.

Answer ;

- c) It reduces the ability to manage multiple matters.
-

0711. What is the main adverse effect of expectation with regards to perception ?

- a) Expectation often guides the focus of attention towards a particular aspect , while possible alternates are neglected.
- b) Expectation is a tool that allows capture of all relevant information.
- c) Expectations always lead to routine errors.
- d) The attention area is enlarged by expectation , however it will therefore lead to uncertainties when a pilot is looking for a decision.

Answer ;

- a) Expectation often guides the focus of attention towards a particular aspect , while possible alternates are neglected.
-

0712. What is the main problem caused by positive (+Gz) accelerations ?

- a) A pooling of blood in the lower portions of the body , and hence less blood available.
- b) An increase in blood pressure in the upper part of the body (above heart-level).
- c) Hyperoxygenation of the blood which may lead to sensory disorders.
- d) An improvement of peripheral vision.

Answer ;

- a) A pooling of blood in the lower portions of the body , and hence less blood available.
-

0713. What is the maximum JAA blood/alcohol limit for flight and cabin crew ?

- a) 40 milligrams per 100 milliliters of blood.
- b) 10 milligrams per 100 milliliters of blood.
- c) 80 milligrams per 100 milliliters of blood.
- d) 20 milligrams per 100 milliliters of blood.

Answer ;

- d) 20 milligrams per 100 milliliters of blood.
-

0714. What is the name of the functional connection between neurones ?

- a) Interconnection.
- b) Occlusion.
- c) Synapse.
- d) By-pass.

Answer ;

- c) Synapse.
-

0715. What is the procedure above 10.000 ft altitude when faced with explosive decompression ?

- a) Don an oxygen mask and descend to below 10.000 ft.
- b) First inform ATC.
- c) Descend to below 10.000 ft and signal an emergency.
- d) Check the cabin altitude , don an oxygen mask and maintain level flight.

Answer ;

- a) Don an oxygen mask and descend to below 10.000 ft.
-

0716. What is the relationship between the perceived level of arousal and the difficulty of a task to be completed ?

- a) All tasks should be completed at the highest arousal regardless of difficulty.
- b) Over-arousal is required to complete difficult tasks.
- c) The arousal level should co-incide with the individual's "Break Point" where the task is particularly difficult.
- d) At the outset , the level of arousal should be within the optimum range and therefore at a level where performance does not suffer.

Answer ;

- d) At the outset , the level of arousal should be within the optimum range and therefore at a level where performance does not suffer.
-

0717. What is the sender's frequent reason to communicate implicitly (between the lines) ?
- a) The receiver grasps quickly what the sender means.
 - b) He/she has not to adjust to the communication style of the communication partner.
 - c) There is no need to make up one's mind before starting to communicate.
 - d) Afterwards he/she always can claim to have been misunderstood.

Answer ;

- d) Afterwards he/she always can claim to have been misunderstood.
-

0718. What is the Time of Useful Consciousness ?
- a) The time taken to become aware of hypoxia due to gradual decompression.
 - b) The length of time during which an individual can act with both mental and physical efficiency , measured from the moment at which he/she loses his/her available oxygen supply.
 - c) The period of time between the start of hypoxia and the moment that the pilot becomes aware of it.
 - d) The pilot's reaction time when faced with hypoxia.

Answer ;

- b) The length of time during which an individual can act with both mental and physical efficiency , measured from the moment at which he/she loses his/her available oxygen supply.
-

0719. What is the "Time of Useful Consciousness" for a rapid decompression at 25.000 ft ?
- a) Between 3 and 5 minutes depending on the physical activities of the subjected pilot.
 - b) About 18 seconds.
 - c) About 30 seconds.
 - d) Between 25 seconds and 1 minute 30 seconds.

Answer ;

- a) Between 3 and 5 minutes depending on the physical activities of the subjected pilot.
-

0720. What is understood by air-sickness ?
- a) An illness caused by an infection of the middle ear.
 - b) An illness caused by reduced air pressure.
 - c) An illness caused by evaporation of gases in the blood.
 - d) A sensory conflict within the vestibular system accompanied by nausea and vomiting.

Answer ;

- d) A sensory conflict within the vestibular system accompanied by nausea and vomiting.
-

0721. What law(s) governs the oxygen transfer at the alveoli ?
- a) Gas Diffusion Laws.
 - b) Henry's.
 - c) Boyle's.
 - d) Charles's.

Answer ;

- a) Gas Diffusion Laws.
-

0722. What may become the main risk of a "laissez-faire" cockpit ?
- a) Disengagement of the co-pilot.
 - b) Inversion of authority.
 - c) Increased captain's authority.
 - d) Appearance of aggressiveness.

Answer ;

- b) Inversion of authority.
-

0723. What may trigger stress in humans ?
- a) Always the reaction to upsetting emotions caused either by physiological or psychological conditions.
 - b) The subjective interpretation an individual gives to a situation experienced.
 - c) Only objective stimulation from environmental factors.
 - d) Only strong stimulation of the sensory organs: a flash of light , noise and the smell of smoke.

Answer ;

- b) The subjective interpretation an individual gives to a situation experienced.
-

0724. What optimises crew co-operation ?

1. Sharing and common task.
 2. Confidence in each others capability.
 3. Precise definition of functions associated with each crew members role.
- a) 1.
b) 2 , 3.
c) 1 , 2 , 3.
d) 1 , 2.

Answer ;

c) 1 , 2 , 3.

0725. What physical symptoms can be attributed to stress ?

- a) Insomnia , loss of appetite and a dry mouth.
b) A false sense of achievement and a severe thirst.
c) A sense of inadequacy and irritability.
d) A sense of well-being and a loss of reality.

Answer ;

a) Insomnia , loss of appetite and a dry mouth.

0726. What role should automation play with respect to flight safety ?

- a) Automation should only be used on flights when weather conditions are poor to enhance flight safety.
b) Automation should be used as an aid to the pilot and not as an end itself.
c) Automation should be used where pilots have to make decisions and manual flying should be compulsory in good weather conditions.
d) Automation should be used as much as possible to replace the pilot who will inevitably make mistakes.

Answer ;

b) Automation should be used as an aid to the pilot and not as an end itself.

0727. What seems to be the main role of Orthodox sleep ?

- a) It includes physical and mental recuperation associate with fatigue.
b) It essentially allows for physical recovery.
c) Via physical recovery, it is characterised by an alternation of dream phases and paradoxical phases.
d) Its main role is associated with activities of memory activities and restoration of attention capabilities.

Answer ;

b) It essentially allows for physical recovery.

0728. What should a pilot do to optimise his/her night vision (scotopic vision) ?

- a) Avoid food containing high amounts of vitamin A.
b) Select meals with high contents of vitamin B and C.
c) Wait at least 60 minutes to night-adapt before he takes off.
d) Not smoke before start and during flight and avoid flash-blindness.

Answer ;

d) Not smoke before start and during flight and avoid flash-blindness.

0729. What strategies can be used to combat human error ?

1. Reducing error-prone mechanisms.
2. Improving the way in which error is taken into account in training.
3. Sanctions against the initiators of error.
4. Improving recovery from errors and its consequences.

The combination of correct statements is ;

- a) 2 , 3 and 4.
b) 1 , 2 and 4.
c) 1 and 2.
d) 3 and 4.

Answer ;

b) 1 , 2 and 4.

0730. What strategy should be put in place when faced with an anticipated period of stress ?

- a) A strategy of preparing decisions.
b) A Laissez-faire strategy.
c) A strategy of no commitment.
d) A non-sequenced strategy.

Answer ;

a) A strategy of preparing decisions.

0731. What would be the effect if , in a tight turn , one bends down to pick up a pencil ?

- a) Coriolis effect.
- b) Barotrauma.
- c) Vertigo.
- d) Inversion illusion.

Answer ;

a) Coriolis effect.

0732. When a pilot is facing a problem during flight he should ;

- a) always make up his mind quickly to give himself as much spare time as possible.
- b) avoid making up his mind until the very last minute.
- c) take as much time as he needs and is available to make up his mind.
- d) make up his mind before consulting other crew members.

Answer ;

c) take as much time as he needs and is available to make up his mind.

0733. When a pilot is staring at an isolated stationary light for several seconds in the dark he might get the illusion that ;

- a) the light is moving.
- b) the colour of the light is varying.
- c) the size of the light is varying.
- d) the intensity of the light is varying.

Answer ;

a) the light is moving.

0734. When a pilot suffers from hypothermia (e.g. after loss of cabin heating) his/her demand for oxygen will be ;

- a) initially increased.
- b) raised leading to an increased tolerance to hypoxia.
- c) reduced giving a better tolerance to hypoxia at higher altitudes.
- d) unaffected

Answer ;

a) initially increased.

0735. When accelerating forward the otoliths in the utricle/sacculus will ;

- a) give the illusion of descending (body tilting downwards , or forwards , nose of the aircraft going down).
- b) give the illusion of climbing (body tilting backwards , nose of the a/c going up).
- c) give the illusion of banking.
- d) give the illusion of straight and level flight.

Answer ;

b) give the illusion of climbing (body tilting backwards , nose of the a/c going up).

0736. When accelerating in level flight we could experience the sensation of a ;

- a) spin.
- b) turn.
- c) descent.
- d) climb.

Answer ;

d) climb.

0737. When assessing an individual's risk of developing coronary artery disease , the following factors may contribute ;

- 1. obesity.
 - 2. chronic stress.
 - 3. smoking.
 - 4. family history.
- a) 1 , 2 and 3 are correct , 4 is false.
 - b) 2 and 3 are correct , 1 and 4 are false.
 - c) Only 3 is correct , 1, 2 and 4 are false.
 - d) 1 , 2 , 3 and 4 are correct.

Answer ;

d) 1 , 2 , 3 and 4 are correct.

0738. When can a system be said to be tolerant to error ? When ;
- a) the consequences of an error will not seriously jeopardise safety.
 - b) its safety system is too permeable to error.
 - c) latent errors do not entail serious consequences for safety.
 - d) its safety system has taken account of all statistically probable errors.

Answer ;

- a) the consequences of an error will not seriously jeopardise safety.
-

0739. When can the General Adaption Syndrome activate ?
- a) Responding to Circadian Disrhythmia.
 - b) Responding to an imaginary threat.
 - c) Responding to boredom.
 - d) Responding to Hypovigilance.

Answer ;

- b) Responding to an imaginary threat.
-

0740. When considering the eating-habits of pilots ;
- a) low fibre and low protein/carbohydratediet is ideal.
 - b) breakfast should bring about 25% of the daily calorie intake.
 - c) high intake of chocolate is recommended before flight to prevent reactional hypoglycemia.
 - d) a heavy supper is best.

Answer ;

- b) breakfast should bring about 25% of the daily calorie intake.
-

0741. When exhaling , the expired air contains ;
- a) more oxygen than the inhaled air.
 - b) more carbon dioxide than the inspired air.
 - c) less water vapour than the inhaled air.
 - d) more nitrogen than the inhaled air.

Answer ;

- b) more carbon dioxide than the inspired air.
-

0742. When faced with sustained cold temperature , how does the body resist this physical stress ?
- a) By intense vasoconstriction.
 - b) By increasing cardiac frequency.
 - c) By speeding up the metabolic rate in the Autonomic Nervous System.
 - d) By vasodilatation which permits a greater flow of blood to the periphery.

Answer ;

- a) By intense vasoconstriction.
-

0743. When flying above 10.000 feet hypoxia arises because ;
- a) the partial oxygen pressure is lower than at sea level.
 - b) the composition of the air is different from sea level
 - c) the percentage of oxygen is lower than at sea level
 - d) the composition of the blood changes

Answer ;

- a) the partial oxygen pressure is lower than at sea level.
-

0744. When flying at night the first sense to be affected by a slight degree of hypoxia is the ;
- a) proprioceptive sensitivity.
 - b) vision.
 - c) sense of balance.
 - d) cochlea.

Answer ;

- b) vision.
-

0745. When flying through a thunderstorm with lightning you can protect yourself from flash blindness by ;
- a. turning up the intensity of cockpit lights.
 - b. looking inside the cockpit.
 - c. wearing sunglasses.
 - d. using blinds or curtains when installed.
- a) a , b , c and d are correct.
b) a , b and c are correct , d is false
c) c and d are correct , a and b are false
d) a and b are correct , c and d are false

Answer ;

a) a , b , c and d are correct.

0746. When focussing on near objects ;
- a) the shape of lens gets flatter.
 - b) the pupil gets larger.
 - c) the cornea gets smaller.
 - d) the shape of lens gets more spherical.

Answer ;

d) the shape of lens gets more spherical.

0747. When hyperventilating you should ;
- a) apply the Valsalva method.
 - b) descend.
 - c) use the oxygen mask.
 - d) control your rate and depth of breathing.

Answer ;

d) control your rate and depth of breathing.

0748. When might a pilot have the sensation of a nose-low attitude ?
- a) When decelerating in straight , horizontal flight.
 - b) During linear acceleration.
 - c) When turning (angular acceleration).
 - d) When climbing at greater than 2.000 ft / min.

Answer ;

a) When decelerating in straight , horizontal flight.

0749. When problem-solving , what determines the transition from rule-based activities to knowledge-based activities ?
- a) Unsuitability of known rules for the problem posed.
 - b) There is never a transition from Rule-based and Knowledge-based behaviour.
 - c) Knowledge of rules that apply for the problem.
 - d) Unsuitability of automated actions.

Answer ;

a) Unsuitability of known rules for the problem posed.

0750. When scanning , how does the eye move and how is smooth vision achieved ?
- a) The eyes move in shory jerky movements which last for several seconds. Smooth vision is achieved by the cerebellum.
 - b) The eyes move in shory jerky movements called sarades. Smooth vision is created by the cerebellum.
 - c) The eyes move in shory jerky movements which last for approximately 1/3 of a second. Smooth vision is achieved by the optic nerve centre.
 - d) The eyes move in shory jerky movements called saccades. Smooth vision is achieved in the visual cortex.

Answer ;

d) The eyes move in shory jerky movements called saccades. Smooth vision is achieved in the visual cortex.

0751. When stopping the rotation of a spin we have the sensation ;
- a) that we are starting a spin in the opposite direction.
 - b) of the sharp dipping of the nose of the aircraft.
 - c) of turning in the same direction.
 - d) of the immediate stabilization of the aircraft.

Answer ;

a) that we are starting a spin in the opposite direction.

0752. When suffering from Hypoxia short-term memory impairment starts at approximately at ;

- a) 12.000 ft.
- b) 18.000 ft.
- c) 10.000 ft.
- d) 25.000 ft.

Answer ;

a) 12.000 ft.

0753. When the optical image forms in front of the retina , this results in ;

- a) hypermetropia.
- b) astigmatism.
- c) myopia.
- d) presbyopia.

Answer ;

c) myopia.

0754. When the pressoreceptors signal a lowering of the blood-pressure there are adaptation mechanisms which result in ;

- 1. an increase of respiratory activity.
 - 2. the arteriols to constrict.
 - 3. an increase of cardiac output.
 - 4. the heart rate to rise.
- a) 1 , 2 and 3 are correct , 4 is false.
 - b) 1 , 2 and 4 are correct , 3 is false.
 - c) 1 , 3 and 4 are correct , 2 is false.
 - d) 2 , 3 and 4 are correct , 1 is false.

Answer ;

d) 2 , 3 and 4 are correct , 1 is false.

0755. When the weather is foggy , on approach , a pilot may get a feeling of ;

- a) his/her eyes focusing on infinity.
- b) aircraft altitude being lower than it actually is.
- c) the airfield being further away than actually is.
- d) his/her visual field shrinking (i.e. tunnel vision).

Answer ;

c) the airfield being further away than actually is.

0756. When turning in IMC , head movements should be avoided as much as possible. This is a prevention against ;

- a) pressure vertigo.
- b) autokinesis.
- c) oculogyral illusion.
- d) coriolis illusion.

Answer ;

d) coriolis illusion.

0757. When using the Body Mass Index (BMI) formula - when is a pilot considered as obese ?

- a) Over 20 for males , 19 for females.
- b) Over 30 for males , 29 for females.
- c) Over 10 for males , 9 for females.
- d) Over 26 for males , 25 for females.

Answer ;

b) Over 30 for males , 29 for females.

0758. When you stare at a single light against the dark (i.e. an isolated star) you will find the light appears to move after some time. This phenomenon is called ;

- a) black hole illusion.
- b) leans.
- c) coriolis illusion.
- d) autokinetic phenomenon.

Answer ;

d) autokinetic phenomenon.

0759. Where on the arousal/performance graph is a pilot best placed to cope with a difficult task without performance degrading ?

- a) below the “breakpoint”.
- b) beyond the “breakpoint”.
- c) at the “breakpoint”.
- d) at any point on the graph.

Answer ;

a) below the “breakpoint”.

0760. Which behaviour is most likely to promote a constructive solution of interpersonal conflicts ?

- a) Responding with counter-arguments.
- b) Staying to the own point of view.
- c) Active listening.
- d) Giving up the own point of view.

Answer ;

c) Active listening.

0761. Which biases relate to human decision making ?

- 1. Personal experience tends to alter the perception of the risk of an event occurring.
 - 2. There is a natural tendency to want to confirm our decision even in the face of facts which contradict it.
 - 3. The group to which an individual belongs tends to influence the particular decision.
 - 4. There is natural tendency to select only objective facts for decision-making purposes.
- a) 1 , 2.
 - b) 1 , 2 , 3.
 - c) 1 , 2 , 4.
 - d) 3 , 4.

Answer ;

b) 1 , 2 , 3.

0762. Which counter-measure can a pilot can use against a barotrauma of the middle ear ?

- a) Stop chewing and swallowing movements.
- b) Use drugs against a cold.
- c) Stop descending , climb again and then descend with reduced sink rate.
- d) Increase the rate of descent.

Answer ;

c) Stop descending , climb again and then descend with reduced sink rate.

0763. Which flight-manoeuvre will most likely induce vertigo ? Turning the head while ;

- a) flying straight and level.
- b) banking.
- c) descending.
- d) climbing.

Answer ;

b) banking.

0764. Which force(s) affect(s) the otoliths in the utricle and saccule ?

- a) Angular acceleration.
- b) Linear acceleration and angular acceleration.
- c) Gravity and linear acceleration.
- d) Gravity alone.

Answer ;

c) Gravity and linear acceleration.

0765. Which gas most readily combines with haemoglobin ?

- a) carbon monoxide.
- b) oxygen.
- c) carbon dioxide.
- d) nitrogen.

Answer ;

a) carbon monoxide.

0766. Which is correct ?

1. Paradoxical (REM) sleep refreshes the brain , memory and body.
 2. Paradoxical (REM) sleep decreases during the night.
- a) 1 only.
b) 2 only.
c) Neither.
d) 1 & 2.

Answer ;

c) Neither.

0767. Which is the audible range to human hearing ?

- a) Between 20 KHz and 20 KHz.
b) Between 20 Hz and 20 MHz.
c) Between 20 MHz and 20.000MHz.
d) Between 20 Hz and 20 KHz.

Answer ;

d) Between 20 Hz and 20 KHz.

0768. Which is the most likely of the following scenarios which might lead to a conflict between status and role ?

- a) A Manager interviewing an experienced Captain of an aircraft.
b) A trainee Air Traffic Officer working with a senior ATC Controller.
c) Two Training Captains flying together.
d) A security officer temporarily standing in for a member of the cabin crew.

Answer ;

c) Two Training Captains flying together.

0769. Which is the procedure to be followed when symptoms of decompression sickness occur ?

- a) Only medical treatment is of use.
b) Only the prompt supply of oxygen is necessary.
c) Descend to the lowest possible level and land as soon as possible.
d) Descend to the lowest possible level and wait for the symptoms to disappear before climbing again.

Answer ;

c) Descend to the lowest possible level and land as soon as possible.

0770. Which measure(s) will help to compensate for hypoxia ?

1. Descend below 10.000 ft.
 2. Breathe 100 % oxygen.
 3. Climb to or above 10.000 ft.
 4. Reduce physical activities.
- a) 1 , 2 and 4 are correct.
b) 1 , 2 and 3 are correct.
c) 1 and 2 are correct , 3 and 4 are false.
d) only 1 is correct.

Answer ;

a) 1 , 2 and 4 are correct.

0771. Which of the following answers are correct concerning the Long Term Memory ?

1. Long Term memory can be affected by expectation and suggestion.
 2. Long Term Memory is also known as the Working Memory.
 3. Long Term Memory is limited to plus or minus 7 pieces of disassociated pieces of information.
 4. Long Term Memory is where Motor Programmes (skills) are kept.
- a) 1 only is correct.
b) 1 , 2 , 3 and 4 are all correct.
c) 1 and 4 only are correct.
d) 2 and 3 only are correct.

Answer ;

a) 1 only is correct.

0772. Which of the following answers is correct ?

- a) Personality is the outward result of behaviour and attitude and is not adaptable.
b) Behaviour is the outward result of personality and attitude and is adaptable.
c) Personality is the outward result of behaviour and attitude and is adaptable.
d) Behaviour is the outward result of personality and attitude and is not adaptable.

Answer ;

b) Behaviour is the outward result of personality and attitude and is adaptable.

0773. Which of the following applies to carbon monoxide poisoning ?

- a) Inhaling carbon monoxide leads to hyperventilation.
- b) Several days are needed to recuperate from a carbon monoxide poisoning.
- c) A very early symptom of carbon monoxide poisoning is euphoria.
- d) The human body shows no sign of carbon monoxide poisoning.

Answer ;

- b)** Several days are needed to recuperate from a carbon monoxide poisoning.
-

0774. Which of the following applies when alcohol has been consumed ?

- a) Acute effects of alcohol cease immediately when 100% oxygen is taken.
- b) Drinking coffee at the same time will increase the elimination rate of alcohol.
- c) Even after the consumption of small amounts of alcohol , normal cautionary attitudes may be lost.
- d) Small amounts of alcohol increase visual performance.

Answer ;

- c)** Even after the consumption of small amounts of alcohol , normal cautionary attitudes may be lost.
-

0775. Which of the following are either cumulative or tend to escalate ?

- 1. Stress factors (stressors).
 - 2. Errors.
 - 3. The effects of carbon monoxide poisoning.
 - 4. Human conflict.
 - 5. Colour blindness.
- a) 1 , 2 , 3 and 4.
 - b) 1 , 2 and 3 only.
 - c) 1 and 2 only.
 - d) 2 , 3 and 5 only.

Answer ;

- a)** 1 , 2 , 3 and 4.
-

0776. Which of the following are included in Maslow's Hierarchy of Needs ?

- 1. Freedom from pain and danger.
 - 2. Expressions of capacities and talents.
 - 3. Self-esteem needs.
 - 4. Self-fulfilment needs.
 - 5. Pysiological needs.
- a) 3 , 4 , and 5 only.
 - b) all except 2.
 - c) 1 , 2 and 4 only.
 - d) all the above.

Answer ;

- d)** all the above.
-

0777. Which of the following are strategies for resolving conflict ?

- 1. Seeking arbitration.
 - 2. Actively listening to other people.
 - 3. Abandoning facts so as to move the conversation to a more emotional level.
 - 4. Becoming aware of cultural influences.
- a) 2 , 4.
 - b) 2 , 3 , 4.
 - c) 1 , 2 , 3.
 - d) 1 , 2 , 4.

Answer ;

- d)** 1 , 2 , 4.
-

0778. Which of the following are the most favourable solutions to manage phases of reduced or low vigilance (hypovigilance) ?

- 1. Healthy living.
 - 2. Use of amphetamines.
 - 3. Reducing the intensity of the light.
 - 4. Organising periods of rest during the flight.
- a) 1 , 4.
 - b) 1 , 2.
 - c) 1 , 3.
 - d) 3 , 4.

Answer ;

- a)** 1 , 4.
-

0779. Which of the following are the most favourable solutions to manage phases of reduced or low vigilance (hypovigilance) ?
1. Keep active open communication between man and machine.
 2. Use of amphetamines.
 3. Reducing the intensity of the light.
 4. Organising periods of rest during the flight.
- a) 1 , 3.
b) 3 , 4.
c) 1 , 2.
d) 1 , 4.

Answer ;

d) 1 , 4.

0780. Which of the following characteristics apply to short-term memory ?

1. It is limited in time and size.
 2. It is unlimited in time and limited in size.
 3. It is stable and insensitive to disturbances.
 4. It is limited in time and unlimited in size.
- a) 2 , 3.
b) 1 , 3.
c) 3 , 4.
d) 1.

Answer ;

d) 1.

0781. Which of the following components belong to the middle ear ?

- a) Endolymph.
b) Ossicles.
c) Semicircular canals.
d) Otoliths.

Answer ;

b) Ossicles.

0782. Which of the following concepts relating to human reliability is true ?

- a) Response to a particular stressful influence does not vary from one person to another.
b) Performance is totally independent of motivation.
c) Expectation has no influence on perception.
d) If equipment is designed in such a way that it can be operated wrongly , then sooner or later , it will be.

Answer ;

d) If equipment is designed in such a way that it can be operated wrongly , then sooner or later , it will be.

0783. Which of the following could a pilot experience when he is hyperventilating ?

1. Dizziness.
 2. Muscular spasms.
 3. Visual disturbances.
 4. Cyanosis.
- a) 1 is false , all others are correct.
b) 2 and 4 are false.
c) 1 , 2 and 3 are correct , 4 is false.
d) 1 , 2 and 4 are correct , 3 is false.

Answer ;

c) 1 , 2 and 3 are correct , 4 is false.

0784. Which of the following describes an autocratic cockpit ?

- a) Each of the members chooses what job to do without telling the others and in the belief that everyone is aware of what he is doing.
b) The atmosphere is relaxed thanks to a captain who leaves complete freedom to the various members of the crew.
c) Despite the overly strong authority of the captain , everything functions correctly owing to his natural leadership.
d) The captain's excessive authority considerably reduces communications and consequently the synergy and cohesion of the crew.

Answer ;

d) The captain's excessive authority considerably reduces communications and consequently the synergy and cohesion of the crew.

0785. Which of the following drawbacks are associated with automation ?

1. Reduced competence in manually controlling the aircraft.
 2. Increased likelihood of slips while programming automatic systems.
 3. Difficulties in adapting to the use of a sidestick.
 4. General decrease in technical reliability.
- a) 1 , 2.
b) 2 , 3 , 4.
c) 1 , 3.
d) 1 , 4.

Answer ;

a) 1 , 2.

0786. Which of the following elements make up the personality of an individual ?

1. Heredity.
 2. Childhood environment.
 3. Upbringing.
 4. Past experience.
- a) 2 , 3.
b) 1 , 2 , 3 , 4.
c) 2 , 3 , 4.
d) 1 , 2 , 4.

Answer ;

b) 1 , 2 , 3 , 4.

0787. Which of the following gases is fundamentally responsible for decompression sickness ?

- a) Oxygen.
b) Carbon dioxide.
c) Sodium.
d) Nitrogen.

Answer ;

d) Nitrogen.

0788. Which of the following human error rates can be described as pretty good realistic and pretty good , after methodical training ;

- a) 1 in 100.000 times.
b) 1 in 1.000 times.
c) 1 in 100 times.
d) 1 in 10.000 times.

Answer ;

b) 1 in 1.000 times.

0789. Which of the following illusions are brought about by conflicts between the visual system and the vestibular system ?

1. Illusions concerning the attitude of the aircraft.
 2. Autokinetic illusion (fixed point viewed as moving).
 3. Illusions when estimating the size and distance of objects.
 4. Illusions of rotation.
- a) 3 , 4.
b) 1 , 4.
c) 2 , 3 , 4.
d) 2.

Answer ;

b) 1 , 4.

0790. Which of the following is an example of human underload ?

- a) A poor landing at an airfield with which the pilot is unfamiliar , in marginal weather conditions.
b) An unexpectedly poor landing at an airfield with which the pilot is familiar in perfect weather conditions.
c) A good landing at an airfield with which the pilot is familiar in poor weather conditions.
d) An unexpectedly good landing at an airfield with which the pilot is unfamiliar in perfect weather conditions.

Answer ;

b) An unexpectedly poor landing at an airfield with which the pilot is familiar in perfect weather conditions.

0791. Which of the following is correct ?

- a) 70%-80% of information processed by man enters the visual channel.
- b) The Kinesthetic channel is the most important channel in flying.
- c) Hearing is the most important sense for man.
- d) 40% of information processed by man enters the visual channel.

Answer ;

- a) 70%-80% of information processed by man enters the visual channel.
-

0792. Which of the following is most likely to be overlooked should a pilot make a rushed decision ?

- a) analysis of the current actual situation and instead applying a decision prepared beforehand.
- b) the skills of air traffic controllers.
- c) the captain's superior knowledge , justified by his/her status.
- d) the need to take account of every possible result or outcome.

Answer ;

- a) analysis of the current actual situation and instead applying a decision prepared beforehand.
-

0793. Which of the following is most true ?

- a) Regular exercise and reduction in calorie consumption are both essential in order to lose weight.
- b) Regular exercise is beneficial to general health , but the most efficient way to lose weight is by reducing calorie consumption.
- c) Regular exercise is beneficial to general health , and is the only effective way to lose weight.
- d) Regular exercise is an impediment to losing weight since it increases the metabolic rate.

Answer ;

- b) Regular exercise is beneficial to general health , but the most efficient way to lose weight is by reducing calorie consumption.
-

0794. Which of the following is the best definition of episodic memory ?

- a) It is the memory of events which are held in the short-term memory and can be influenced by suggestion.
- b) It is the memory of events which are held in the long-term memory and can be influenced by suggestion.
- c) It is the memory of events which are held in the long-term memory and can be influenced by the meaning of words and language.
- d) It is the memory of events which are held in the short-term memory and can be influenced by the meaning of words and language.

Answer ;

- b) It is the memory of events which are held in the long-term memory and can be influenced by suggestion.
-

0795. Which of the following is true concerning carbon monoxide ?

- a) It is always present in the lungs.
- b) It has no physiological effect when mixed with oxygen.
- c) It is to be found in the smoke of cigarettes lifting up a smoker's "physiological altitude".
- d) It combines 5 times faster to the haemoglobin than oxygen.

Answer ;

- c) It is to be found in the smoke of cigarettes lifting up a smoker's "physiological altitude".
-

0796. Which of the following is true with reference to semantic memory ?

- a) It is the meaning of words and general knowledge of the world , it lasts for a longer period of time and tends to be more accurate than episodic memory.
- b) It is memory of events , it lasts for a shorter period of time than episodic memory and tends to be less accurate.
- c) It is the meaning of words and general knowledge of the world , it lasts for a shorter period of time than episodic memory and tends to be less accurate.
- d) It is memory of events , it lasts for a longer period of time than episodic memory and tends to be more accurate.

Answer ;

- a) It is the meaning of words and general knowledge of the world , it lasts for a longer period of time and tends to be more accurate than episodic memory.
-

0797. Which of the following is true with regards to sleep ?

1. metabolic rate falls.
 2. arterial blood pressure falls.
 3. the pulse rate decreases.
 4. the sensitivity level of the senses is increased.
- a) 1, 3 and 4 only are correct.
b) 1 and 4 only are correct.
c) 1, 2, 3 and 4 are correct.
d) 1, 2 and 3 only are correct.

Answer ;

d) 1, 2 and 3 only are correct.

0798. Which of the following is/are the symptom(s) of hypoxia ?

- a) Lack of concentration , fatigue , euphoria.
b) Pain in the joints.
c) Low blood pressure.
d) Excessive rate and depth of breathing combined with pains in the chest area.

Answer ;

a) Lack of concentration , fatigue , euphoria.

0799. Which of the following laws explains bubbles of nitrogen coming out of solution in body tissues due to a decrease in atmospheric pressure ?

- a) Gay Lussac's law.
b) Henry's law.
c) Boyle's law.
d) Dalton's law.

Answer ;

b) Henry's law.

0800. Which of the following list are symptoms of fatigue ?

1. Diminished accommodation.
 2. Slowed reactions.
 3. Long-term memory access problems.
 4. Being over-talkative.
 5. Diminished motor skills.
- a) 1, 2 & 4.
b) 1, 2, 3 & 5.
c) 2 & 5 only.
d) All of the above.

Answer ;

b) 1, 2, 3 & 5.

0801. Which of the following list are symptoms of fatigue ?

1. Tiredness.
 2. Faster reactions.
 3. Diminished motor skills.
 4. Tunneled concentration.
 5. Increased long term memory capacity.
- a) 1 and 3 only.
b) 1, 2, 3, 4 and 5.
c) 3, 4 and 5 only.
d) 1, 3 and 4 only.

Answer ;

d) 1, 3 and 4 only.

0802. Which of the following list can cause fatigue ?

1. Sleep loss.
 2. Jet lag.
 3. Boredom.
 4. Low levels of external stimulation.
- a) 1, 3 and 4 only.
b) 1, 2 and 4 only.
c) 1 and 3 only.
d) 1, 2, 3 and 4.

Answer ;

d) 1, 2, 3 and 4.

0803. Which of the following list may stress effect ?

1. Attention.
 2. Concentration.
 3. Memory.
 4. Judgment.
- a) 1, 2, 3 and 4 are correct.
b) 1 and 2 are correct.
c) 2, 3 and 4 are correct.
d) 1 and 3 are correct.

Answer ;

- a) 1, 2, 3 and 4 are correct.
-

0804. Which of the following list of statements concerning stress are correct ?

1. Stress is an inevitable part of human life and in small amounts is necessary to achieve optimum performance.
 2. Each individual has a stress limit which, if exceeded, could result in inability to handle even a moderate workload.
 3. Anxiety creates worry which may lead to stress.
 4. Stress is often perceived by the brain as some form of threat which may cause adrenaline to be released into the blood stream.
- a) 1, 3 & 4 only.
b) 1, 2, 3 & 4.
c) 1, 2 & 3 only.
d) 2, 3 & 4 only.

Answer ;

- b) 1, 2, 3 & 4.
-

0805. Which of the following mechanisms regulate body temperature when exposed to extreme high environmental temperatures ?

1. Shivering.
 2. Vaso-constriction of peripheral blood vessels.
 3. Sweating.
 4. Vaso-dilation of peripheral blood vessels.
- a) 1.
b) 3, 4.
c) 2, 3.
d) 1, 3, 4.

Answer ;

- b) 3, 4.
-

0806. Which of the following occurs in man if the internal body temperature increases to 39°C ?

- a) Nothing significant happens at this temperature. The first clinical signs only start to appear at 39°C.
b) Impairment of physical and mental performance.
c) Considerable dehydration.
d) Apathy.

Answer ;

- b) Impairment of physical and mental performance.
-

0807. Which of the following operations are performed more effectively by automatic systems than by people ?

1. Waiting for an infrequent phenomenon.
 2. Long term controlling of a set value (e.g holding of trajectory).
 3. Monitoring to ensure that certain values are not exceeded (e.g. holding of flightpath).
 4. Qualitative decision-making.
- a) 1, 2, 3.
b) 2, 3, 4.
c) 3, 4.
d) 2, 4.

Answer ;

- a) 1, 2, 3.
-

0808. Which of the following operations are performed more effectively by people than by automatic systems ?

1. Qualitative decision-making.
 2. Waiting for an infrequent phenomenon.
 3. Monitoring to ensure that certain values are not exceeded.
 4. Detection of unusual conditions (smell , noise , etc.).
- a) 3 , 4.
b) 2 , 3 , 4.
c) 1 , 4.
d) 1 , 2.

Answer ;

c) 1 , 4.

0809. Which of the following physical stimuli may cause stress reactions ?

1. noise.
2. conflict.
3. temperature.
4. an administrative problem.
5. hunger.

The combination of correct statements is ;

- a) 3 , 4 , 5.
b) 1 , 3 , 4.
c) 1 , 3 , 5.
d) 2 , 3 , 5.

Answer ;

c) 1 , 3 , 5.

0810. Which of the following provides the basis of all perceptions ?

- a) The aural or visual significance attributed in long term memory.
b) The intensity of the stimuli.
c) The separation of figure and background.
d) The aural or visual significance attributed in short term memory.

Answer ;

b) The intensity of the stimuli.

0811. Which of the following responses is an example of “habit reversion” (negative habit transfer) ;

- a) A pilot who has flown many hours in an aircraft in which the fuel lever points forward for the ON position , may unintentionally turn the fuel lever into the false position , when flying a different aircraft , where the fuel lever has to point aft to be in the ON position.
b) Incorrect anticipation of an air traffic controller’s instructions.
c) Turning the aircraft to the left when intending to turn it to the right.
d) Habitually missing an item on the checklist or missing the second item when two items are on the same line.

Answer ;

a) A pilot who has flown many hours in an aircraft in which the fuel lever points forward for the ON position , may unintentionally turn the fuel lever into the false position , when flying a different aircraft , where the fuel lever has to point aft to be in the ON position.

0812. Which of the following statements about hyperthermia is correct ?

- a) Vasodilatation is the only mechanism which is capable of reducing body temperature.
b) Performance is not impaired by an increase in body temperature to 40°C or more.
c) Evaporation is more effective when ambient humidity is high.
d) Complete adaptation to the heat in a hot country takes about a fortnight.

Answer ;

d) Complete adaptation to the heat in a hot country takes about a fortnight.

0813. Which of the following statements about long-term memory are correct ?

1. Information is stored in the Semantic, Episodic and Procedural memories.
2. The period of time for which information is retained is limited by the frequency with which this same information is used.
3. It processes information quickly and has an effective mode of access in real time.
4. Ease and speed of access is dependent upon , to a large extent , the frequency with which the information is recalled.

- a) 2 and 4 are correct.
- b) 1 and 2 are correct.
- c) 1 and 4 are correct.
- d) 2 , 3 and 4 are correct.

Answer ;

c) 1 and 4 are correct.

0814. Which of the following statements are correct ?

1. Hypothermia affects physical and mental abilities.
2. Man has effective natural protection against intense cold.
3. Shivering makes it possible to combat the cold to a certain extent , but uses up a lot of energy.
4. Disorders associated with hypothermia appear at a body temperature of less than 35°C.

- a) 2 , 4.
- b) 1 , 3 , 4.
- c) 2 , 3 , 4.
- d) 1 , 2 , 3.

Answer ;

b) 1 , 3 , 4.

0815. Which of the following statements are correct ?

1. Interesting information is easier to take into consideration for creating a mental picture than boring information.
2. The sequence in which information is presented is also important for the use the pilot makes of it.

- a) 1 and 2 are both not correct.
- b) 1 is correct , 2 is not correct.
- c) 1 is not correct , 2 is correct.
- d) 1 and 2 are both correct.

Answer ;

d) 1 and 2 are both correct.

0816. Which of the following statements are correct ?

1. Modern aircraft allow for 50 - 60% relative humidity in the cabin air under any conditions of flight , which is satisfactory for the body.
2. Thirst is a symptom of dehydration.
3. Dehydration may lead to clinical manifestations such as dizziness and fatigue.
4. Drinking excessive quantities of water must be avoided since resistance to periods of low hydration will otherwise be lost.

- a) 1 , 4.
- b) 1 , 2 , 4.
- c) 2 , 3 , 4.
- d) 2 , 3.

Answer ;

d) 2 , 3

0817. Which of the following statements are correct ;

1. Scuba diving may be practiced without restriction.
2. Many medicines have effects which are incompatible with flight safety.
3. An adequate amount of fluid should be drunk when flying.
4. Alcohol has no effect on the inner ear.

- a) 1 , 2 and 3 are correct.
- b) 2 and 3 are correct.
- c) 1 , 3 and 4 are correct.
- d) 2 , 3 and 4 are correct.

Answer ;

b) 2 and 3 are correct.

0818. Which of the following statements are correct ?

1. The first information received determines how subsequent information will be evaluated.
2. If one has made up one's mind , contradictory information may not get the attention it really needs.
3. With increasing stress , attention is limited thereby reducing the flow of information to the central decision maker.

- a) 1 and 3 are correct.
- b) 2 and 3 are correct.
- c) 1 and 2 are correct.
- d) 1 , 2 and 3 are correct.

Answer ;

d) 1 , 2 and 3 are correct.

0819. Which of the following statements are correct with regard to alcohol ?

- a) Eating while drinking alcohol reduces the amount of alcohol absorbed into the body.
- b) It affects Orthodox sleep more than it affects REM sleep.
- c) The maximum JAA limit for Flight and Cabin Crew is 40 milligrammes per 100 milliliters of blood.
- d) It is absorbed directly from the stomach and intestines into the bloodstream.

Answer ;

d) It is absorbed directly from the stomach and intestines into the bloodstream.

0820. Which of the following statements are correct with regard to the design of a check list ?

1. The longer a check list , the more it must be subdivided into logical parts.
2. The trickiest points must be placed in the middle of the check list.
3. Check lists must be designed in such a way that they can be lumped together with other tasks.
4. Whenever possible , a panel scan sequence should be applied.
5. Critical points should have redundancies.

The combination of correct statements is ;

- a) 1 , 2 and 5 are correct.
- b) 1 , 4 and 5 are correct.
- c) 1 , 3 and 5 are correct.
- d) 1 , 2 and 3 are correct.

Answer ;

b) 1 , 4 and 5 are correct.

0821. Which of the following statements are correct with regard to the design of a check list ?

1. The longer a check list , the more it must be subdivided into logical parts.
2. The trickiest points must be placed in the middle of the check list.
3. Check lists must be designed in such a way that they can be lumped together with other tasks.
4. Whenever possible , a panel scan sequence should be applied.
5. Use should be made of upper case/"bold"/italics/colour with care to emphasize critical items or points.

The combination of correct statements is ;

- a) 1 , 3 and 5 are correct.
- b) 1 , 2 and 3 are correct.
- c) 1 , 2 and 5 are correct.
- d) 1 , 4 and 5 are correct.

Answer ;

d) 1 , 4 and 5 are correct.

0822. Which of the following statements best characterise a self-centered cockpit ?

- a) The egoistic and self-centered personality of the captain often leads to a synergetic cockpit.
- b) Without taking note of what the other members are doing , each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on.
- c) While decreasing communication , the independence of each member bolsters the crew's synergy.
- d) The communication between crew members always increases when the captain takes charge of a situation.

Answer ;

b) Without taking note of what the other members are doing , each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on.

0823. Which of the following statements best characterise a synergetic cockpit ?
1. Decisions are taken by the captain , but prepared by the crew.
 2. There is little delegating of tasks.
 3. Communications are few in number but precise and geared purely to the flight.
 4. Fluid , consensual boundaries exist with regard to leadership-style.
- a) 2 , 3.
b) 2 , 4.
c) 1 , 4.
d) 1 , 3 , 4.

Answer ;

c) 1 , 4.

0824. Which of the following statements best fits the definition of an active error ? Active error is ;
- a) produced by the operator and can be rapidly detected.
 - b) rare in front-line actions and difficult to detect owing to the fact that it usually occurs in a complex system of uncontrolled and involuntary deviations.
 - c) produced either by a front-line operator or by a remote operator and results in a hidden or latent consequence at a specific moment of the action.
 - d) essentially results from the application of a bad rule or the poor application of a good rule by aeroplane designers.

Answer ;

a) produced by the operator and can be rapidly detected.

0825. Which of the following statements concerning barotrauma are correct ?
- a) It is caused by an increase in the partial pressure of oxygen associated with a decrease in altitude.
 - b) It is caused by pressure differentials between gases in hollow cavities of the body and the ambient pressure.
 - c) It is mainly associated with a sink rate which exceeds the ability of the body to balance its internal pressures.
 - d) It is more likely to occur during ascent then during a rapid descent.

Answer ;

b) It is caused by pressure differentials between gases in hollow cavities of the body and the ambient pressure.

0826. Which of the following statements concerning check list is correct ?
- a) All the items of a check list are equally important , their sequence is of no importance.
 - b) The most important items must be placed in the middle of check list so that they come to be examined once attention is focused but before concentration starts to wane.
 - c) The most important items should be placed at the beginning of a check list since attention is usually focused here.
 - d) The most important items must be placed at the end of check list , allowing them to be kept near at hand so that they are quickly available for any supplementary check.

Answer ;

c) The most important items should be placed at the beginning of a check list since attention is usually focused here.

0827. Which of the following statements concerning communication is valid ?
- a) Word order is of little importance to its success. Only the words uttered are important.
 - b) Professional communication means : using a restricted and specific language , tailored to minimize misunderstandings.
 - c) Communication must take priority over any other flight activity under all circumstances.
 - d) Professional communication means to exchange information as little as possible.

Answer ;

b) Professional communication means : using a restricted and specific language , tailored to minimize misunderstandings.

0828. Which of the following statements concerning conflicts is correct ?
- a) Conflicts are negative in themselves and can only lead to a general detachment of involved parties.
 - b) The emergence of a conflict always results from calling into question the general abilities of one of the involved parties.
 - c) Conflict management involves the participation of all involved parties in finding an acceptable collective solution.
 - d) Whatever the cause of the conflict , its resolution must necessarily involve an additional party if it is to be effective.

Answer ;

c) Conflict management involves the participation of all involved parties in finding an acceptable collective solution.

0829. Which of the following statements concerning hypovigilance is correct ?

Hypovigilance ;

- a) essentially occurs several minutes after the intense take-off phase.
- b) tends to occur at the end of the mission as a result of a relaxation in the operators's attention.
- c) may occur at any moment of the flight.
- d) only affects certain personality types.

Answer ;

c) may occur at any moment of the flight.

0830. Which of the following statements concerning hypovigilance is correct ?

Hypovigilance ;

- a) usually occurs a number of minutes after a stressful take-off.
- b) only affects certain personality types.
- c) may occur at any time in the flight.
- d) tends to occur at the end of the mission as a result of a relaxation in the crew's attention.

Answer ;

c) may occur at any time in the flight.

0831. Which of the following statements concerning hypoxia is correct ?

- a) It is a potential threat to safety.
- b) It has little effect on the body , because the body can always compensate for it.
- c) It activates the senses and makes them function better.
- d) It is never a problem at altitudes below 25.000 ft.

Answer ;

a) It is a potential threat to safety.

0832. Which of the following statements concerning information is correct ?

- a) Hearing is the sense which collects most information in man.
- b) 40% of information processed by man enters via the visual channel.
- c) The kinesthetic channel provides the most important information for flying.
- d) 70% of information processed by man enters via the visual channel.

Answer ;

d) 70% of information processed by man enters via the visual channel.

0833. Which of the following statements concerning stress are true ?

- 1. Adaptation is a new state of equilibrium after having coped with a stressful situation.
 - 2. An individual's anticipation of the situation and his/her perceived abilities to cope with it will determine the type and strength of stress.
- a) 1 and 2 are both false.
 - b) 1 is false , 2 is true.
 - c) 1 is true , 2 is false.
 - d) 1 and 2 are both true.

Answer ;

d) 1 and 2 are both true.

0834. Which of the following statements concerning tiredness is correct ?

- a) Tiredness always effect people of the same age in a similar way.
- b) Tiredness is the consequence of poor performance.
- c) Tiredness is always the result of an intellectual overload.
- d) Tiredness is a subjective sensation which may result in hypovigilance.

Answer ;

d) Tiredness is a subjective sensation which may result in hypovigilance.

0835. Which of the following statements fits best the concept of latent error ?

- a) Latent errors are mainly associated with the behaviour of front-line operators and are only detected after advanced problem-solving.
- b) Latent errors are detected rapidly.
- c) Latent errors are rarely made by front-line operators , and are consequently readily identified and detected by the monitoring , detection and warning links.
- d) Latent errors may have been present in the system for a certain length of time and are difficult to identify as a result of the time lag between the generation and the occurrence of the error.

Answer ;

d) Latent errors may have been present in the system for a certain length of time and are difficult to identify as a result of the time lag between the generation and the occurrence of the error.

0836. Which of the following statements fits best the concept of latent error ? Latent errors ;
- a) have been present in the system for a certain length of time and are difficult to identify as a result of the time lag between the generation and the occurrence of the error.
 - b) are rarely made by front-line operators , and are consequently readily identified and detected by the monitoring , detection and warning links.
 - c) rapidly may be detected via their immediate consequences on the action in progress.
 - d) are mainly associated with the behaviour of front-line operators and are only detected after advanced problem-solving.

Answer ;

- a) have been present in the system for a certain length of time and are difficult to identify as a result of the time lag between the generation and the occurrence of the error.
-

0837. Which of the following statements in regard to motivation is correct ?

- a) Low motivation will guarantee adequate attention management capabilities.
- b) Extremely high motivation in combination with excessive stress will limit attention management capabilities.
- c) Too much motivation may result in hypovigilance and thus in a decrease in attention.
- d) Motivation will reduce the task automation process.

Answer ;

- b) Extremely high motivation in combination with excessive stress will limit attention management capabilities.
-

0838. Which of the following statements is correct ?

The blood-pressure which is measured during flight medical checks is the pressure ;

- a) in the artery of the upper arm (representing the pressure at heart level).
- b) in the muscles of the upper arm.
- c) in the veins of the upper arm.
- d) in all the blood-vessels of the body (representing the pressure in the whole body).

Answer ;

- a) in the artery of the upper arm (representing the pressure at heart level)
-

0839. Which of the following statements is correct ?

- a) 40% of information processed by man enters via the visual channel.
- b) The kinesthetic channel provides the most important information for flying.
- c) Hearing is the sense which collects most information in man.
- d) 70% of information processed by man enters via the visual channel.

Answer ;

- d) 70% of information processed by man enters via the visual channel.
-

0840. Which of the following statements is correct ?

- a) Short-term memory is affected by interruptions.
- b) All motivators (both physical and mental) are subjective.
- c) Motivation does not affect performance.
- d) Extreme “achievement” motivation in a Captain is an ideal attribute for a Captain of an aircraft.

Answer ;

- a) Short-term memory is affected by interruptions.
-

0841. Which of the following statements is correct ?

- a) Scotopic Vision is vision through the operation of the Rods.
- b) Scotopic Vision is the same as Photopic Vision.
- c) Photopic Vision is the same as Peripheral Vision.
- d) Scotopic Vision is vision through the operation of the Cones.

Answer ;

- a) Scotopic Vision is vision through the operation of the Rods.
-

0842. Which of the following statements is correct ?

- 1. Psychosomatic means that mental and/or emotional stressors can be manifested in physical reactions.
 - 2. Psychosomatic means that a physical problem is always followed by psychological stress.
- a) 1 is correct , 2 is false.
 - b) 1 is false , 2 is correct.
 - c) 1 and 2 are both correct.
 - d) 1 and 2 are both false.

Answer ;

- a) 1 is correct , 2 is false.
-

0843. Which of the following statements is correct concerning flight in an environment of low contrast (fog , snow , darkness , haze) ?

- a) It is impossible to detect objects.
- b) It is unlikely that visual illusions occur.
- c) It is difficult to estimate the correct speed and size of approaching objects.
- d) There is no problem to estimate the correct speed and size of approaching objects.

Answer ;

c) It is difficult to estimate the correct speed and size of approaching objects.

0844. Which of the following statements is correct regarding decision making ?

- a) Deciding means imposing one's point of view.
- b) Deciding means being able to come up with original solutions.
- c) Deciding means choosing between alternatives.
- d) Deciding means applying an automatic procedure.

Answer ;

c) Deciding means choosing between alternatives.

0845. Which of the following statements is false ?

- a) Human conflict tends to escalate.
- b) The initial effects of carbon monoxide poisoning are not immediately alarming.
- c) Behaviour is generally the outcome of personality and attitude.
- d) Stress has essentially negative effects and the human will operate efficiently with an absence of stress.

Answer ;

d) Stress has essentially negative effects and the human will operate efficiently with an absence of stress.

0846. Which of the following statements is true ?

- a) People are capable of living without stress.
- b) Stressors accumulate thus increasing the likelihood of exhaustion.
- c) Stress should always be avoided under any circumstances.
- d) Stressors are independent from each other.

Answer ;

b) Stressors accumulate thus increasing the likelihood of exhaustion.

0847. Which of the following statements is true concerning regular physical exercise ;

- 1. it increases pulse rate and stroke volume in the short term.
 - 2. it reduces blood pressure and heart rate in the long term.
 - 3. it reduces the human tolerance to hypoxia.
- a) 1 , 3.
 - b) 1 , 2 , 3.
 - c) 1 , 2.
 - d) 1.

Answer ;

b) 1 , 2 , 3.

0848. Which of the following statements is true concerning the pulse rate ?

- a) It is the only indicator of hypoxia.
- b) It is not affected by emotion.
- c) When multiplied by the blood pressure gives the cardiac output.
- d) It has a normal range of 60-100 beats per minute.

Answer ;

d) It has a normal range of 60-100 beats per minute.

0849. Which of the following statements is/are correct ?

- 1. A person experiencing sleep loss is unlikely to be aware of personal performance degradation.
 - 2. Performance loss may be present up to 20 minutes after awaking from a short sleep (nap).
- a) 1 and 2 are both correct.
 - b) 1 is correct , 2 is false.
 - c) 1 and 2 are both false.
 - d) 1 is false , 2 is correct.

Answer ;

a) 1 and 2 are both correct.

0850. Which of the following statements is/are correct ?

1. REM-sleep becomes shorter with any repeated sleep cycle during the night.
 2. REM-sleep is more important for the regeneration of mental functions than all the other sleep stages are.
- a) 1 is false , 2 is correct.
b) 1 and 2 are both correct.
c) 1 and 2 are false.
d) 1 is correct , 2 is false.

Answer ;

- a) 1 is false , 2 is correct.
-

0851. Which of the following statement(s) is/are correct ?

1. The retina has rods in its peripheral zone and cones in its central zone.
 2. The retina has cones and the crystalline lens has rods.
 3. The rods allow for night-vision.
 4. The cones are located on the peripheral zone of the retina.
- a) 2 , 3.
b) 1 , 3.
c) 1.
d) 4.

Answer ;

- b) 1 , 3.
-

0852. Which of the following statements regarding interpersonal interactions are correct ?

1. If the sender perceives that the receiver is incompetent , he/she will increase the length of the message.
2. If the receiver is of non-native tongue , the sender will reinforce what he is saying by using more complicated words so as to optimize understanding.
3. If the sender considers the receiver incompetent , he/she tends to simplify the content of the sentences.
4. It is much safer to fly with a crew who know each other well because communication will always be excellent.

The correct statement(s) is (are) ;

- a) 2 and 3 are correct.
b) 1 and 2 are correct.
c) 3 only is correct.
d) 3 and 4 are correct.

Answer ;

- c) 3 only is correct.
-

0853. Which of the following statements summarises the impact that motivation may have on attention ?

- a) It stimulates attention but may lead to phases of low arousal.
b) It only facilitates attention in extreme cases (risk of death).
c) Motivation has only a small effect on attention , but it facilitates alertness.
d) It increases alertness and attention.

Answer ;

- d) It increases alertness and attention.
-

0854. Which of the following statements with regard to motivation is correct ?

- a) Excessive motivation in combination with high levels of stress will limit attention management capabilities.
b) Motivation will reduce risk.
c) Low motivation will guarantee adequate attention management capabilities.
d) Too much motivation may result in hypovigilance and thus in a decrease in attention.

Answer ;

- a) Excessive motivation in combination with high levels of stress will limit attention management capabilities.
-

0855. Which of the following symptoms can indicate hypoxia ?

1. Blue lips and finger nails.
 2. Euphoria.
 3. Flatulence.
 4. Unconsciousness..
- a) 1 , 3 and 4 are correct.
b) 1 , 2 and 3 are correct.
c) 2 , 3 and 4 are correct.
d) 1 , 2 and 4 are correct.

Answer ;

- d) 1 , 2 and 4 are correct.
-

0856. Which of the following symptoms can mark the onset of hyperventilation ?

- a) Slow heart beat.
- b) Cyanosis (blueing of lips and finger nails).
- c) Slow rate of breathing.
- d) Dizzy feeling.

Answer ;

d) Dizzy feeling.

0857. Which of the following symptoms may a pilot experience when subjected to hypoxia ?

- 1. Fatigue.
 - 2. Euphoria.
 - 3. Lack of concentration.
 - 4. Pain in the joints.
 - 5. Pain in the inner ear.
- a) Only 5 is false.
 - b) 4 and 5 are correct.
 - c) 1 , 2 , 3 and 4 are correct.
 - d) 1 , 2 and 3 are correct.

Answer ;

d) 1 , 2 and 3 are correct.

0858. Which of the following systems are involved in motion sickness ?

- 1. Hearing.
 - 2. The vestibular system.
 - 3. Vision.
 - 4. The proprioceptive senses "Seat-of-the-Pants-Sense".
 - 5. The gastrointestinal system.
- a) 1 , 2 , 3.
 - b) 1 , 2 , 5.
 - c) 2 , 3 , 4 , 5.
 - d) 2 , 3 , 4.

Answer ;

c) 2 , 3 , 4 , 5.

0859. Which of the following tasks are possible to do simultaneously without mutual interference ?

- a) Read and listen attentively.
- b) Talk and rehearse a frequency in working memory.
- c) Maintain manual straight and level flight and solve a problem.
- d) Listen attentively and solve a problem.

Answer ;

c) Maintain manual straight and level flight and solve a problem.

0860. Which part of the ear could be affected by air pressure changes during climb and/or descent ?

- a) The cochlea.
- b) The saccules and utricles.
- c) The semicircular canals.
- d) The Eustachian tube and the tympanic membrane (ear drum).

Answer ;

d) The Eustachian tube and the tympanic membrane (ear drum).

0861. Which part of the inner ear is responsible for the perception of sound ?

- a) The cochlea.
- b) The Eustachian tube.
- c) The semicircular canals.
- d) The sacculus and utriculus.

Answer ;

a) The cochlea.

0862. Which part of the vestibular apparatus is affected by changes in gravity and linear acceleration ?

- a) The semicircular canals.
- b) The cochlea.
- c) The sacculus and utriculus.
- d) The Eustachian tube.

Answer ;

c) The sacculus and utriculus.

0863. Which part of the vestibular apparatus is responsible for the detection of angular acceleration ?

- a) The sacculus and utriculus.
- b) The semicircular canals.
- c) The cochlea.
- d) The Eustachian tube.

Answer ;

b) The semicircular canals.

0864. Which problem may occur , when flying in an environment of low contrast (fog , snow , darkness , haze) ? Under these conditions it is ;

- a) difficult to estimate the correct speed and size of approaching objects.
- b) impossible to detect objects.
- c) no problem to estimate the correct speed and size of approaching objects.
- d) unlikely that visual illusions occur.

Answer ;

a) difficult to estimate the correct speed and size of approaching objects.

0865. Which procedure is recommended to prevent or overcome spatial disorientation ?

- a) Get adapted to low levels of illumination before flying and use off-center vision all the time.
- b) Tilt your head to the side to get better information from the semicircular canals.
- c) Rely on the Seat-of-the-Pants-Sense.
- d) Rely entirely on the indications of the flight instruments.

Answer ;

d) Rely entirely on the indications of the flight instruments.

0866. Which scanning technique should be used when flying at night ?

- a) Blink your eyes.
- b) Look directly at the object.
- c) Look with one eye.
- d) Look to the side (10 - 15 deg) of the object.

Answer ;

d) Look to the side (10 - 15 deg) of the object.

0867. Which sensations does a pilot get , when he is rolling out of a prolonged level turn ?

- a) Flying straight and level.
- b) Turning into the original direction.
- c) Turning in the opposite direction.
- d) Climbing.

Answer ;

c) Turning in the opposite direction.

0868. Which statement about acute or chronic fatigue is INCORRECT ?

- a) Acute fatigue is felt after a period of significant exertion or emotional excitement.
- b) Acute fatigue generally has psychological roots.
- c) Chronic fatigue may be caused by inadequate recuperation from periods of acute fatigue.
- d) Chronic fatigue may lead to a person being totally apathetic and indifferent to what goes on around them.

Answer ;

b) Acute fatigue generally has psychological roots.

0869. Which statement applies to hypoxia ?

- a) It is possible to predict when , how and where hypoxia reaction starts to set in.
- b) Carbon monoxide increases the tolerance of the brain to oxygen deficiency.
- c) Sensitivity and reaction to hypoxia varies from person to person.
- d) You may become immune to hypoxia when exposed repeatedly to hypoxia.

Answer ;

c) Sensitivity and reaction to hypoxia varies from person to person.

0870. Which statement is correct ?

- a) Inconsistent communication improves flight safety.
- b) Personal conflict that takes place prior to take-off should wait to be addressed until the end of the flight.
- c) There is no relation between inadequate communication and incidents or accidents.
- d) Problems in the personal relationships between crew members are likely to hamper the communication process.

Answer ;

d) Problems in the personal relationships between crew members are likely to hamper the communication process.

0871. Which statement is correct ?

- a) Oxygen diffusion from the blood into the cells depends on the partial oxygen pressure gradient.
- b) Oxygen is mainly transported in plasma.
- c) Oxygen diffusion from the lungs into the blood does not depend on partial oxygen pressure.
- d) The gradient of diffusion is higher at altitude than it is at sea-level.

Answer ;

- a) Oxygen diffusion from the blood into the cells depends on the partial oxygen pressure gradient.
-

0872. Which statement is correct ?

- 1. Smokers have a greater chance of suffering from coronary heart disease.
 - 2. Smoking tobacco will raise the individuals physiological altitude during flight.
 - 3. Smokers have a greater chance of contracting lung cancer.
- a) 1 , 2 and 3 are correct.
 - b) 1 and 2 are correct , 3 is false.
 - c) 2 and 3 are correct , 1 is false.
 - d) 1 and 3 are correct , 2 is false.

Answer ;

- a) 1 , 2 and 3 are correct.
-

0873. Which statement is correct regarding alcohol in the human body ?

- a) Judgement and decision making can be affected even by a small amount of alcohol.
- b) A small amount of alcohol increases visual acuity.
- c) An increase of altitude decreases the adverse effect of alcohol.
- d) When drinking coffee , the human body metabolizes alcohol at a faster rate than normal.

Answer ;

- a) Judgement and decision making can be affected even by a small amount of alcohol.
-

0874. Which symptom does not belong to the following list ;

Bends , Chokes , Creeps , Leans

- a) Leans
- b) Chokes
- c) Creeps
- d) Bends

Answer ;

- a) Leans
-

0875. Which symptom of hypoxia is the most dangerous for conducting safe flight ?

- a) The degradation of reasoning and perceptive functions.
- b) Lack of adaptation.
- c) Dizziness.
- d) Lack of accommodation.

Answer ;

- a) The degradation of reasoning and perceptive functions.
-

0876. While flying VFR “wings level” on top of clouds at 10.000 feet , your artificial horizon indicates a 20 degree bank.

- a) No attitude correction is needed since you know that your vision is more reliable than the instrument with regards to spatial orientation.
- b) You trust the instrument and level off according to the instrument read-out.
- c) You take the proper action by “leaning” toward the lower end of the artificial horizon , thus compensating for the perceptual illusion.
- d) You pull the circuit breaker to the artificial horizon knowing that perceptual illusions called “leans” might severely impair your control of the aircraft.

Answer ;

- b) You trust the instrument and level off according to the instrument read-out.
-

0877. Whilst flying a coordinated turn , most of your activity is ;

- a) skill based behaviour.
- b) coping behaviour.
- c) knowledge based behaviour.
- d) rule based behaviour.

Answer ;

- a) skill based behaviour.
-

0878. Who in the Aviation Industry is responsible for flight safety ?

- a) Aircrew.
- b) Aircrew and Groundcrew.
- c) Everyone involved.
- d) Management , Aircrew , Groundcrew and ATC.

Answer ;

- c) Everyone involved.**
-

0879. Why are Mnemonics are used in aviation ?

- a) Because they help to increase retention of information.
- b) Because they increase response time.
- c) Because they are a tool to avoid illusions.
- d) Because they increase motivation.

Answer ;

- a) Because they help to increase retention of information.**
-

0880. Why is hypoxia especially dangerous for pilots flying solo ?

- a) Difficulties in controlling the aircraft.
- b) Since the first signs of hypoxia are generally hard to detect.
- c) A solo pilot is more prone to illusions.
- d) Hypoxia improves night vision therefore the pilot will have no indication of danger.

Answer ;

- b) Since the first signs of hypoxia are generally hard to detect.**
-

0881. Why must flight safety considerations consider the human error mechanism ?

1. It is analysis of an incident or accident which will make it possible to identify what error has been committed and by whom. It is the process whereby the perpetrator is made responsible which may lead to elimination of the error.
2. If we have a better understanding of the cognitive error mechanism , it will be possible to adapt procedures , aircraft interfaces , etc.
3. It is error management procedure which enables us to continuously adjust our actions. The better we understand the underlying mechanism of an error , the better will be our means for detecting and reducing future errors.
4. Since error is essentially human , once it has been identified by the use of procedures , a person will be able to anticipate and deal with it automatically in the future.

The correct statement(s) is(are) ;

- a) 1 and 4.
- b) 2 and 3.
- c) 3 and 4.
- d) 2 and 4.

Answer ;

- b) 2 and 3.**
-

0882. With a heart rate of 72 beats per minute and a stroke volume of 70 ml the cardiac output is about ;

- a) 8 litres/min.
- b) 5 litres/min.
- c) 6 litres/min.
- d) 7 litres/min.

Answer ;

- b) 5 litres/min.**
-

0883. With hyperventilation , caused by high levels of arousal or overstress ;

- a) finger nails and lips will turn blue ("cyanosis").
- b) more oxygen will reach the brain.
- c) peripheral and scotopic vision will be improved.
- d) an increased amount of carbon dioxide is exhaled causing muscular spasms and even unconsciousness.

Answer ;

- d) an increased amount of carbon dioxide is exhaled causing muscular spasms and even unconsciousness.**
-

0884. With reference to the relationship between arousal and performance , the highest level of performance is achieved when the arousal level is ;

- a) lowest.
- b) half way up the arousal curve.
- c) half way down the arousal curve.
- d) optimum.

Answer ;

d) optimum.

0885. With reference to the SHELL Model , S represents ;

- a) Security (which includes safety).
- b) Software (which includes check lists).
- c) Symbolology (which includes safety).
- d) Self-awareness (which includes safety).

Answer ;

b) Software (which includes check lists)

0886. With regard to central vision , which of the following statements are correct ?

- 1. It is due to the functioning of rods.
 - 2. It enables details , colours and movement to be seen.
 - 3. Its very active both during the day and at night.
 - 4. It represents a zone where about 150.000 cones per mm are located to give high resolution capacity.
- a) 1 , 2 , 4.
 - b) 1 , 3.
 - c) 2 , 4.
 - d) 2 , 3 , 4.

Answer ;

c) 2 , 4.

0887. With regard to communication in a cockpit , we can say that ;

- a) all the characteristics of communication , namely output , duration , precision , clarity , etc. are stable and are not much affected by changes in workload.
- b) communication uses up resources , thus limiting the resources allocated to work in progress.
- c) communication is always sufficiently automated to enable an activity with a high workload element to be carried out at the same time.
- d) communication is only effective if messages are kept short and sufficiently precise to limit their number.

Answer ;

b) communication uses up resources , thus limiting the resources allocated to work in progress.

0888. With regard to decompression sickness associated with flight , we know that ;

- a) physical activity after decompression reduces the risks of decompression sickness symptoms
- b) scuba diving does not pose any problem for a subsequent flight
- c) age , obesity and scuba diving are risk factors
- d) gender is the prime risk factor, with two out of every three women being sensitive to it

Answer ;

c) age , obesity and scuba diving are risk factors

0889. With regard to short-term memory , we can say that ;

- a) it is made up of everyday information for immediate use , and is limited in its capacity for storing and retaining data.
- b) it is made up of everyday information for immediate use , and is limited in terms of the time for which it retains data but not in its storage capacity.
- c) it is a stable form of working memory , and thus not very sensitive to any disturbance.
- d) it stores mainly procedural knowledge (skills).

Answer ;

a) it is made up of everyday information for immediate use , and is limited in its capacity for storing and retaining data.

0890. With regard to short-term memory , we can say that ;

- a) it holds information for immediate use , and is limited in terms of the time for which it retains data but not in its storage capacity.
- b) it is a stable form of working memory , and thus not very sensitive to any disturbance.
- c) it holds information for immediate use , and is limited in its capacity for storing and retaining data.
- d) it stores mainly procedural knowledge (skills).

Answer ;

c) it holds information for immediate use , and is limited in its capacity for storing and retaining data.

-
0891. With regard to stress , as it affects human beings , which of the following statements is correct ?
- a) Self imposed obligations will not create stress.
 - b) Stressors relate purely to a pilot's physical condition.
 - c) All forms of stress should be avoided.
 - d) "Stress" is a term used to describe how a person reacts to demands placed upon him/her.

Answer ;

- d) "Stress" is a term used to describe how a person reacts to demands placed upon him/her.
-

0892. With regard to the average influence of age on pilot performance , it may be said that age ;
- a) has little impact when the pilot is able to compensate for it by his/her flight experience.
 - b) sharply reduces performance without , however , affecting cognitive capabilities.
 - c) increases in impact as speed of thought and memory deteriorate.
 - d) has a major impact owing to the impairment of memory.

Answer ;

- a) has little impact when the pilot is able to compensate for it by his/her flight experience.
-

0893. With regard to the humidity of air in current in a pressurized cabin , we know that it ;
- 1. varies between 40 and 60%.
 - 2. varies between 5 and 15%.
 - 3. may cause dehydration effecting the performance of the crew.
 - 4. has no special effects on crew members.

- a) 2 , 3 , 4.
- b) 1 , 3.
- c) 1 , 4.
- d) 2 , 3.

Answer ;

- d) 2 , 3.
-

0894. With regard to the practice of English , which of the following statements is correct ?
- a) It is necessary and sufficient to have a command of any of the official languages of the ICAO.
 - b) Be familiar with normal procedures in English since only this allows for effective management of any flight's communication.
 - c) The composition of every crew should be geared to a command of the official aeronautical language of the destination country.
 - d) All pilots should master it because the aeronautical world needs one common language.

Answer ;

- d) All pilots should master it because the aeronautical world needs one common language.
-

0895. With too much cohesion , Groupthink can be ;
- a) negative.
 - b) positive and far-reaching.
 - c) popular and positive.
 - d) positive.

Answer ;

- a) negative.
-

0896. Within communication , what element suggests that a message has been received and understood ?
- a) Feedback.
 - b) Coding.
 - c) Encoding.
 - d) Synchronization.

Answer ;

- a) Feedback.
-

0897. Without visual reference , what illusion could the pilot get , when he is stopping the rotation to recover from a spin ? He will get the illusion of ;

- a) climbing and turning into the original direction of the spin.
- b) straight and level flight.
- c) spinning into the same direction.
- d) spinning into the opposite direction.

Answer ;

- d) spinning into the opposite direction.
-

0898. Working memory ;

- a) is unlimited in size.
- b) is unlimited in duration.
- c) is sensitive to interruptions which may erase all or some of its contents.
- d) varies considerably in size between an expert pilot and a novice pilot.

Answer ;

- c) is sensitive to interruptions which may erase all or some of its contents.
-

0899. Working memory enables us , for example ;

- a) to remember our own name.
- b) to remember a clearance long enough to write it down.
- c) to store a large amount of visual information for about 0.5 seconds.
- d) to ignore messages for other aircraft.

Answer ;

- b) to remember a clearance long enough to write it down.
-

0900. Workload essentially depends on ;

- a) the current situation , the pilot's expertise and the ergonomics of the system.
- b) the pilot's experience and the ergonomics of the system.
- c) the pilot's knowledge.
- d) the task and the day's parameters (weather report , aircraft load , type of flight , etc).

Answer ;

- a) the current situation , the pilot's expertise and the ergonomics of the system.
-

0901. You are crossing the Alps in a non-pressurised aircraft at an altitude of 15.000 feet. You do not use the oxygen mask because you feel fine. This is unsafe , because ;

- a) the blood-pressure can get too low.
- b) you will get the bends.
- c) your judgement could be impaired.
- d) the blood-pressure can get too high.

Answer ;

- c) your judgement could be impaired.
-

0901. You can overcome hyperventilation by breathing into a plastic or paper bag. The intention is ;

- a) to reduce blood pressure.
- b) to prevent you from exhaling too much oxygen.
- c) to increase the amount of nitrogen in the lungs.
- d) to raise the level of CO₂ in the blood as fast as possible.

Answer ;

- d) to raise the level of CO₂ in the blood as fast as possible.
-

0902. You can survive at any altitude , provided that ;

- a) enough oxygen , pressure and heat is available.
- b) the temperature in the cabin does not drop below 10°C.
- c) pressure respiration is guaranteed for that altitude.
- d) 21% oxygen is available in the air you breath in.

Answer ;

- a) enough oxygen , pressure and heat is available.
-

0903. You climb from 0 to 50.000 ft and measure the decrease of the pressure per 5.000 ft. The absolute difference in barometric pressure is greatest between ;

- a) 45.000 and 50.000 feet.
- b) 10.000 and 15.000 feet.
- c) 5.000 and 10.000 feet.
- d) 0 and 5.000 feet.

Answer ;

- d) 0 and 5.000 feet.
-

0904. You fly VFR from your home base (runway width 27 m) , to an international airport (runway width 45 m). On reaching your destination there is a risk of performing a ;

- a) low approach with overshoot.
- b) high approach with overshoot.
- c) high approach with undershoot.
- d) low approach with undershoot.

Answer ;

- b) high approach with overshoot.

0905. You fly VFR from your home base (runway width 45 m) to a small airfield (runway width 27 m). On reaching your destination there is a risk of performing a ;

- a) high approach with overshoot.
- b) low approach with overshoot.
- c) low approach with undershoot.
- d) high approach with undershoot.

Answer ;

c) low approach with undershoot.

0906. You suffered a rapid decompression without the appearance of any decompression sickness symptoms. What should you do before flying ?

- a) seek aeromedical advice only if the symptoms become apparent.
- b) delay your next flight for 6 hours.
- c) nothing , you may fly immediately.
- d) seek prompt aeromedical advice.

Answer ;

d) seek prompt aeromedical advice.
